Supporting Information for Spline-Fitting with a Genetic Algorithm: a Method for Developing Classification Structure-Activity Relationships, J.J. Sutherland, L.A. O'Brien and D.F. Weaver.

A. Datasets

Table 1.1. Structures (families) of COX-2 inhibitors

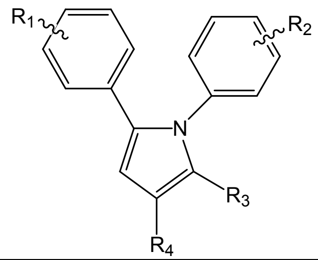
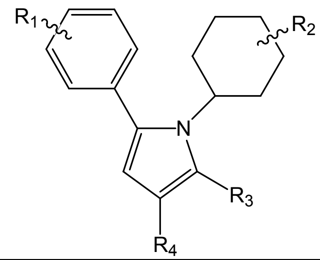
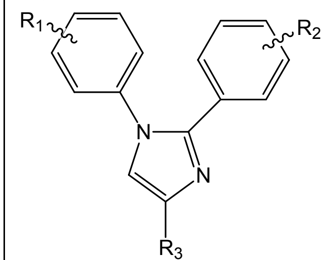
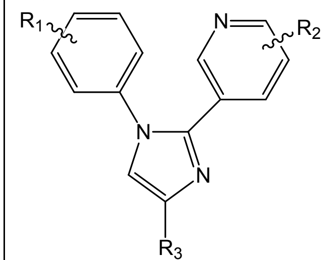
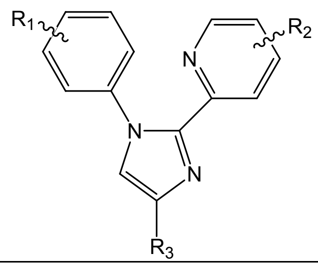
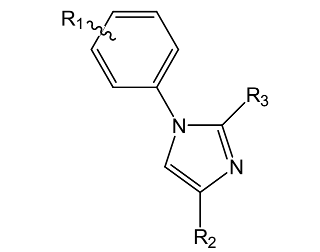
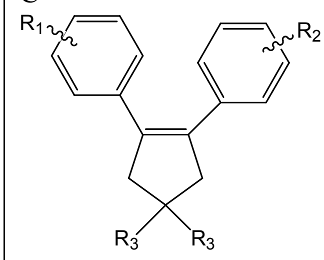
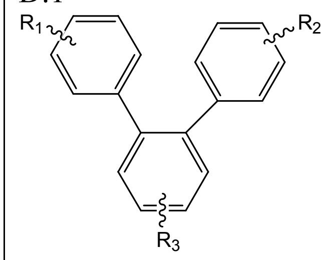
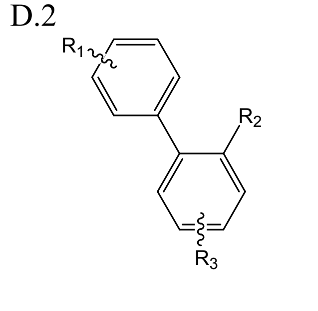
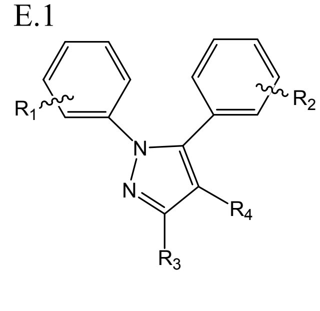
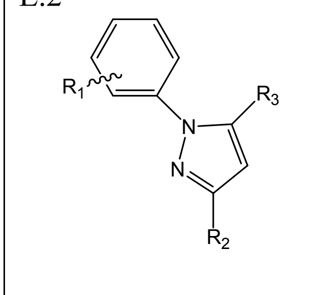
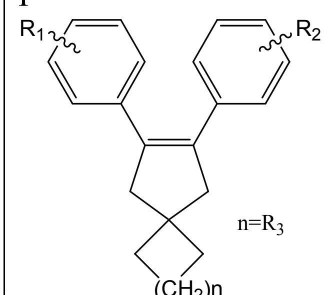
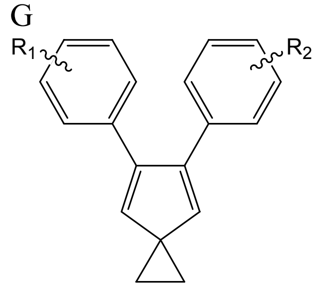
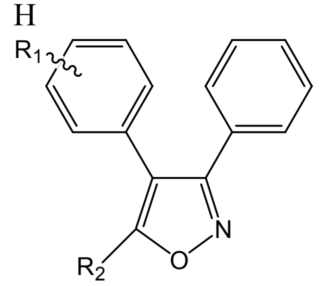
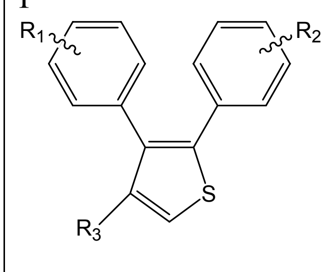
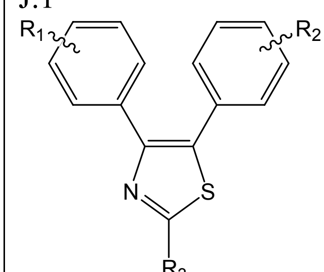
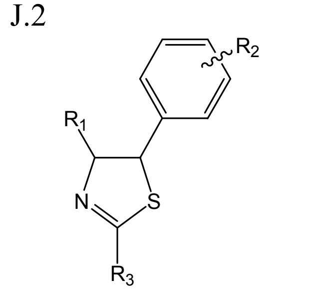
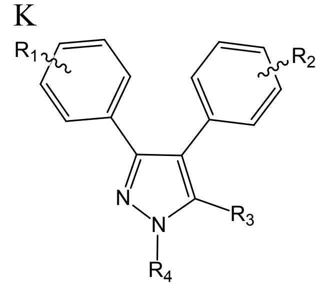
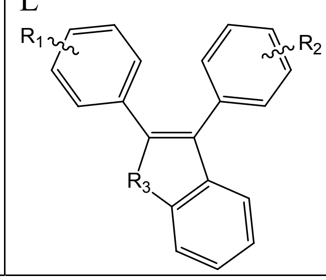
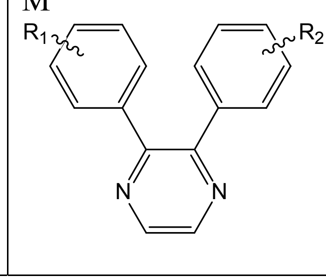
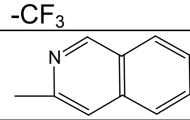
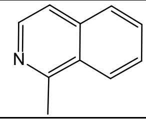
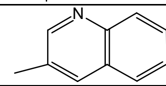
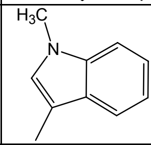
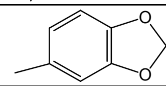
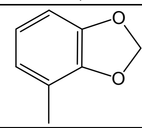
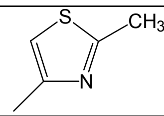
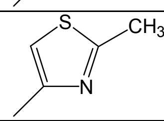
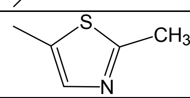
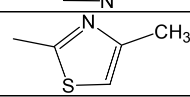
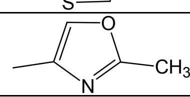
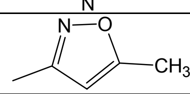
A.1 	A.2 	B.1 	B.2 
B.3 	B.4 	C 	D.1 
D.2 	E.1 	E.2 	F 
G 	H 	I 	J.1 
J.2 	K 	L 	M 

Table 1.2. COX-2 inhibitors

Name[a]	R_1	R_2	R_3	R_4	Family	IC_{50} (μM)	set[b]
1-1	-4-SO_2Me	-4-F	-Me	-H	A.1	0.06	1
1-3	-4-F	-4-SO_2Me	-H	-H	A.1	0.51	2
1-4	-4-SO_2Me	-4-F	-H	-H	A.1	10.2	1
1-5	-4-SO_2Me	-H	-Me	-H	A.1	0.06	1
1-6	-4-SO_2Me	-4-CF_3	-Me	-H	A.1	0.08	1
1-7	-4-SO_2Me	-4-Me	-Me	-H	A.1	0.04	0
1-8	-4-SO_2Me	-4COMe	-Me	-H	A.1	2.87	2
1-9	-4-SO_2Me	-3,4-F	-Me	-H	A.1	0.25	1
1-10	-4-SO_2Me	-4-F	-Et	-H	A.1	>100	1
1-16	-4-SO_2Me	-3-SO_2Me	-H	-H	A.1	>100	2
1-17	-4-SO_2Me	-4SO_2Et	-H	-H	A.1	>100	1
1-18	-4-SO_2Me	-4SO_2Ph	-H	-H	A.1	>100	1
1-19	-4-SO_2Me	-3-Cl-4-SO_2Me	-H	-H	A.1	>100	2
1-20	-4-F	-4-SO_2NH_2	-H	-H	A.1	0.014	2
1-21	-4-SO_2Me	-4-SO_2NHMe	-H	-H	A.1	>100	1
1-22	-4-SO_2Me	-4-SO_2NMe2	-H	-H	A.1	>100	2
1-26	-4-SO_2Me	-4-F	-$CH_2CH_2CO_2H$	-H	A.1	>100	2
1-27	-4-SO_2Me	-4-F	-CF_3	-H	A.1	>100	2
1-28	-4-SO_2Me	-4-F	-Me	-$COCF_3$	A.1	0.12	1
1-29	-4-SO_2Me	-4-F	-Me	-COMe	A.1	1.61	1
1-30	-4-SO_2Me	-4-F	-Me	-COPh	A.1	1.02	1
1-31	-4-SO_2Me	-4-F	-Me	-SO_2CF_3	A.1	0.06	2
1-32	-4-SO_2Me	-4-F	-Me	-CHO	A.1	3.23	2
1-33	-4-SO_2Me	-4-F	-Me	-CN	A.1	0.75	2
1-34	-4-SO_2Me	-4-F	-Me	-Br	A.1	0.02	1
1-35	-4-SO_2Me	-4-F	-Me	-Cl	A.1	0.05	0
1-36	-4-SO_2Me	-4-F	-Me	-CH_2NMe_2	A.1	>100	1
1-37	-4-SO_2Me	-4-F	-Me	-CH_2OAc	A.1	0.47	2
1-38	-4-SO_2Me	-4-F	-Me	-CH_2OH	A.1	3.88	1
1-39	-4-SO_2Me	-4-F	-Me	-$CH_2OC_6H_4$-4-Cl	A.1	0.03	2
1-40	-4-SO_2Me	-4-F	-Me	-$CH_2OC_6H_4$-3-Cl	A.1	0.08	0
1-41	-4-SO_2Me	-4-F	-Me	-CH(OH)CF_3	A.1	1.44	1
1-42	-4-SO_2Me	-4-F	-Me	-CH_2CF_3	A.1	0.14	2
1-43	-4-SO_2Me	-4-F	-Me	-CH(CF_3)OC_6H_4-3-Cl	A.1	>100	1
2-20	-4-SO_2Me	-4-F	-Me	-H	A.2	0.52	2
3-5	-4-SO_2Me	-4-Cl	-Me		B.1	0.24	1
3-6	-4-SO_2Me	-4-Cl	-CF_3		B.1	0.11	1
3-9	-4-SO_2Me	-4-F	-CF_3		B.1	0.1	0
3-10	-4-SO_2Me	-H	-CF_3		B.1	0.12	0
3-11	-4-SO_2Me	-4-Me	-CF_3		B.1	0.16	0
3-12	-4-SO_2Me	-4-OMe	-CF_3		B.1	0.57	1
3-13	-4-SO_2Me	-4-NHMe	-CF_3		B.1	1.47	1
3-14	-4-SO_2Me	-4-NMe_2	-CF_3		B.1	0.7	1
3-15	-4-SO_2Me	-4-SMe	-CF_3		B.1	0.16	1
3-16	-4-SO_2Me	-4-SOMe	-CF_3		B.1	>100	1
3-17	-4-SO_2Me	-4-SO_2Me	-CF_3		B.1	5.7	0
3-18	-4-SO_2NH_2	-4-Cl	-CF_3		B.1	0.01	0
3-19	-4-SO_2NH_2	-4-F	-CF_3		B.1	0.01	0
3-20	-4-SO_2NH_2	-H	-CF_3		B.1	0.04	0
3-21	-4-SO_2NH_2	-4-Me	-CF_3		B.1	0.04	1
3-22	-4-SO_2Me	-3-Cl	-CF_3		B.1	0.06	0
3-23	-4-SO_2Me	-3-F	-CF_3		B.1	0.12	0

3-24	-4-SO_2Me	-3-Br	-CF_3		B.1	0.08	0
3-25	-4-SO_2Me	-3-Me	-CF_3		B.1	0.06	0
3-26	-4-SO_2Me	-3-CF_3	-CF_3		B.1	0.21	0
3-27	-4-SO_2Me	-3-OMe	-CF_3		B.1	0.35	1
3-28	-4-SO_2Me	-3-SMe	-CF_3		B.1	0.35	1
3-29	-4-SO_2Me	-3-CH_2OMe	-CF_3		B.1	68.1	2
3-30	-4-SO_2Me	-3-NMe_2	-CF_3		B.1	3.2	1
3-31	-4-SO_2Me	-3-NHMe	-CF_3		B.1	0.92	1
3-32	-4-SO_2Me	-3-NH_2	-CF_3		B.1	5.89	1
3-33	-4-SO_2Me	-3-NO_2	-CF_3		B.1	0.58	1
3-34	-4-SO_2NH_2	-3-Cl	-CF_3		B.1	0.008	1
3-35	-4-SO_2NH_2	-3-F	-CF_3		B.1	0.03	0
3-36	-4-SO_2NH_2	-3-Br	-CF_3		B.1	0.007	0
3-37	-4-SO_2NH_2	-3-Me	-CF_3		B.1	0.03	0
3-38	-4-SO_2Me	-2-Cl	-CF_3		B.1	0.9	0
3-39	-4-SO_2Me	-2-F	-CF_3		B.1	0.4	0
3-40	-4-SO_2Me	-2-Me	-CF_3		B.1	0.8	1
3-41	-4-SO_2Me	-2-OMe	-CF_3		B.1	>100	1
3-42	-4-SO_2NH_2	-2-F	-CF_3		B.1	0.1	1
3-43	-4-SO_2NH_2	-2-Me	-CF_3		B.1	0.2	0
3-44	-4-SO_2Me	-4-OMe-3-F	-CF_3		B.1	0.15	2
3-45	-4-SO_2Me	-4-OMe-3-Cl	-CF_3		B.1	0.13	0
3-46	-4-SO_2Me	-4-SMe-3-Cl	-CF_3		B.1	0.04	1
3-47	-4-SO_2Me	-4-NMe_2-3-Cl	-CF_3		B.1	0.32	2
3-48	-4-SO_2Me	-4-NMe_2-3-F	-CF_3		B.1	0.33	0
3-49	-4-SO_2Me	-4-NHMe-3-Cl	-CF_3		B.1	0.66	1
3-50	-4-SO_2Me	-4-Me-3-Cl	-CF_3		B.1	0.03	1
3-51	-4-SO_2Me	-4-Me-3-F	-CF_3		B.1	0.11	0
3-52	-4-SO_2Me	-3-Me-4-F	-CF_3		B.1	0.17	0
3-53	-4-SO_2Me	-3-Me-4-Cl	-CF_3		B.1	0.09	0
3-54	-4-SO_2Me	-3-OMe-4-Cl	-CF_3		B.1	0.25	1
3-55	-4-SO_2Me	-3-NMe_2-4-Cl	-CF_3		B.1	1.04	1
3-56	-4-SO_2Me	-3,4-OCH_2O-	R_2		B.1	0.17	2
3-57	-4-SO_2Me	-3,4-F	-CF_3		B.1	0.12	0
3-58	-4-SO_2Me	-3,4-Me	-CF_3		B.1	0.33	0
3-59	-4-SO_2Me	-3-Me-5-Cl	-CF_3		B.1	0.08	0
3-60	-4-SO_2Me	-3-Me-5-F	-CF_3		B.1	0.11	0
3-61	-4-SO_2Me	-3-OMe-5-F	-CF_3		B.1	0.96	1
3-62	-4-SO_2Me	-3-CF_3-5-F	-CF_3		B.1	>100	0
3-63	-4-SO_2Me	-3,5-Cl	-CF_3		B.1	0.17	0
3-64	-4-SO_2Me	-2-Me-5-F	-CF_3		B.1	>100	1
3-65	-4-SO_2Me	-2-Me-6-Cl	-CF_3		B.1	>100	0
3-66	-4-SO_2NH_2	-4-OMe-3-F	-CF_3		B.1	0.03	1
3-67	-4-SO_2NH_2	-4-OMe-3-Cl	-CF_3		B.1	0.02	0
3-68	-4-SO_2NH_2	-4-OMe-3-Br	-CF_3		B.1	0.03	0
3-69	-4-SO_2NH_2	-4-SMe-3-Cl	-CF_3		B.1	0.01	1
3-70	-4-SO_2NH_2	-4-Me-3-Cl	-CF_3		B.1	0.003	0
3-71	-4-SO_2NH_2	-3-OMe-4-Cl	-CF_3		B.1	0.02	2
3-72	-4-SO_2NH_2	-3,4-F	-CF_3		B.1	0.03	0
3-73	-4-SO_2NH_2	-3-Me-5-Cl	-CF_3		B.1	0.04	0
3-74	-4-SO_2NH_2	-3-Me-5-F	-CF_3		B.1	0.03	0
3-75	-4-SO_2NH_2	-3-OMe-5-Cl	-CF_3		B.1	0.46	1
3-76	-4-SO_2Me	-4-OMe-3,5-F	-CF_3		B.1	0.17	0
3-77	-4-SO_2Me	-4-OMe-3,5-Cl	-CF_3		B.1	0.14	0
3-78	-4-SO_2Me	-4-OMe-3,5-Br	-CF_3		B.1	0.09	0
3-79	-4-SO_2Me	-4-OMe-3,5-Me	-CF_3		B.1	0.72	1

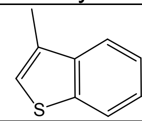
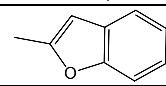
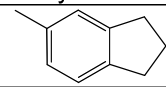
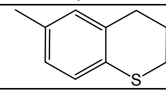
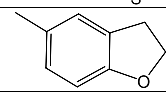
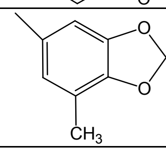
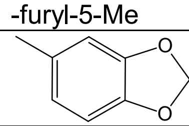
3-80	-4-SO_2Me	-4-OMe-2,5-Me	-CF_3		B.1	12.2	1
3-81	-4-SO_2Me	-4-NMe_2-3,5-Cl	-CF_3		B.1	0.14	0
3-82	-4-SO_2NH_2	-4-OMe-3,5-F	-CF_3		B.1	0.03	0
3-83	-4-SO_2Me	-4-Cl	-H		B.1	>100	2
3-87	-4-SO_2Me	4-Cl	-CHF_2		B.1	0.61	0
3-88	-4-SO_2Me	-4-Cl	-CH_2F		B.1	0.41	2
3-89	-4-SO_2Me	-4-Cl	-CHO		B.1	1.6	2
3-90	-4-SO_2Me	-4-Cl	-CN		B.1	0.23	2
3-91	-4-SO_2Me	-4-Cl	-CO_2Et		B.1	5.7	1
3-92	-4-SO_2Me	-4-Cl	-CO_2H		B.1	>100	1
3-94	-4-SO_2Me	-4-Cl	-CO_2NEt_2		B.1	>100	2
3-95	-4-SO_2Me	-4-Cl	-CH_2NH_2		B.1	>100	1
3-96	-4-SO_2Me	-4-Cl	-Ph		B.1	0.24	0
3-97	-4-SO_2Me	-4-Cl	-$CH_2OC_6H_4$-4-Cl		B.1	0.03	2
3-98	-4-SO_2Me	-4-Cl	-$CH_2SC_6H_4$-4-Cl		B.1	0.05	1
3-99	-4-SO_2Me	-4-Cl	-CH_2OMe		B.1	3.72	1
3-100	-4-SO_2Me	-4-Cl	-CH_2OH		B.1	8.35	1
3-101	-4-SO_2Me	-4-Cl	-CH_2SMe		B.1	0.32	1
3-102	-4-SO_2Me	-4-Cl	-CH_2SO_2Me		B.1	>100	1
3-103	-4-SO_2Me	-4-Cl	-CH_2CN		B.1	1.54	1
4-25	-4-SO_2Me	-H	-CF_3		B.2	1.69	0
4-36	-4-SO_2Me	-2-Me	-CF_3		B.2	9.6	0
4-37	-4-SO_2Me	-6-Me	-CF_3		B.2	1.8	0
4-38	-4-SO_2Me	-5-Me	-CF_3		B.2	1.8	1
4-39	-4-SO_2Me	-4-Me	-CF_3		B.2	54	1
4-40	-4-SO_2Me	-6-OMe	-CF_3		B.2	1.2	1
4-41	-4-SO_2Me	-5-OMe	-CF_3		B.2	37.6	2
4-42	-4-SO_2Me	-5-Br	-CF_3		B.2	0.95	1
4-43	-4-SO_2NH_2	-H	-CF_3		B.2	0.44	1
4-44	-4-SO_2NH_2	-2-Me	-CF_3		B.2	2.8	0
4-45	-4-SO_2NH_2	-6-Me	-CF_3		B.2	0.29	0
4-46	-4-SO_2NH_2	-5-Me	-CF_3		B.2	0.51	0
4-47	-4-SO_2NH_2	-4-Me	-CF_3		B.2	51	0
4-48	-4-SO_2NH_2	-5-OMe	-CF_3		B.2	>100	2
4-49	-4-SO_2NH_2	-5-Br	-CF_3		B.2	0.34	2
4-58	-4-SO_2Me	-H	-CHF_2		B.2	20.7	1
4-59	-4-SO_2Me	-H	-CN		B.2	24.4	2
4-60	-4-SO_2Me	-H	-Me		B.2	79	1
4-61	-4-SO_2Me	-H	-CO_2Et		B.2	>100	2
4-62	-4-SO_2Me	-H	-CH_2OH		B.2	93.4	2
4-63	-4-SO_2NH_2	-H	-CHF_2		B.2	1.83	0
4-50	-4-SO_2Me	-6-Me	-CF_3		B.3	2.9	0
4-51	-4-SO_2Me	-5-Me	-CF_3		B.3	1.3	1
4-52	-4-SO_2Me	-4-Me	-CF_3		B.3	0.53	1
4-53	-4-SO_2Me	-3-Me	-CF_3		B.3	5.8	0
4-54	-4-SO_2NH_2	-6-Me	-CF_3		B.3	0.42	1
4-55	-4-SO_2NH_2	-5-Me	-CF_3		B.3	0.73	0
4-56	-4-SO_2NH_2	-4-Me	-CF_3		B.3	0.44	0
4-57	-4-SO_2NH_2	-3-Me	-CF_3		B.3	1.54	1
4-64	-4-SO_2Me	-CF_3			B.4	1.2	1

4-65	-4-SO_2Me	-CF_3			B.4	1.7	0
4-66	-4-SO_2Me	-CF_3			B.4	0.63	1
4-67	-4-SO_2Me	-CF_3			B.4	1.1	2
4-68	-4-SO_2Me	-CF_3			B.4	0.28	0
4-69	-4-SO_2Me	-CF_3			B.4	0.47	1
4-70	-4-SO_2Me	-CF_3	-3-thienyl		B.4	51	1
4-71	-4-SO_2Me	-CF_3	-2-thienyl		B.4	0.47	1
4-72	-4-SO_2NH_2	-CF_3	-2-thienyl		B.4	0.07	1
4-73	-4-SO_2Me	-CF_3	-4-Br-2-thienyl		B.4	0.026	1
4-74	-4-SO_2Me	-CF_3	-3-Me-2-thienyl		B.4	1.1	1
4-75	-4-SO_2Me	-CF_3			B.4	0.94	2
4-76	-4-SO_2NH_2	-CF_3			B.4	0.52	1
4-77	-4-SO_2NH_2	-CF_3			B.4	0.43	2
4-78	-4-SO_2NH_2	-CF_3			B.4	0.11	0
4-79	-4-SO_2Me	-CF_3			B.4	4.15	1
4-80	-4-SO_2NH_2	-CF_3			B.4	0.41	2
5-1a	-4-SO_2Me	-4-F	-H		C	0.026	1
5-1b	-4-SO_2Me	-4-OMe	-H		C	0.005	1
5-1c	-4-SO_2Me	-4-Cl	-H		C	0.003	0
5-1d	-4-SO_2Me	-4-Me	-H		C	0.003	1
5-1e	-4-SO_2Me	-H	-H		C	2.25	0
5-1f	-4-SO_2Me	-4-CF_3	-H		C	0.865	2
5-1g	-4-SO_2Me	-2,4-Cl	-H		C	0.053	1
5-1h	-4-SO_2Me	-4-CN	-H		C	77.9	2
5-1i	-4-SO_2Me	-4-CH_2OH	-H		C	3.2	1
5-1j	-4-SO_2Me	-4-CH_2OCH_3	-H		C	6.6	2
5-1k	-4-SO_2Me	-4-SMe	-H		C	0.221	1
5-1l	-4-SO_2Me	-4-F-2-Me	-H		C	0.075	1
5-7a	-4-SO_2Me	-4-F	-Me		C	0.015	1
5-7b	-4-SO_2Me	-4-Cl	-Me		C	0.007	0
5-7c	-4-SO_2Me	-4-F	-CF_3		C	0.067	1
5-7d	-4-SO_2Me	-4-F	-C_2H_5		C	65	2
5-7e	-4-SO_2Me	-4-F	-CH_2F		C	0.051	1
6-7b	-4-SO_2Me	-3,4-F	-H		C	0.051	1

6-7c	-4-SO_2Me	-4-F-3-Cl	-H		C	0.03	0
6-7d	-4-SO_2Me	-3-CF_3-4-F	-H		C	>100	2
6-7e	-4-SO_2Me	-3,4,5-F	-H		C	>100	0
6-7g	-4-SO_2Me	-3,4-Cl	-H		C	0.01	0
6-7i	-4-SO_2Me	-4-OMe-3-F	-H		C	0.12	1
6-7j	-4-SO_2Me	-4-OMe-3-Cl	-H		C	0.14	0
6-7k	-4-SO_2Me	-4-OMe-3,5-Cl	-H		C	0.017	0
6-7l	-4-SO_2Me	-3,4-OCH_2O-	-H		C	0.021	2
6-7m	-4-SO_2Me	-4-NMe_2-3-Cl	-H		C	0.005	2
6-7p	-4-SO_2Me	-4-CF_3-3-F	-H		C	0.76	1
6-8a	-4-SO_2NH_2	-4-F	-H		C	0.007	1
6-8b	-4-SO_2NH_2	-3,4-F	-H		C	0.018	1
6-8c	-4-SO_2NH_2	-4-F-3-Cl	-H		C	0.01	0
6-8e	-4-SO_2NH_2	-3,4,5-F	-H		C	2.9	0
6-8f	-4-SO_2NH_2	-4-Cl	-H		C	0.003	0
6-8g	-4-SO_2NH_2	-3,4-Cl	-H		C	0.002	0
6-8h	-4-SO_2NH_2	-4-OMe	-H		C	0.002	1
6-8i	-4-SO_2NH_2	-4-OMe-3-F	-H		C	0.016	1
6-8j	-4-SO_2NH_2	-4-OMe-3-Cl	-H		C	0.009	0
6-8k	-4-SO_2NH_2	-4-OMe-3,5-Cl	-H		C	0.009	0
6-8l	-4-SO_2NH_2	-3,4-OCH_2O-	-H		C	0.002	2
6-8m	-4-SO_2NH_2	-4-NMe_2-3-Cl	-H		C	0.002	2
6-8o	-4-SO_2NH_2	-4-CF_3	-H		C	0.15	2
6-8p	-4-SO_2NH_2	-4-CF_3-3-F	-H		C	0.17	2
7-16a	-4-SO_2Me	-4-F	-H		D.1	0.26	1
7-16b	-4-SO_2Me	-4-F-3-Cl	-H		D.1	0.36	1
7-16c	-4-SO_2Me	-4-Cl	-H		D.1	0.14	0
7-16d	-4-SO_2Me	-4-OMe-3-F	-H		D.1	>100	0
7-16e	-4-SO_2Me	-4-OMe-3-Cl	-H		D.1	11.4	0
7-17a	-4-SO_2NH_2	-4-F	-H		D.1	0.061	2
7-17b	-4-SO_2NH_2	-4-F-3-Cl	-H		D.1	0.017	2
7-17c	-4-SO_2NH_2	-4-Cl	-H		D.1	0.006	0
7-17d	-4-SO_2NH_2	-4-OMe-3-F	-H		D.1	0.033	0
7-17e	-4-SO_2NH_2	-4-OMe-3-Cl	-H		D.1	0.019	0
7-20a	-4-SO_2Me	-4-F	-3,4-F		D.1	0.014	0
7-20b	-4-SO_2Me	-4-F-3-Cl	-3,4-F		D.1	0.01	0
7-20c	-4-SO_2Me	-4-F-3-Me	-3,4-F		D.1	0.005	0
7-20d	-4-SO_2Me	-4-OMe-3-F	-3,4-F		D.1	0.021	1
7-20e	-4-SO_2Me	-4-OMe-3-Cl	-3,4-F		D.1	0.019	0
7-20f	-4-SO_2Me	-4-OMe-3,5-Cl	-3,4-F		D.1	>100	0
7-20g	-4-SO_2Me	-4-OMe-3-Me	-3,4-F		D.1	0.013	1
7-20h	-4-SO_2Me	-3,4-OMe	-3,4-F		D.1	0.34	2
7-20i	-4-SO_2Me	-3,4-OEtO-	-3,4-F		D.1	0.34	2
7-20j	-4-SO_2Me	-3,4-OMeO-	-3,4-F		D.1	0.012	1
7-20k	-4-SO_2Me	-4-Me	-3,4-F		D.1	0.007	0
7-20l	-4-SO_2Me	-4-Me-3-Cl	-3,4-F		D.1	0.013	0
7-20m	-4-SO_2Me	-3,4-Me	-3,4-F		D.1	0.023	0
7-20n	-4-SO_2Me	-4-Cl-3-Me	-3,4-F		D.1	0.006	0
7-20o	-4-SO_2Me	-4-NMe_2-3-Cl	-3,4-F		D.1	0.008	2
7-21a	-4-SO_2NH_2	-4-F	-3,4-F		D.1	0.004	0
7-21b	-4-SO_2NH_2	-4-F-3-Cl	-3,4-F		D.1	0.002	0
7-21c	-4-SO_2NH_2	-4-F-3-Me	-3,4-F		D.1	0.002	0
7-21d	-4-SO_2NH_2	-4-OMe-3-F	-3,4-F		D.1	0.013	2
7-21e	-4-SO_2NH_2	-4-OMe-3-Cl	-3,4-F		D.1	0.013	0
7-21f	-4-SO_2NH_2	-4-OMe-3,5-Cl	-3,4-F		D.1	0.021	0
7-21g	-4-SO_2NH_2	-4-OMe-3-Me	-3,4-F		D.1	0.005	1
7-21h	-4-SO_2NH_2	-3,4-OMe	-3,4-F		D.1	0.065	1

7-21i	-4-SO_2NH_2	-3,4-OEtO-	-3,4-F		D.1	0.032	1
7-21j	-4-SO_2NH_2	-3,4-OMeO-	-3,4-F		D.1	0.004	2
7-21k	-4-SO_2NH_2	-4-Me	-3,4-F		D.1	0.004	0
7-21l	-4-SO_2NH_2	-4-Me-3-Cl	-3,4-F		D.1	0.003	0
7-21m	-4-SO_2NH_2	-3,4-Me	-3,4-F		D.1	0.005	0
7-21n	-4-SO_2NH_2	-4-Cl-3-Me	-3,4-F		D.1	0.003	0
7-21o	-4-SO_2NH_2	-4-NMe_2-3-Cl	-3,4-F		D.1	0.006	2
7-30	-4-SO_2Me	-4-F	-3,4-Cl		D.1	0.24	0
7-32	-4-SO_2Me	-4-F	-3,4-OMeO-		D.1	0.083	2
7-34	-4-SO_2Me	-4-F	-2,3,4,5-F		D.1	>100	2
7-36	-4-SO_2Me	-4-F	-3,4-$(CH)_4$-		D.1	>100	2
7-20q	-4-SO_2Me	-6-Me-3-pyridyl	-3,4-F		D.2	0.17	2
7-21q	-4-SO_2NH_2	-6-Me-3-pyridyl	-3,4-F		D.2	0.051	1
7-20p	-4-SO_2Me	-5-Me-2-pyridyl	-3,4-F		D.2	52.3	1
7-21p	-4-SO_2NH_2	-5-Me-2-pyridyl	-3,4-F		D.2	0.33	2
8-1a	-4-SO_2NH_2	-H	-CF_3	-H	E.1	0.032	1
8-1b	-4-SO_2NH_2	-2-F	-CF_3	-H	E.1	0.058	0
8-1c	-4-SO_2NH_2	-3-F	-CF_3	-H	E.1	7.73	0
8-1d	-4-SO_2NH_2	-4-F	-CF_3	-H	E.1	0.041	0
8-1e	-4-SO_2NH_2	-2-Cl	-CF_3	-H	E.1	0.056	0
8-1f	-4-SO_2NH_2	-4-Cl	-CF_3	-H	E.1	0.01	0
8-1g	-4-SO_2NH_2	-2-Me	-CF_3	-H	E.1	0.069	0
8-1h	-4-SO_2NH_2	-3-Me	-CF_3	-H	E.1	0.11	0
8-1i	-4-SO_2NH_2	-4-Me	-CF_3	-H	E.1	0.04	0
8-1j	-4-SO_2NH_2	-4-Et	-CF_3	-H	E.1	0.86	0
8-1k	-4-SO_2NH_2	-4-CF_3	-CF_3	-H	E.1	8.23	0
8-1l	-4-SO_2NH_2	-4-NO_2	-CF_3	-H	E.1	2.63	1
8-1m	-4-SO_2NH_2	-4-OH	-CF_3	-H	E.1	>100	0
8-1n	-4-SO_2NH_2	-2-OMe	-CF_3	-H	E.1	0.29	1
8-1o	-4-SO_2NH_2	-4-OMe	-CF_3	-H	E.1	0.008	1
8-1p	-4-SO_2NH_2	-4-OEt	-CF_3	-H	E.1	0.64	2
8-1q	-4-SO_2NH_2	-4-SMe	-CF_3	-H	E.1	0.009	0
8-1r	-4-SO_2NH_2	-4-NH_2	-CF_3	-H	E.1	0.34	0
8-1s	-4-SO_2NH_2	-2-NMe_2	-CF_3	-H	E.1	14.3	0
8-1t	-4-SO_2NH_2	-4-NHMe	-CF_3	-H	E.1	0.016	1
8-1u	-4-SO_2NH_2	-4-NMe_2	-CF_3	-H	E.1	0.0047	1
8-1v	-4-SO_2NH_2	-4-CH_2OH	-CF_3	-H	E.1	93.3	2
8-1w	-4-SO_2NH_2	-4-CO_2H	-CF_3	-H	E.1	11.2	0
8-1x	-4-SO_2NH_2	-4-OMe-3-Me	-CF_3	-H	E.1	0.0093	0
8-1y	-4-SO_2NH_2	-4-OMe-3-Et	-CF_3	-H	E.1	0.43	1
8-1z	-4-SO_2NH_2	-3,4-OMe	-CF_3	-H	E.1	0.6	1
8-1aa	-4-SO_2NH_2	-3-Me-4-SMe	-CF_3	-H	E.1	0.0037	1
8-1ab	-4-SO_2NH_2	-3-F-4-NMe_2	-CF_3	-H	E.1	0.0057	2
8-1ac	-4-SO_2NH_2	-4NHMe-3-Cl	-CF_3	-H	E.1	0.027	1
8-1ad	-4-SO_2NH_2	-5-Me-4-OMe-3-Cl	-CF_3	-H	E.1	0.066	1
8-1ae	-4-SO_2NH_2	-3,4-Cl	-CF_3	-H	E.1	0.015	1
8-1af	-4-SO_2NH_2	-2,4-Cl	-CF_3	-H	E.1	0.056	1
8-1ag	-4-SO_2NH_2	-2,5-Cl	-CF_3	-H	E.1	>100	1
8-1ah	-4-SO_2NH_2	-2,4-Me	-CF_3	-H	E.1	0.12	0
8-2a	-4-SO_2NH_2	-4-Cl	-CHF_2	-H	E.1	0.01	1
8-29	-4-SO_2NHCH_3	-4-Cl	-CF_3	-H	E.1	>100	0
8-2b	-4-SO_2NH_2	-4-Me	-CHF_2	-H	E.1	0.013	0
8-2c	-4-SO_2NH_2	-4-CN	-CHF_2	-H	E.1	29.7	2
8-2d	-4-SO_2NH_2	-4-SO_2Me	-CHF_2	-H	E.1	>100	2
8-2e	-4-SO_2NH_2	-4-$CONH_2$	-CHF_2	-H	E.1	>100	1
8-2f	-4-SO_2NH_2	-4-CO_2H	-CHF_2	-H	E.1	46.8	1

8-2g	-4-SO_2NH_2	-4-OMe	-CHF_2	-H	E.1	0.015	0
8-2h	-4-SO_2NH_2	-4-OMe-3-F	-CHF_2	-H	E.1	0.05	2
8-2i	-4-SO_2NH_2	-4-OMe-3-Cl	-CHF_2	-H	E.1	0.027	0
8-2j	-4-SO_2NH_2	-4-OMe-3,5-Cl	-CHF_2	-H	E.1	0.021	0
8-2k	-4-SO_2NH_2	-4-OMe-3,5-F	-CHF_2	-H	E.1	0.35	0
8-2l	-4-SO_2NH_2	-2,5-Me	-CHF_2	-H	E.1	>100	1
8-2o	-4-SO_2NH_2	-H	-CHF_2	-H	E.1	0.13	0
8-3a	-4-SO_2NH_2	-4-F	-CO_2Me	-H	E.1	100	2
8-8a	-4-SO_2NH_2	-H	-Me	-H	E.1	62.8	1
8-9a	-4-SO_2NH_2	-4-Cl	-C_6H_4-4-OMe	-H	E.1	0.1	2
8-9b	-4-SO_2NH_2	-4-Cl	-5-Cl-2-thienyl	-H	E.1	0.052	2
8-10a	-4-SO_2NH_2	-4-F	-CO_2H	-H	E.1	>100	1
8-11a	-4-SO_2NH_2	-4-F	-CO_2NH_2	-H	E.1	>100	1
8-12a	-4-SO_2NH_2	-4-F	-CONHC_6H_4-3-Cl	-H	E.1	0.056	2
8-13a	-4-SO_2NH_2	-4-F	-CN	-H	E.1	0.34	1
8-14a	-4-SO_2NH_2	-4-Cl	-CH_2OH	-H	E.1	0.83	2
8-15a	-4-SO_2NH_2	-4-Cl	-CH_2OCH_2Ph	-H	E.1	0.029	1
8-16a	-4-SO_2NH_2	-H	-CH_2F	-H	E.1	0.2	1
8-17a	-4-SO_2NH_2	-4-Cl	-CH_2CN	-H	E.1	0.12	2
8-19	-4-SO_2NH_2	-H	-OMe	-H	E.1	>100	2
8-20a	-4-SO_2NH_2	-4-Cl	-CF_3	-Cl	E.1	0.0053	0
8-20b	-4-SO_2NH_2	-H	-Me	-Cl	E.1	0.028	2
8-20c	-4-SO_2NH_2	-4-Cl	-CH_2OH	-Cl	E.1	0.34	1
8-20d	-4-SO_2NH_2	-4-Cl	-CN	-Cl	E.1	0.01	2
8-20e	-4-SO_2NH_2	-4-Cl	-CO_2H	-Cl	E.1	70	2
8-20f	-4-SO_2NH_2	-4-Cl	-CO_2Me	-Cl	E.1	0.16	1
8-20g	-4-SO_2NH_2	-4-Cl	-$CONH_2$	-Cl	E.1	1.09	1
8-21a	-4-SO_2NH_2	-H	-H	-Cl	E.1	0.049	0
8-21b	-4-SO_2NH_2	-4-Cl	-H	-Br	E.1	0.031	1
8-22a	-4-SO_2NH_2	-4-F	-H	-H	E.1	>100	2
8-22b	-4-SO_2NH_2	-H	-H	-F	E.1	4.66	0
8-22c	-4-SO_2NH_2	-H	-H	-Me	E.1	47.1	1
8-22d	-4-SO_2NH_2	-4-Me	-H	-CN	E.1	0.076	2
8-22e	-4-SO_2NH_2	-H	-H	-NO_2	E.1	0.29	2
8-22f	-4-SO_2NH_2	-4-Cl	-H	-SO_2Me	E.1	19.8	2
8-23	-4-SO_2NH_2	-H	-H	-NH_2	E.1	29.7	1
8-24	-4-SO_2NH_2	-H	-CF_3	-F	E.1	0.0017	1
8-25a	-4-SO_2NH_2	-4-Cl	-CF_3	-Me	E.1	0.022	1
8-25b	-4-SO_2NH_2	-4-Cl	-CF_3	-Et	E.1	0.028	0
8-25c	-4-SO_2NH_2	-4-F	-CF_3	-n-Pr	E.1	>100	2
8-25d	-4-SO_2NH_2	-H	-CF_3	-OMe	E.1	0.08	2
8-26	-4-SO_2NH_2	-H	-CF_3	-OH	E.1	3.58	1
8-28a	-4-F	-4-SO_2NH_2	-CF_3	-H	E.1	0.01	1
8-28b	-4-OMe	-4-SO_2NH_2	-CF_3	-H	E.1	0.0067	0
8-30	-4-$SO_2N(CH_3)_2$	-4-Cl	-CF_3	-H	E.1	>100	2
8-31	-4-$NHSO_2CH_3$	-4-Cl	-CF_3	-H	E.1	>100	1
8-32	-4-NO_2	-4-Cl	-CF_3	-H	E.1	>100	1
8-33	-4-$COCF_3$	-4-Cl	-CF_3	-H	E.1	>100	2
8-34	-H	-H	-CF_3	-H	E.1	>100	2
8-35	-4-Cl	-4-Cl	-CF_3	-H	E.1	4.79	1
8-36	-4-OMe	-4-OMe	-CF_3	-H	E.1	0.75	1
8-37	-4-Cl	-4-OMe	-CF_3	-H	E.1	0.75	2
8-38	-4-OMe	-4-Cl	-CF_3	-H	E.1	74.9	1
8-39	-4-SO_2Me	-4-F	-CF_3	-H	E.1	0.1	2

8-celecoxib	-4-SO_2NH_2	-4-Me	-CF_3	-H	E.1	0.04	0
8-1ai	-4-SO_2NH_2	-CF_3	-2-pyridyl		E.2	45.6	1
8-1aj	-4-SO_2NH_2	-CF_3	-3-pyridyl		E.2	45	2
8-1ak	-4-SO_2NH_2	-CF_3	-4-pyridyl		E.2	64.7	1
8-1al	-4-SO_2NH_2	-CF_3	-thienyl-5-Br		E.2	0.012	1
8-1am	-4-SO_2NH_2	-CF_3	-thienyl-5-Cl		E.2	0.026	0
8-1an	-4-SO_2NH_2	-CF_3			E.2	0.35	2
8-1ao	-4-SO_2NH_2	-CF_3			E.2	0.89	1
8-1ap	-4-SO_2NH_2	-CF_3	-1-cyclohexene		E.2	0.084	2
8-1aq	-4-SO_2NH_2	-CF_3			E.2	0.031	2
8-1ar	-4-SO_2NH_2	-CF_3			E.2	0.23	1
8-1as	-4-SO_2NH_2	-CF_3			E.2	0.021	1
8-1at	-4-SO_2NH_2	-CF_3			E.2	0.052	1
8-2m	-4-SO_2NH_2	-CHF_2	-furyl-5-Me		E.2	3.29	2
8-2n	-4-SO_2NH_2	-CHF_2			E.2	0.024	2
9-3	-4-SO_2Me	-4-F	0		F	0.0075	2
9-20	-4-SO_2Me	-4-OMe-3-Cl	0		F	0.017	0
9-24	-4-SO_2NH_2	-4-OMe-3-Cl	0		F	0.002	0
9-32	-4-SO_2NH_2	-4-F	0		F	0.003	0
9-33	-4-SO_2Me	-4-Cl	0		F	0.001	0
9-34	-4-SO_2NH_2	-4-Cl	0		F	0.001	0
9-35	-4-SO_2Me	-4-Me	0		F	0.0015	1
9-36	-4-SO_2Me	-4-OMe	0		F	0.005	1
9-37	-4-SO_2NH_2	-4-OMe	0		F	0.001	2
9-38	-4-SO_2Me	-4-OCF_3	0		F	0.135	2
9-39	-4-SO_2NH_2	-4-OCF_3	0		F	0.13	1
9-40	-4-SO_2Me	-4-CF_3	0		F	0.002	1
9-41	-4-SO_2NH_2	-4-CF_3	0		F	0.001	1
9-42	-4-SO_2Me	-4-OMe-3-F	0		F	0.01	0
9-43	-4-SO_2NH_2	-4-OMe-3-F	0		F	0.002	0
9-44	-4-SO_2Me	-4-OMe-3-Br	0		F	0.013	0
9-45	-4-SO_2NH_2	-4-OMe-3-Br	0		F	0.002	0
9-46	-4-SO_2Me	-3,4-OCH_2O-	0		F	0.0025	1
9-47	-4-SO_2Me	-3,4-F	0		F	0.0033	2
9-48	-4-SO_2NH_2	-3,4-F	0		F	0.003	2
9-49	-4-SO_2Me	-3,4-Cl	0		F	0.003	0
9-50	-4-SO_2NH_2	-3,4-Cl	0		F	0.001	0
9-51	-4-SO_2Me	-4-F-3-Cl	0		F	0.007	0
9-52	-4-SO_2NH_2	-4-F-3-Cl	0		F	0.0015	0
9-53	-4-SO_2Me	-2,4-F	0		F	0.022	1
9-54	-4-SO_2Me	-2,4-Cl	0		F	0.033	0
9-55	-4-SO_2Me	-4-OMe-3,5-Cl	0		F	0.006	1

9-56	-4-SO_2NH_2	-4-OMe-3,5-Cl	0		F	0.004	2
9-25	-4-SO_2Me	-4-F	1		F	0.004	2
9-30	-4-SO_2Me	-4-F	3		F	0.062	0
9-31	-4-SO_2Me	-4-F	4		F	>100	0
9-19	-4-SO_2Me	-4-F			G	0.026	1
9-29	-4-SO_2Me	-4-OCH_3-3-Cl			G	0.027	2
10-2	-4-SO_2NH_2	-Me			H	0.005	1
10-3	-4-SO_2NH_2	-CH_2OH			H	0.18	2
6-Dup697	-4-SO_2Me	-4-F	-Br		I	0.01	2
11-7	-4-SO_2Me	-4-F	-CF_3		J.1	0.006	1
11-8	-4-SO_2Me	-4-F	-Me		J.1	0.12	0
11-9	-4-SO_2Me	-4-F	-Ph		J.1	0.54	0
11-10	-4-F	-4-SO_2Me	-CF_3		J.1	0.023	0
11-11	-4-F	-4-SO_2Me	-CH_3		J.1	0.11	0
11-12	-4-F	-4-SO_2Me	-Ph		J.1	>30	0
11-13	-4-F	-4-SO_2Me	-Et		J.1	0.023	0
11-14	-4-F	-4-SO_2Me	-t-Bu		J.1	0.11	1
11-15	-4-F	-4-SO_2Me	-CO_2Et		J.1	0.61	2
11-16	-4-F	-4-SO_2Me	-CH_2CN		J.1	0.35	1
11-17	-4-F	-4-SO_2Me	-CH_2Ph		J.1	0.018	1
11-18	-4-F	-4-SO_2Me	-CH_2CH_2Ph		J.1	0.032	1
11-19	-4-F	-4-SO_2Me	-nPr-Ph		J.1	0.52	2
11-20	-4-F	-4-SO_2Me	-nPr-C_6H_4-4-Br		J.1	0.007	0
11-21	-4-F	-4-SO_2Me	-C_6H_4-2-Cl		J.1	0.05	0
11-22	-4-F	-4-SO_2Me	-C_6H_4-2-Cl		J.1	0.05	0
11-23	-4-F	-4-SO_2Me	-C_6H_4-4-OMe		J.1	0.27	2
11-24	-4-F	-4-SO_2Me	-2-thienyl		J.1	0.021	1
11-25	-4-F	-4-SO_2Me	-3-pyridyl		J.1	0.43	2
11-26	-4-F	-4-SO_2Me	-4-pyridyl		J.1	1.6	1
11-27	-4-F	-4-SO_2Me	-C_6H_4-4-t-Bu		J.1	>100	1
11-28	-H	-4-SO_2Me	-C_6H_4-2-Cl		J.1	0.01	1
11-29	-2-F	-4-SO_2Me	-C_6H_4-2-Cl		J.1	0.01	0
11-30	-3-F	-4-SO_2Me	-C_6H_4-2-Cl		J.1	0.016	0
11-31	-2,5-F	-4-SO_2Me	-C_6H_4-2-Cl		J.1	>100	2
11-32	-2,4-F	-4-SO_2Me	-C_6H_4-2-Cl		J.1	0.021	1
12-3a	-4-F	-4-SO_2NH_2	-C_6H_4-2-Cl		J.1	0.013	2
12-3b	-4-F	-4-SO_2NH_2	-OMe-C_6H_3-3,5-Cl		J.1	0.009	2
12-5a	-H	-4-SO_2NH_2	-CH_3		J.1	0.038	0
12-5b	-4-Cl	-4-SO_2NH_2	-CH_3		J.1	0.005	1
12-7a	-4-SO_2NH_2	-4-Br	-CH_3		J.1	0.024	0
12-7b	-4-SO_2NH_2	-H	-CH_3		J.1	0.029	0
12-7c	-4-SO_2NH_2	-4-Cl	-CH_3		J.1	0.018	0
12-10a	-4-OMe-3-F	-4-SO_2NH_2	-$NHCH_2$-Ph		J.1	0.059	1
12-10b	-4-OMe-3-F	-4-SO_2NH_2	-C_6H_4-2-Cl		J.1	2.48	1
12-10c	-4-OMe-3-F	-4-SO_2NH_2	-Me		J.1	0.028	2
11-33	-2-thienyl	-4-SO_2Me	-C_6H_4-2-Cl		J.2	0.057	1
11-34	-3-thienyl	-4-SO_2Me	-C_6H_4-2-Cl		J.2	0.079	2
13-4	-4-SO_2Me	-4-F	-H	-H	K	>100	2
13-5	-4-SO_2Me	-4-F	-CF_3	-H	K	>100	2
13-6a	-4-SO_2Me	-4-F	-H	-CH_2CH=CH_2	K	>100	2
13-6b	-4-SO_2Me	-4-F	-H	-CH_2CH_2-Ph	K	0.53	1
13-7a	-4-SO_2Me	-4-F	-CF_3	-CH_2CH=CH_2	K	0.075	1
13-7b	-4-SO_2Me	-4-F	-CF_3	-CH_2CH_2-Ph	K	0.045	2
13-7c	-4-SO_2Me	-4-F	-CF_3	-Et	K	0.13	1
13-7d	-4-SO_2Me	-4-F	-CF_3	-CH_2CO_2Et	K	0.44	2
13-7e	-4-SO_2Me	-4-F	-CF_3	-$CH_2CONHPh$	K	0.48	1

13-10	-4-SO_2NH_2	-4-F	-CF_3	-Et	K	0.033	2
14-3	-4-SO_2Me	-4-F	C		L	0.011	2
14-7	-4-SO_2Me	-4-F	O		L	0.02	2
14-13	-4-SO_2NH_2	-4-F	C		L	0.005	2
14-12	-4-SO_2NH_2	-4-F	O		L	0.007	2
14-14	-4-SO_2Me	-3-F-4-OMe	O		L	0.093	2
14-15	-4-SO_2NH_2	-3-F-4-OMe	O		L	0.005	2
14-16	-4-SO_2Me	-3-Cl-4-OMe	O		L	0.1	0
14-17	-4-SO_2NH_2	-3-Cl-4-OMe	O		L	0.02	0
7-39	-4-SO_2Me	-4-F			M	>100	2

[a] names are formed by hyphenating the reference number and the label given to the compound in the reference; e.g. 14-9a is compound 9a in reference 14.
[b] training set, test set and unused compounds are indicated as 1, 2, and 0, respectively.

COX-2 references

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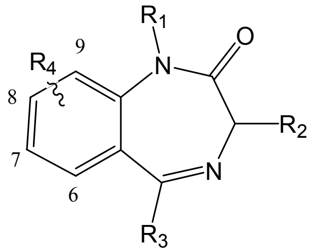
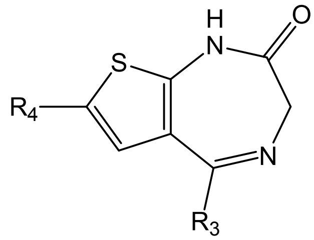
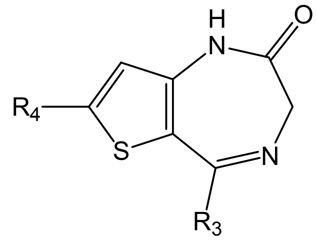
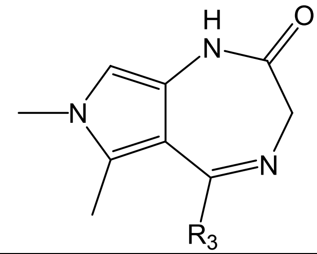
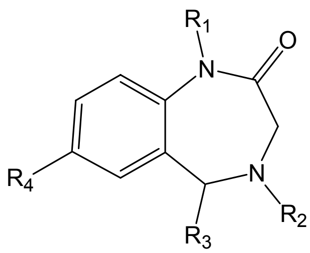
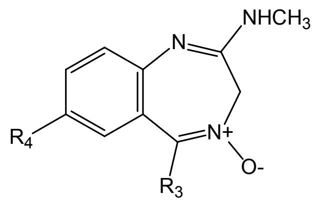
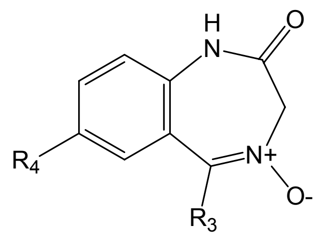
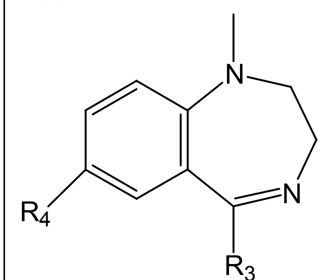
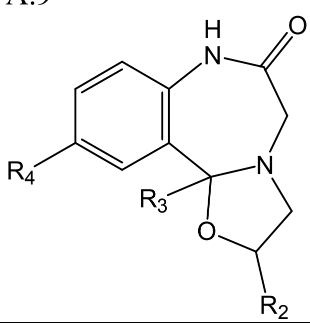
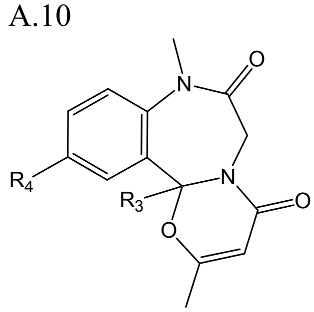
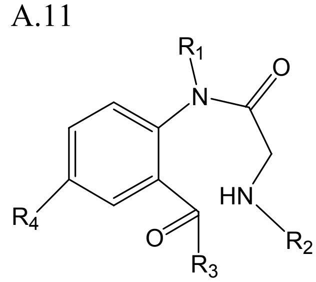
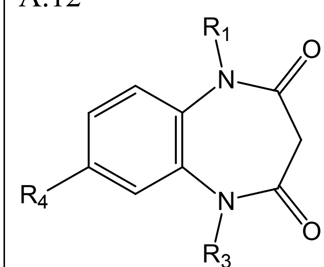
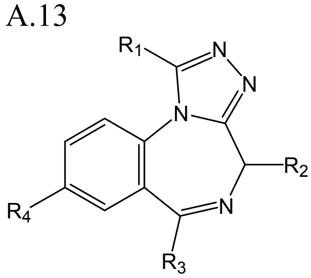
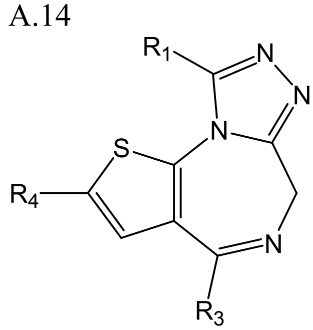
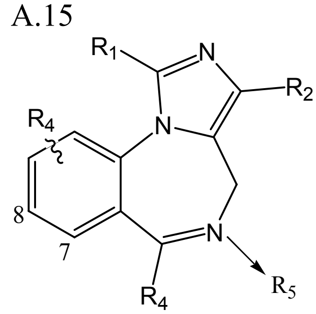
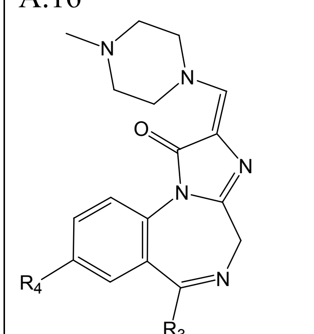
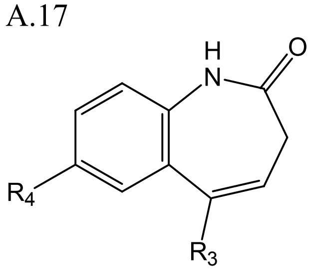
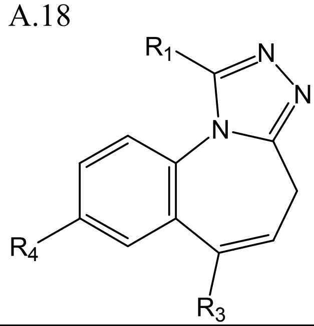
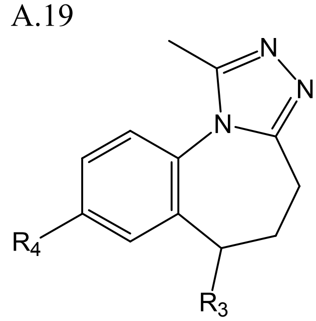
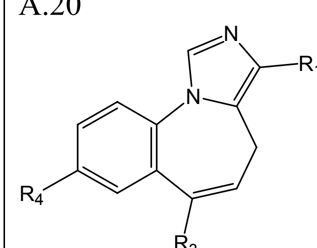
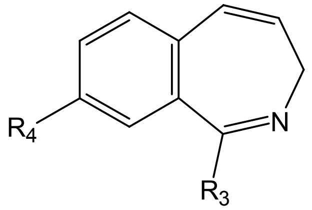
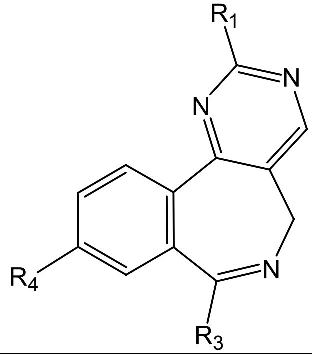
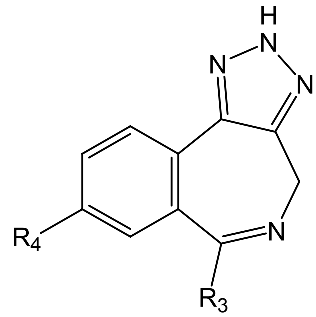
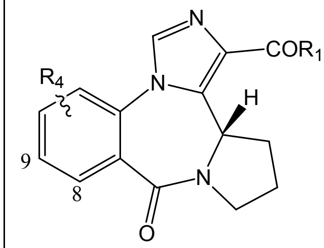
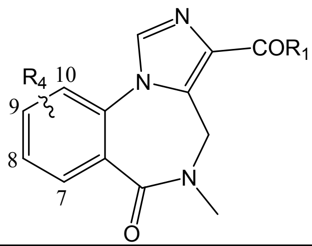
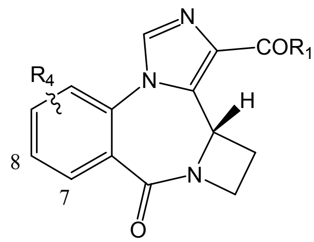
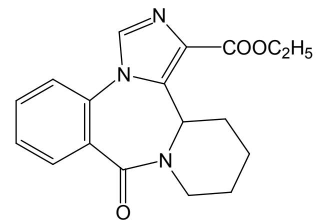
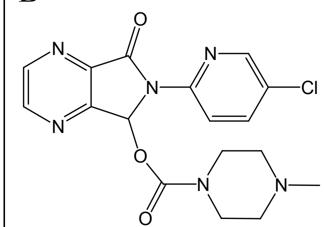
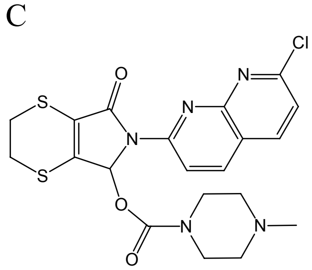
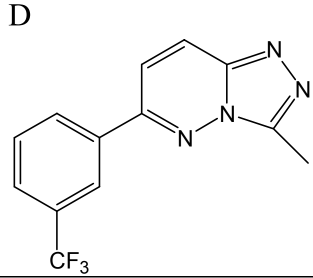
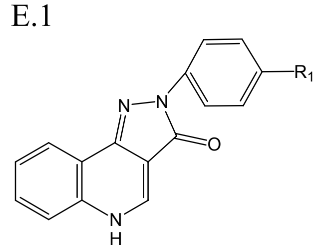
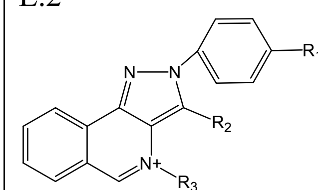
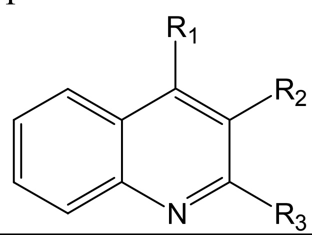
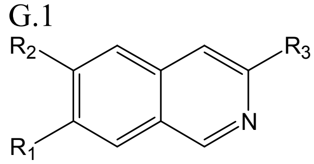
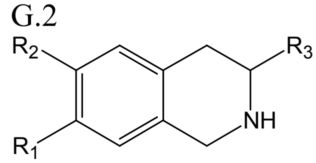
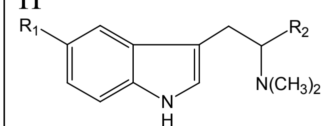
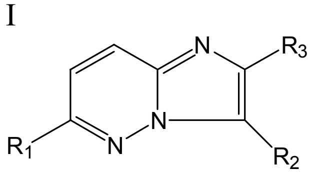
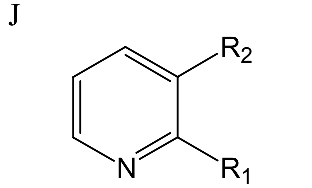
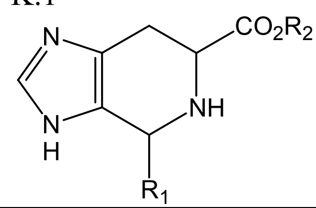
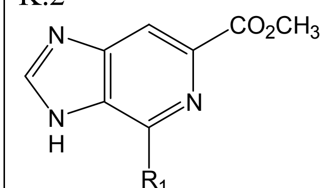
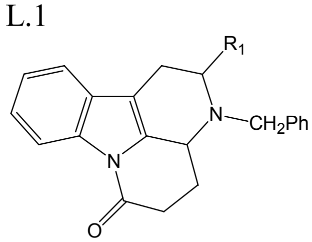
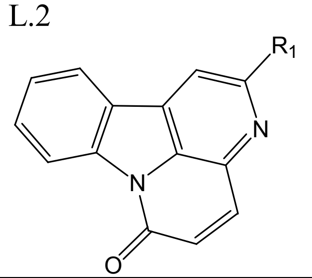
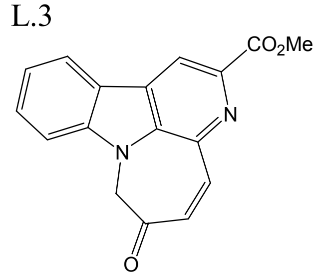
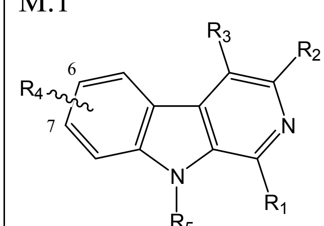
Table 2.1. Structures (families) of BZR ligands

A.1 	A.2 	A.3 	A.4 
A.5 	A.6 	A.7 	A.8 
A.9 	A.10 	A.11 	A.12 
A.13 	A.14 	A.15 	A.16 
A.17 	A.18 	A.19 	A.20 

A.21 	A.22 	A.23 	A.24 
A.25 	A.26 	A.27 	B 
C 	D 	E.1 	E.2 
F 	G.1 	G.2 	H 
I 	J 	K.1 	K.2 
L.1 	L.2 	L.3 	M.1 

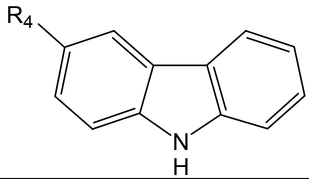
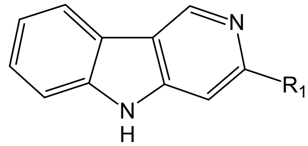
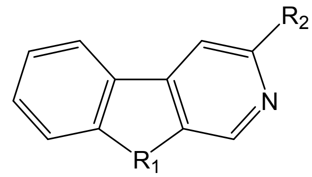
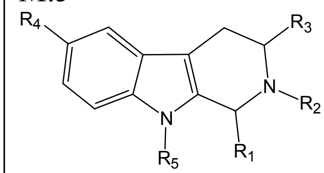
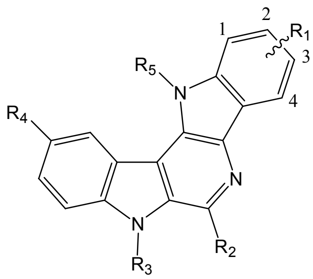
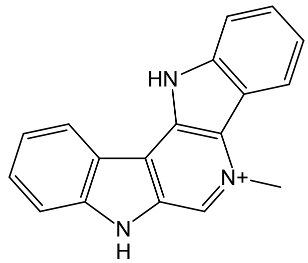
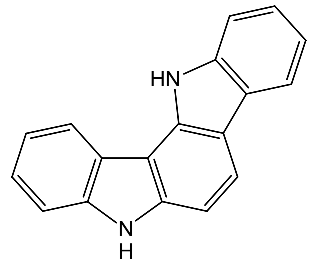
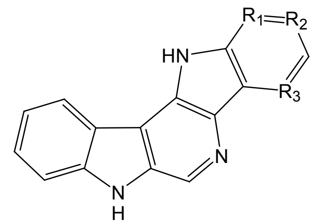
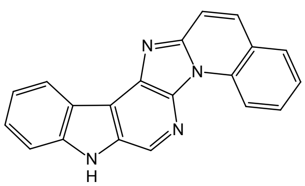
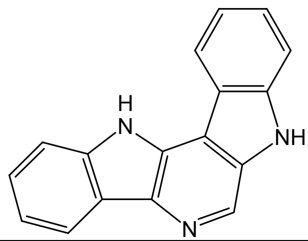
M.2 	M.3 	M.4 	M.5 
N.1 	N.2 	N.3 	N.4 
N.5 	N.6 		

Table 2.2. BZR ligands

Name[a]	R_1	R_2	R_3	R_4	R_5	Fam-ily	IC_{50} (μM)[b]	set[c]
Ro05-3061	-H	-H	-Ph	-7-F		A.1	0.04	1
Ro05-4865	-Me	-H	-Ph	-7-F		A.1	0.017	1
Ro05-6820	-H	-H	-C_6H_4-2-F	-7-F		A.1	0.0074	2
Ro05-6822	-Me	-H	-C_6H_4-2-F	-7-F		A.1	0.0051	1
Nordazepam	-H	-H	-Ph	-7-Cl		A.1	0.0094	0
Diazepam	-Me	-H	-Ph	-7-Cl		A.1	0.0081	0
Ro05-3367	-H	-H	-C_6H_4-2-F	-7-Cl		A.1	0.002	0
Ro07-3953	-H	-H	-C_6H_3-2,6-F	-7-Cl		A.1	0.0016	0
Ro07-4065	-Me	-H	-C_6H_3-2,6-F	-7-Cl		A.1	0.0041	0
Delorazepam	-H	-H	-C_6H_4-2-Cl	-7-Cl		A.1	0.0018	0
Ro07-5193	-H	-H	-C_6H_3-2-Cl-6-F	-7-Cl		A.1	0.003	0
Ro22-3294	-H	-H	-C_6H_3-2,6-Cl	-7-Cl		A.1	0.007	0
Ro07-5220	-Me	-H	-C_6H_3-2,6-Cl	-7-Cl		A.1	0.0055	0
Ro13-3780	-Me	-H	-C_6H_3-2,6-F	-7-Br		A.1	0.0024	0
Ro07-9957	-Me	-H	-C_6H_4-2-F	-7-I		A.1	0.0029	0
Ro05-2904	-H	-H	-Ph	-7-CF_3		A.1	0.013	1
Ro14-3074	-H	-H	-C_6H_4-2-F	-7-N_3		A.1	0.0053	1
Nitrazepam	-H	-H	-Ph	-7-NO_2		A.1	0.01	2
Ro05-4435	-H	-H	-C_6H_4-2-F	-7-NO_2		A.1	0.0015	2
Flunitrazepam	-Me	-H	-C_6H_4-2-F	-7-NO_2		A.1	0.0038	1
Clonazepam	-H	-H	-C_6H_4-2-Cl	-7-NO_2		A.1	0.0018	0

Ro05-4082	-Me	-H	$-C_6H_4$-2-Cl	$-7-NO_2$		A.1	0.0022	0
Ro05-3590	-H	-H	$-C_6H_4-2-CF_3$	$-7-NO_2$		A.1	0.0035	1
Ro20-7736	-Me	-H	$-C_6H_4$-2-F	-7-NHOH		A.1	0.096	1
Ro05-3072	-H	-H	-Ph	$-7-NH_2$		A.1	0.386	1
Ro05-4318	-Me	-H	-Ph	$-7-NH_2$		A.1	0.46	2
Ro20-1815	-Me	-H	$-C_6H_4$-2-F	$-7-NH_2$		A.1	0.065	1
Ro05-4619	-H	-H	$-C_6H_4$-2-Cl	$-7-NH_2$		A.1	0.075	1
Ro05-3308	-H	-H	-Ph	$-7-NHCOCH_3$		A.1	>1	2
Ro12-6377	-Me	-H	$-C_6H_4$-2-F	$-7-NHCONHCH_3$		A.1	0.455	2
Ro05-9090	-Me	-H	-Ph	$-7-CH_2NH_2$		A.1	>1	1
Ro05-4528	-Me	-H	-Ph	-7-CN		A.1	0.38	1
Ro20-2541	-Me	-H	$-C_6H_4$-2-F	-7-CN		A.1	0.03	1
Ro20-2533	-H	-H	-Ph	-7-Et		A.1	0.036	1
Ro20-5747	-H	-H	-Ph	$-7-CH=CH_2$		A.1	0.024	1
Ro20-5397	-H	-H	-Ph	-7-CHO		A.1	0.043	2
Ro20-3053	-H	-H	$-C_6H_4$-2-F	$-7-COCH_3$		A.1	0.018	1
Ro05-3343	-H	-H	-Ph	$-7-SO_2N(CH_3)_2$		A.1	>1	2
Ro05-2921	-H	-H	-Ph			A.1	0.35	1
Ro05-4336	-H	-H	$-C_6H_4$-2-F			A.1	0.021	0
Ro07-4419	-H	-H	$-C_6H_3$-2,6-F			A.1	0.019	1
Ro05-4520	-Me	-H	$-C_6H_4$-2-F			A.1	0.014	1
Ro05-4608	-Me	-H	$-C_6H_4$-2-Cl			A.1	0.0038	0
Ro05-3546	-H	-H	-Ph	-6-Cl		A.1	0.32	0
Ro13-0699	-Me	-H	$-C_6H_4$-2-F	-6-Cl		A.1	0.15	0
Ro07-6198	-H	-H	$-C_6H_3$-2,6-F	-8-Cl		A.1	0.028	0
Ro20-8895	-H	-H	$-C_6H_4$-2-F	$-8-CH_3$		A.1	0.019	1
Ro13-0593	-Me	-H	$-C_6H_4$-2-F	-9-Cl		A.1	0.072	1
Ro13-0882	-Me	-H	$-C_6H_4$-2-F	-6,8-Cl		A.1	0.3	1
Ro22-6762	-Me	-H	-Ph	-6,7-Cl		A.1	0.04	1
Ro20-8065	-H	-H	$-C_6H_4$-2-F	-6,7-Cl		A.1	0.0036	1
Ro20-8552	-H	-H	$-C_6H_4$-2-F	-6-Me,7-Cl		A.1	0.014	1
Ro05-2750	-H	-H	-Ph	-6,8-Cl		A.1	0.037	0
Ro07-3609	-H	-H	-Ph	-6-Cl,8-Me		A.1	0.052	2
Ro14-2312	-Me	-H	$-C_6H_4$-2-F	$-6-NH_2$,8-Cl		A.1	>1	2
Ro17-2221	$-CH_2CH_2NH_2$	-H	-Ph			A.1	0.26	2
Ro08-3026	$-CH_2OCH_2CH_2NH_2$	-H	-Ph	-7-Cl		A.1	0.063	1
Halazepam	$-CH_2CF_3$	-H	-Ph	-7-Cl		A.1	0.092	1
Pinazepam	$-CH_2C{\equiv}CH$	-H	-Ph	-7-Cl		A.1	0.0925	1
Prazepam	$-CH_2$-cycloPr	-H	-Ph	-7-Cl		A.1	0.11	1
Ro06-9098	$-CH_2OCH_3$	-H	-Ph	$-7-NO_2$		A.1	0.43	1
Ro20-1310	-t-Bu	-H	-Ph	-7-Cl		A.1	0.62	1
Ro05-7094	$-CH(CH_3)CONHCH_3$	-H	-Ph	-7-Cl		A.1	>1	1
Ro07-1986	$-CH_2CH_2NH_2$	-H	$-C_6H_4$-2-F	-7-Cl		A.1	0.0083	2
Flurazepam	$^-CH_2CH_2N(Et)_2$	-H	$-C_6H_4$-2-F	-7-Cl		A.1	0.0148	1
Ro07-2750	$-CH_2CH_2OH$	-H	$-C_6H_4$-2-F	-7-Cl		A.1	0.0245	1
Irazepine	$-CH_2CH_2NCS$	-H	$-C_6H_4$-2-F	-7-Cl		A.1	0.033	1
Ro08-9013	$-CH_2CH_2OCH_2CONH_2$	-H	$-C_6H_4$-2-F	-7-Cl		A.1	0.0425	2
Ro10-3580	$-CH_2CH(OH)CH_2OH$	-H	$-C_6H_4$-2-F	-7-Cl		A.1	0.14	1
Ro07-9238	$-CH_2CH_2OPO_3^-$	-H	$-C_6H_4$-2-F	-7-Cl		A.1	0.13	1
Ro22-4683	-t-Bu	-H	$-C_6H_4$-2-Cl	$-7-NO_2$		A.1	0.3	2
Ro07-5096	$-CH_2CO_2H$	-H	$-C_6H_4$-2-F	-7-Cl		A.1	>1	1

Ro05-3663	-H	-H	-Me			A.1	>1	1
Desmethyl-tetrazepam	-H	-H	-1-cyclohexene	-7-Cl		A.1	0.034	2
Tetrazepam	-Me	-H	-1-cyclohexene	-7-Cl		A.1	0.034	2
Ro05-3328	-H	-H	-cyclohexane	-7-Cl		A.1	0.087	1
Bromazepam	-H	-H	-2-pyridyl	-7-Br		A.1	0.018	1
Ro05-5605	-H	-H	-1-napthyl	-7-Cl		A.1	0.29	0
Ro05-3580	-H	-H	-CH_2Ph	-7-Cl		A.1	>1	2
Ro11-4878	-H	-α-Me	-C_6H_4-2-F	-7-Cl		A.1	0.0035	2
Meclonazepam	-H	-α-Me	-C_6H_4-2-Cl	-7-NO_2		A.1	0.0012	1
Ro11-6896	-Me	-α-Me	-C_6H_4-2-F	-7-NO_2		A.1	0.007	2
Ro07-4532	-Me	-$(CH_3)_2$	-Ph	-7-Cl		A.1	>1	2
Ro06-7263	-Cl	-Me	-Ph	-7-Cl		A.1	0.049	1
Oxazepam	-H	-OH	-Ph	-7-Cl		A.1	0.018	1
Temazepam	-Me	-OH	-Ph	-7-Cl		A.1	0.016	1
Lorazepam	-H	-OH	-C_6H_4-2-Cl	-7-Cl		A.1	0.0035	1
Oxazepam-hemisuccinate	-H	-$OCOCH_2CH_2CO_2H$	-Ph	-7-Cl		A.1	0.33	2
Camazepam	-Me	-$OCON(CH_3)_2$	-Ph	-7-Cl		A.1	0.9	2
Ro20-7078	-Me	-Cl	-C_6H_4-2-F	-7-Cl		A.1	0.0053	1
Clorazepate	-H	-CO_2H	-Ph	-7-Cl		A.1	0.059	2
Ro11-8125			-C_6H_4-2-Cl	-H		A.2	0.037	2
Ro08-6739			-Ph	-Cl		A.2	0.07	2
Ro08-9212			-C_6H_4-2-Cl	-Cl		A.2	0.0039	1
Ro10-2643			-C_6H_4-2-Cl	-Cl		A.3	0.0094	1
Premazepam			-Ph			A.4	0.17	2
Ro05-3395	-H	-H	-Ph	-H		A.5	>1	1
Ro05-2181	-H	-H	-Ph	-Cl		A.5	>5	1
Ro05-2881	-Me	-H	-Ph	-Cl		A.5	>1	2
Ro05-3636	-Me	-Me	-C_6H_4-2-F	-Cl		A.5	>1	1
Ro15-8852	-H	-Me	-C_6H_4-2-Cl	-NO_2		A.5	0.726	2
Chlordiaz-epoxide			-Ph	-Cl		A.6	0.352	2
Demoxepam			-Ph	-Cl		A.7	0.31	1
Medazepam			-Ph	-Cl		A.8	0.87	1
Oxazolam		-Me	-Ph	-Cl		A.9	>1	2
Cloxazolam		-H	-C_6H_4-2-Cl	-Cl		A.9	>1	1
Ketazolam			-Ph	-Cl		A.10	>1	2
Ro07-4668	-H	-H	-C_6H_3-2,6-Cl	-Cl		A.11	>1	1
Ro11-3129	-H	-Me	-C_6H_4-2-Cl	-NO_2		A.11	0.0065	2
Ro03-7355	-Me	-$COCH(NH_2)(CH_2)_4NH_2$	-Ph	-Cl		A.11	0.42	1
Ro31-0214	-H	-CO-Pyrrolidine	-2-pyridyl	-Br		A.11	>1	1
Clobazam	-Me		-Ph	-Cl		A.12	0.13	1
Desmethyl-clobazam	-H		-Ph	-Cl		A.12	0.21	1
U-35005	-Me	-H	-C_6H_4-2-Cl	-H		A.13	0.0043	1
Estazolam	-H	-H	-Ph	-Cl		A.13	0.0085	1
Alprazolam	-Me	-H	-Ph	-Cl		A.13	0.02	1
Triazolam	-Me	-H	-C_6H_4-2-Cl	-Cl		A.13	0.004	2
alpha-hydroxy-triazolam	-CH_2OH	-H	-Ph	-Cl		A.13	0.0042	2
Adinazolam	-$CH_2N(CH_3)_2$	-H	-Ph	-Cl		A.13	0.135	1
U-43465F	-$(CH_2)_2N(CH_3)_2$	-H	-Ph	-Cl		A.13	0.0114	1
Ro11-5073	-Me	-α-Me	-C_6H_4-2-F	-Cl		A.13	0.0033	1
Ro11-6679	-Me	-α-Me	-C_6H_4-2-F	-NO_2		A.13	0.004	1
Ro11-5074	-Me	-α-Me	-C_6H_4-2-Cl	-NO_2		A.13	0.0034	0

Ro17-4582	-Me		-C_6H_4-2-Cl	-H		A.14	0.0035	1
Etizolam	-Me		-C_6H_4-2-Cl	-Et		A.14	0.0031	2
Ro11-1465	-Me		-C_6H_4-2-Cl	-Cl		A.14	0.0014	1
Brotizolam	-Me		-C_6H_4-2-Cl	-Br		A.14	0.0012	0
Ro14-1636	-Me		-C_6H_4-2-Cl	-I		A.14	0.0015	0
Ro11-7800	-CH_2NH_2		-C_6H_4-2-Cl	-Cl		A.14	0.0029	0
Midazolam	-Me	-H	-C_6H_4-2-F	-8-Cl		A.15	0.0048	2
alpha-hydroxy-midazolam	-CH_2OH	-H	-C_6H_4-2-F	-8-Cl		A.15	0.0045	1
Ro15-8670	-H	-CO_2Et	-Ph	-8-Cl		A.15	0.015	2
Ro16-0529	-H	-$CO_2C(CH_3)_3$	-Ph	-7-Cl		A.15	0.014	1
Ro21-5205	-H	-CO_2CH_3	-C_6H_4-2-F	-8-Cl		A.15	0.0074	1
Ro22-1892	-H	-$CO_2CH(CH_3)_2$	-C_6H_4-2-F	-8-Cl	O	A.15	0.012	2
Ro22-0992	-H	-CO_2H	-C_6H_4-2-Cl	-8-Cl		A.15	0.013	1
Ro21-8137	-H	-$CONH_2$	-C_6H_4-2-F	-8-Cl		A.15	0.0035	2
Ro21-8384	-H	-$CONH_2$	-C_6H_4-2-Cl	-8-Cl		A.15	0.0038	0
Ro21-8482	-$CH_2N(CH_3)_2$	-$CONH_2$	-C_6H_4-2-Cl	-8-Cl		A.15	0.026	1
Loprazolam			-C_6H_4-2-Cl	-NO_2		A.16	0.0063	2
Ro14-1359			-C_6H_4-2-F	-Cl		A.17	0.07	2
Ro15-8867			-C_6H_4-2-Cl	-NO_2		A.17	0.025	2
Ro14-7187	-Me		-Ph	-H		A.18	0.41	2
Ro13-9868	-Me		-Ph	-Cl		A.18	0.042	2
Ro14-5921	-H		-C_6H_4-2-F	-Cl		A.18	0.019	2
Ro14-0304	-Me		-C_6H_4-2-F	-Cl		A.18	0.0065	1
Ro14-2652	-CH_2NH_2		-C_6H_4-2-F	-Cl		A.18	0.0056	1
Ro15-9270	-Me		-C_6H_4-2-Cl	-NO_2		A.18	0.005	2
Ro14-0609			-C_6H_4-2-F	-Cl		A.19	0.0244	2
Ro15-2201	-CO_2CH_3		-Ph	-H		A.20	0.0015	2
Ro15-0791	-CO_2Et		-Ph	-H		A.20	0.0025	1
Ro15-2200	-CO_2Et		-C_6H_4-2-F	-H		A.20	0.0042	0
Ro15-3929	-CO_2Et		-C_6H_4-2-F	-Cl		A.20	0.016	2
Ro14-3930	-$CONH_2$		-C_6H_4-2-F	-Cl		A.20	0.015	2
Ro14-5568	-H		-C_6H_4-2-F	-Cl		A.20	0.23	2
Ro22-1274			-C_6H_4-2-F	-Cl		A.21	0.075	2
Ro22-1251	-Me		-Ph	-Cl		A.22	0.011	2
Ro22-1366	-Me		-C_6H_4-2-F	-Cl		A.22	0.004	1
Ro22-2038	-NH_2		-C_6H_4-2-F	-Cl		A.22	0.0028	1
Ro22-3245	-H		-C_6H_4-2-Cl	-Cl		A.22	0.0028	2
Ro22-3148			-Ph	-H		A.23	0.42	1
Ro22-3147			-C_6H_4-2-Cl	-H		A.23	0.0052	2
Ro22-0780			-Ph	-Cl		A.23	0.011	2
Ro22-2466			-C_6H_4-2-F	-Cl		A.23	0.0019	1
Ro22-2468			-C_6H_4-2-Cl	-Cl		A.23	0.0025	0
Ro14-7181	-NH_2					A.24	>1	1
Ro16-4234	-NH_2			-8-Cl		A.24	3	2
Ro14-5974	-OEt					A.24	0.0064	0
Ro15-4941	-OEt			-8-Cl		A.24	0.0017	1
Ro14-5975	-OEt			-9-Cl		A.24	0.062	1
Ro16-3607	-O-t-Bu			-8-OMe		A.24	0.046	1
Ro16-6624	-O-t-Bu			-8-Et		A.24	0.01	1
Ro16-3774	-O-t-Bu			-8-Me		A.24	0.0032	0
Ro16-0071	-O-t-Bu					A.24	0.0032	1
Ro16-5824	-O-t-Bu			-8-SMe		A.24	0.0034	0
Ro16-8912	-O-t-Bu			-8-F		A.24	0.0062	0
Ro16-4261	-O-t-Bu			-9-F		A.24	0.0077	0
Ro16-0075	-O-t-Bu			-8-Cl		A.24	0.0025	0
Ro16-6127	-O-t-Bu			-8-Cl,9-F		A.24	0.0031	1

Ro16-3058	-O-t-Bu			-9-Cl		A.24	0.09	0
Ro16-6028	-O-t-Bu			-8-Br		A.24	0.0022	0
Ro16-6605	-O-t-Bu			-8-I		A.24	0.0021	0
Ro16-6048	-O-t-Bu			-8-CF_3		A.24	0.0033	1
Ro16-6950	-O-t-Bu			-8-NO_2		A.24	0.0028	2
Ro16-4019	-O-n-Pr			-8-Cl		A.24	0.0014	2
Ro16-3031	-O-i-Pr			-8-Cl		A.24	0.0025	0
Ro16-9906	-O-CH_2CH=CH_2			-8-Cl		A.24	0.0017	0
Ro16-7081	-O-n-Bu			-8-Cl		A.24	0.0026	0
Ro16-7082	-O-i-Bu			-8-Cl		A.24	0.0063	0
Ro16-7083	-O-sec-Bu			-8-Cl		A.24	0.0029	0
Ro16-9918	-O-CH_2-cycloPr			-8-Cl		A.24	0.0023	1
Ro16-4020	-O-n-hexyl			-8-Cl		A.24	0.0026	0
Ro16-6654	-O-cyclohexyl			-8-Cl		A.24	0.004	1
Ro17-3206	-O-cycloheptyl			-8-Cl		A.24	0.0034	0
Ro17-3207	-O-cyclooctyl			-8-Cl		A.24	0.0052	0
Ro16-4021	-O-CH_2-Ph			-8-Cl		A.24	0.0015	0
Ro17-1302	-O-Ph			-8-Cl		A.24	0.0053	1
Ro15-2427	-NH_2					A.25	>1	2
Ro14-7437	-OEt					A.25	0.003	0
Ro15-3505	-OEt			-7-Cl		A.25	0.0027	0
Ro15-1310	-OEt			-8-Cl		A.25	0.0068	2
Ro15-1746	-OEt			-9-Cl		A.25	>1	1
Ro15-3237	-OEt			-10-Cl		A.25	>1	0
Ro15-5623	-OEt			-7-F		A.25	0.002	0
Ro15-1788	-OEt			-8-F		A.25	0.0025	0
Ro17-4896	-OEt			-9-F		A.25	>1	0
Ro17-9741	-OMe			-7-Cl		A.26	0.0024	2
Ro16-0858	-OEt			-7-Cl		A.26	0.0013	0
Ro16-0153	-O-t-Bu			-7-Cl		A.26	0.0053	0
Ro17-1812	-OCH_2-cycloPr			-7-Cl		A.26	0.0013	0
Ro17-3211	-O-cyclohexyl			-7-Cl		A.26	0.0024	0
Ro17-3401	-OCH_2Ph			-7-Cl		A.26	0.0009	0
Ro17-2620	-O-t-Bu			-8-F		A.26	0.0051	0
Ro16-9351	-O-t-Bu			-7-Cl,8-F		A.26	0.0037	0
Ro16-9905	-O-t-Bu			-7-Br		A.26	0.0012	0
Ro17-1338	-O-t-Bu			-7-I		A.26	0.001	0
Ro14-7462						A.27	1	0
Zopiclone						B	0.031	2
Suriclone						C	0.0021	2
CL218-872						D	0.12	2
CGS8216	-H					E.1	0.0005	2
CGS9895	-OMe					E.1	0.0003	1
CGS9896	-Cl					E.1	0.0007	1
1-6a	-H	-OH	-CH_2Ph			E.2	>2	1
1-6b	-OMe	-OH	-CH_2Ph			E.2	>2	1
1-6c	-Cl	-OH	-CH_2Ph			E.2	>2	2
1-6d	-F	-OH	-CH_2Ph			E.2	>2	0
1-7a	-H	-OH				E.2	>2	1
1-7b	-OMe	-OH				E.2	>2	2
1-7c	-Cl	-OH				E.2	>2	2
1-7d	-F	-OH				E.2	>2	0
1-8	-H	-SH				E.2	>2	1
1-9	-H	-Cl				E.2	>2	2
PK8165	-$(CH_2)_2$-4-piperidine	-H	-Ph			F	0.1	2

PK9084	-$(CH_2)_2$-3-piperidine	-H	-Ph			F	0.38	1
2-37	-H	-CO_2Me	-H			F	>100	1
2-38	-H	-CO_2H	-H			F	>100	1
2-39	-H	-CHO	-H			F	>100	1
2-31a	-H	-H	-CO_2Me			G.1	13.2	2
2-31b	-H	-H	-H			G.1	>100	2
3-3b	-OMe	-OMe	-CO_2Me			G.1	10	2
2-35	-OH	-OH	-H			G.2	>100	2
2-36	-OMe	-OMe	-H			G.2	>100	1
2-29	-H	-H	-CO_2H			G.2	>100	1
2-30	-H	-H	-CO_2Me			G.2	>100	1
3-2b	-OMe	-OMe	-CO_2Me			G.2	32	2
3-2c	-OH	-OH	-CO_2H			G.2	>50	2
2-32	-OH	-H				H	>100	1
2-33	-H	-H				H	>100	2
2-34	-H	-CO_2Me				H	>100	2
4-GBLD_137	-SCH_2Ph	-OMe	-Ph			I	0.025	1
4-GBLD_150	-$SCH_2C_6H_4$-2-Cl	-OMe	-Ph			I	0.054	1
4-GBLD_167	-OC_6H_4-2-OMe	-OMe	-Ph			I	0.07	2
4-GBLD_168	-OC_6H_4-3-OMe	-OMe	-Ph			I	0.463	1
4-GBLD_177	-$SCH_2C_6H_4$-4-Me	-OMe	-Ph			I	0.074	0
4-GBLD_190	-$SCH_2C_6H_4$-3-OMe	-OMe	-Ph			I	0.011	0
4-GBLD_214	-$SCH_2C_6H_4$-2-OMe	-OMe	-Ph			I	0.009	1
4-GBLD_219	-$SCH_2C_6H_4$-4-NO_2	-OMe	-Ph			I	0.024	2
4-GBLD_220	-$SCH_2C_6H_4$-4-NH_2	-OMe	-Ph			I	0.009	0
4-GBLD_221	-$OCH_2C_6H_4$-3-OMe	-OMe	-Ph			I	0.0064	1
4-GBLD_230	-$SCH_2C_6H_4$-2-NH_2	-OMe	-Ph			I	0.049	1
4-GBLD_231	-$NHCH_2C_6H_4$-3-OMe	-OMe	-Ph			I	0.0029	1
4-GBLD_233	-$SCH_2C_6H_4$-3-OMe	-OMe	-C_6H_4-4-F			I	0.011	2
4-GBLD_251	-F	-OMe	-C_6H_4-4-Me			I	0.03	1
4-GBLD_254	-$NHCH_2C_6H_4$-3-OMe	-OMe	-3-pyridyl			I	0.0021	0
4-GBLD_266	-$SCH_2C_6H_4$-3-OMe	-OMe	-C_6H_4-3-F			I	0.01	0
4-GBLD_268	-OC_6H_4-2-OMe	-OMe	-C_6H_4-3-F			I	0.065	1
4-GBLD_274	-$NHCH_2C_6H_4$-3-OMe	-OMe	-C_6H_4-4-F			I	0.0015	0
4-GBLD_293	-$SCH_2C_6H_4$-3-OMe	-OMe	-C_6H_4-3-CF3			I	0.163	1
4-GBLD_305	-OC_6H_4-2-OMe	-CH_2NHAc	-Ph			I	10	1
4-GBLD_307	-$NHCH_2C_6H_4$-3-OMe	-OMe	-C_6H_4-3-NO_2			I	0.0025	2

4-GBLD_308	-NHCH$_2$C$_6$H$_4$-3-OMe	-OMe	-C$_6$H$_4$-4-Me			I	0.0032	0
4-GBLD_312	-NHCH$_2$C$_6$H$_4$-2-OMe	-OMe	-C$_6$H$_4$-4-F			I	0.0022	2
4-GBLD_313	-NHCH$_2$C$_6$H$_4$-3-OMe	-OMe	-C$_6$H$_4$-3-NH$_2$			I	0.0018	0
4-GBLD_345	-NHCH$_2$C$_6$H$_4$-3-OMe	-OMe	-C$_6$H$_4$-4-NH$_2$			I	0.001	0
4-GBLD_318	-Cl	-CH$_2$NHAc	-Ph			I	0.474	1
4-GBLD_322	-F	-CH$_2$NHCOPh	-C$_6$H$_4$-4-Me			I	0.0082	0
4-GBLD_331	-H	-CH$_2$NHCOPh	-Ph			I	0.214	2
2-40	-CHO	-H				J	>100	1
2-41a	-H	-CHO				J	>100	1
2-41b	-COMe	-H				J	>100	2
3-5a	-H	-H				K.1	>50	2
3-7a	-Ph	-H				K.1	>50	2
3-8a	-Ph	-Me				K.1	>50	1
3-6a	-H					K.2	>50	1
3-6b	-Ph					K.2	10	2
3-12a	-H					L.1	11.2	1
3-12b	-CO$_2$Me					L.1	>50	1
3-13a	-H					L.2	10	1
3-13b	-CO$_2$Me					L.2	0.101	2
5-32						L.3	0.1	1
beta-CMA	-H	-CONHCH$_3$	-H		-H	M.1	0.25	2
DMCM	-H	-CO$_2$Me	-Et	-6,7-diOMe	-H	M.1	0.0094	2
8-2_betaCCM	-H	-CO$_2$Me	-H		-H	M.1	0.005	1
8-6	-H	-OCH(CH$_3$)CH$_2$CH$_3$	-H		-H	M.1	0.471	1
8-7	-H	-OCH$_2$CH(CH$_3$)$_2$	-H		-H	M.1	0.093	1
8-8	-H	-OCH$_2$CH$_2$CH(CH$_3$)$_2$	-H		-H	M.1	0.535	2
8-9	-H	-OCH$_2$C(CH$_3$)$_3$	-H		-H	M.1	0.104	1
8-10	-H	-OCH$_2$Ph	-H		-H	M.1	>1	1
8-12	-H	-COC(CH$_3$)$_3$	-H		-H	M.1	0.358	1
8-14_betaCCE	-H	-CO$_2$Et	-H		-H	M.1	0.005	1
8-19	-H	-CO$_2$CH$_2$C(CH$_3$)$_3$	-H		-H	M.1	0.75	0
8-20	-H	-CO$_2$-t-Bu	-H		-H	M.1	0.01	1
2-15	-Et	-CO$_2$Me	-H	-6-OH	-H	M.1	5.78	1
2-12	-Et	-CO$_2$Me	-H		-H	M.1	7.54	2
2-12a	-Ph	-CO$_2$Me	-H		-H	M.1	3.89	2
2-44	-Et	-H	-H		-H	M.1	250	2
2-42	-Me	-H	-H		-H	M.1	12.4	1
2-4	-H	-CO$_2$H	-H		-H	M.1	24	1
2-27	-H	-COMe	-H		-H	M.1	0.058	2
2-25	-H	-CHO	-H		-H	M.1	0.0623	1
2-9	-H	-CO$_2$Me	-H	-6-OH	-H	M.1	0.0027	1
2-5c	-H	-CO$_2$-n-Pr	-H		-H	M.1	0.001	2
2-5a	-H	-CO$_2$Me	-H		-Me	M.1	>50	2
2-24	-H	-CH$_2$OH	-H		-H	M.1	1.47	1
2-26	-H	-CH(OH)CH$_3$	-H		-H	M.1	3.16	1
5-6	-H	-OCH(CH$_3$)$_2$	-H		-H	M.1	0.5	1
5-7	-H	-O-n-Bu	-H		-H	M.1	0.098	1
5-11	-H	-CO-n-Pr	-H		-H	M.1	0.0028	1
5-12	-H	-n-Bu	-H		-H	M.1	0.245	2

3-16	-H	$-CONH_2$	-H		-H	M.1	0.068	1
3-17	-H	$-CH_2OC(=O)$ Me	-H		-H	M.1	0.75	1
6-5	-H	-Cl	-H		-H	M.1	0.045	2
6-6	-H	$-NO_2$	-H		-H	M.1	0.125	1
6-7	-H	-NCS	-H		-H	M.1	0.008	2
6-11	-H	-H	-H		-H	M.1	1.62	0
6-2	-H	-OMe	-H		-H	M.1	0.124	1
6-3	-H	-OEt	-H		-H	M.1	0.024	2
6-4	-H	-O-n-Pr	-H		-H	M.1	0.011	1
6-8	-H	$-NH_2$	-H		-H	M.1	25	1
6-9	-H	-OH	-H		-H	M.1	4	2
6-12	-H	-H	-H	$-6-NHCH_2PH$	-H	M.1	0.106	1
6-13	-H	$-NHCH_2PH$	-H		-H	M.1	>1.28	2
7-1a	-H	$-CO_2Me$	-H	$-6-NHCH_2PH$	-H	M.1	0.01	2
7-1b	-H	$-CH_2OH$	-H	$-6-NHCH_2PH$	-H	M.1	0.076	2
7-5	-H	-H	-H	$-6-NH_2$	-H	M.1	1.3	0
7-6				$-NHCH_2PH$		M.2	>5	2
5-13	$-CO_2Me$					M.3	0.653	1
6-18	S	$-CO_2Et$				M.4	0.98	1
5-35	C=O	$-CO_2Et$				M.4	26	2
5-36	C=NOH	$-CO_2Et$				M.4	5	1
5-37	O	$-CO_2Et$				M.4	9.2	2
5-38	CH_2	$-CO_2Et$				M.4	0.68	1
2-49	-Ph	-H	$-CO_2Me$	-H	-H	M.5	54	1
2-21	-H	-H	$-CO_2Me$	-H	-Me	M.5	4.5	1
2-50	-Ph	-H	$-CO_2Me$	-H	-Me	M.5	>100	2
2-59	-Ph	-H	$-CH_2OH$	-H	-H	M.5	6.11	2
2-57	-Me	-H	$-CO_2Me$	-H	-H	M.5	>50	1
2-17	$-CH_2OH$	-H	$-CO_2Me$	-H	-H	M.5	5	2
2-11a	-Et	-H	$-CO_2Me$	-H	-H	M.5	>100	1
2-14a	-Et	-H	$-CO_2Me$	-OH	-H	M.5	39	1
2-8	-H	-H	$-CO_2Me$	-OH	-H	M.5	0.573	2
2-61	-H	-H	-H	-OMe	-H	M.5	>500	1
2-62	-H	-H	-H	-H	-H	M.5	>100	1
2-63	-H	-Me	-H	-H	-H	M.5	>100	2
2-64	-H	-H	-H	-OH	-H	M.5	>100	2
2-55	-3-pyridyl	-H	$-CO_2Me$	-H	-H	M.5	6.25	1
2-22	-Et	-H	$-CO_2Me$	-H	-Me	M.5	>100	2
10-2		-H	-H	-H	-H	N.1	0.004	1
10-4a	-1-F	-H	-H	-H	-H	N.1	0.012	1
10-4c	-2-F	-H	-H	-H	-H	N.1	0.007	1
10-4e	-2-OMe	-H	-H	-H	-H	N.1	0.008	2
10-4f	-3-F	-H	-H	-H	-H	N.1	0.016	1
10-4h	-3-OMe	-H	-H	-H	-H	N.1	0.89	1
6-14		-H	-Me	-H	-H	N.1	1.163	1
5-18	-1-Me	-H	-H	-H	-H	N.1	0.083	2
5-19	-2-Me	-H	-H	-H	-H	N.1	0.01	1
5-20	-3-Me	-H	-H	-H	-H	N.1	0.224	0
5-21	-4-Me	-H	-H	-H	-H	N.1	6.86	0
5-24		-H	-H	-H	-Me	N.1	0.157	0
5-25		-H	-Me	-H	-Me	N.1	1.917	0
5-44	-4-OMe	-H	-H	-H	-H	N.1	0.25	0
5-45	-4-Cl	-H	-H	-H	-H	N.1	0.715	0
5-28		-Et	-H	-H	-H	N.1	0.285	1
9-5	-3-Cl	-H	-H	-H	-H	N.1	2.15	0
9-6	-3-Br	-H	-H	-H	-H	N.1	0.75	0
9-9	$-3-OCH_2Ph$	-H	-H	-H	-H	N.1	1.6	1

9-10	-3-OH	-H	-H	-H	-H	N.1	0.114	1
9-12	-1-Cl	-H	-H	-H	-H	N.1	0.08	0
9-13	-1-Br	-H	-H	-H	-H	N.1	0.028	0
9-16a	-2-Cl	-H	-H	-H	-H	N.1	0.01	0
9-17a	-2-Br	-H	-H	-H	-H	N.1	0.019	0
9-20a	-2-OCH_2Ph	-H	-H	-H	-H	N.1	>4	2
9-21a	-2-OH	-H	-H	-H	-H	N.1	0.006	1
9-20b	-4-OCH_2Ph	-H	-H	-H	-H	N.1	>4	0
9-21b	-4-OH	-H	-H	-H	-H	N.1	0.578	0
9-23		-H	-H	NO_2	-H	N.1	0.004	2
9-24	-3-Cl	-H	-H	NO_2	-H	N.1	1.19	1
9-25	-2-Cl	-H	-H	NO_2	-H	N.1	0.126	1
9-26		-H	-H	NH_2	-H	N.1	0.044	1
9-27	-3-Cl	-H	-H	NH_2	-H	N.1	>0.35	1
9-28	-2-Cl	-H	-H	NH_2	-H	N.1	0.098	2
9-29		-H	-H	Br	-H	N.1	0.006	0
9-30	-3-Br	-H	-H	Br	-H	N.1	>0.32	0
9-31	-1-Cl	-H	-H	Cl	-H	N.1	0.31	0
9-32	-3-SO_3^-	-H	-H	SO_3^-	-H	N.1	5	2
11-11a	-1-$(CH)_4$-2-	-H	-H	-H	-H	N.1	>0.125	0
11-11b	-3-$(CH)_4$-4-	-H	-H	-H	-H	N.1	>0.125	0
5-34						N.2	4.66	0
6-17						N.3	1.92	2
11-1-azadiindole	N	CH	CH			N.4	0.0106	2
11-2-azadiindole	CH	N	CH			N.4	0.0515	0
11-4-azadiindole	CH	CH	N			N.4	0.0102	0
11-11c						N.5	>3	1
7-2						N.6	0.005	0

[a] names that begin with a number are formed by hyphenating the reference number and the label given to the compound in the reference. Names that begin with a letter are from reference 12.
[b] K_i values are given in references 2 and 3. Reference 2 indicates a factor of 1.36 for converting K_i to IC_{50} values. In this work, we take the K_i values to be IC_{50} values; for 4 compounds that also appear in references 1, 5-11, the IC_{50} values given are equal to the K_i values in 2-3. Thus, it seems the authors use the terms K_i and IC_{50} interchangeably, although this is not correct (all references except 4 and 12 are from Cook *et al.*).
[c] training set, test set and unused compounds are indicated as 1, 2, and 0, respectively.

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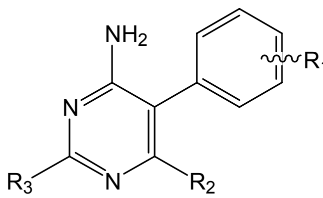
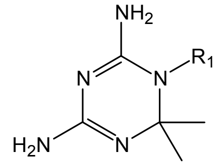
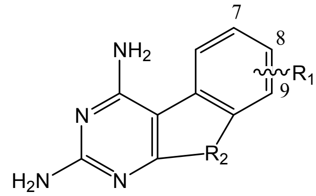
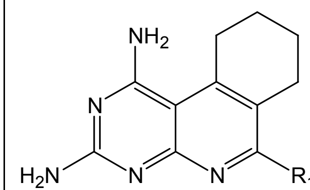
Table 3.1.1. Structures (families) of DHFR inhibitors

A 	B 	C 	D 

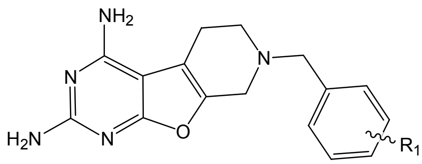
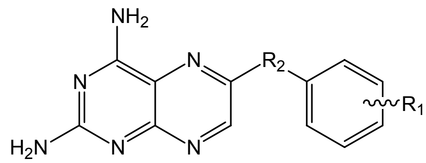
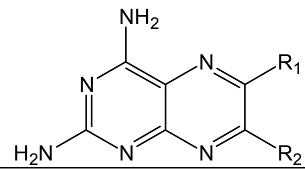
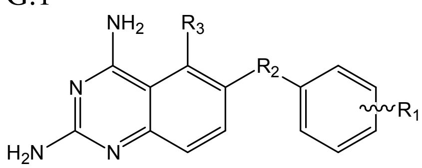
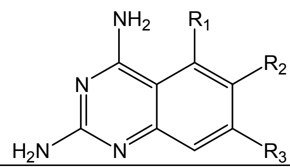
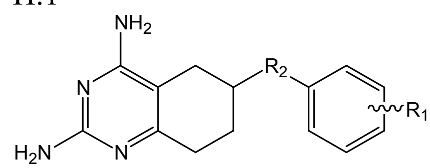
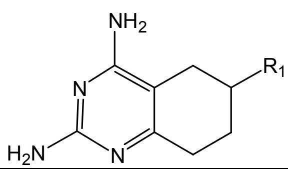
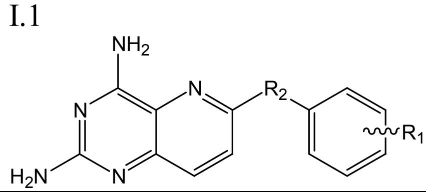
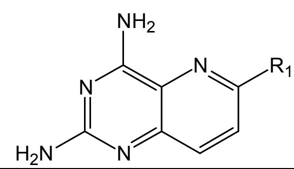
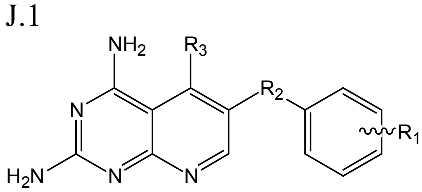
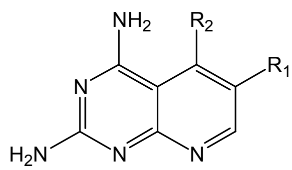
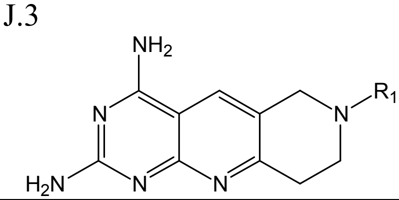
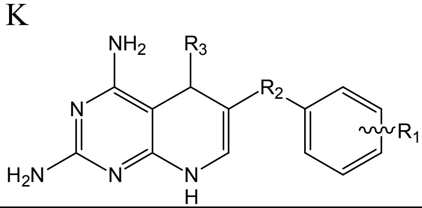
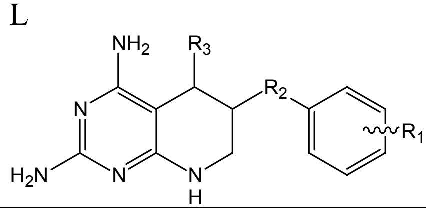
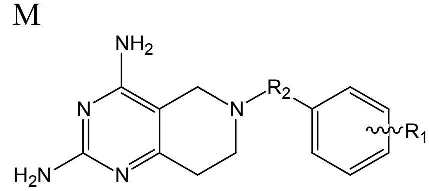
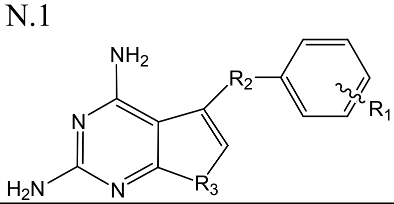
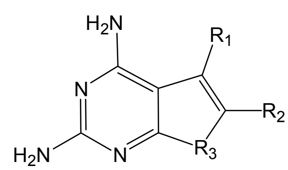
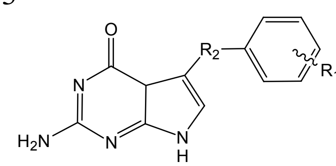
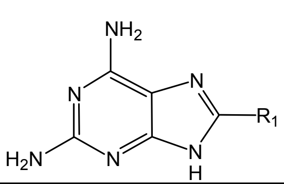
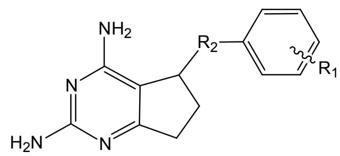
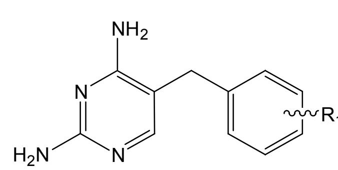
E 	F.1 	F.2 
G.1 	G.2 	H.1 
H.2 	I.1 	I.2 
J.1 	J.2 	J.3 
K 	L 	M 
N.1 	N.2 	N.3 
O 	P 	Q 

Table 3.1.2. Miscellaneous DHFR inhibitors

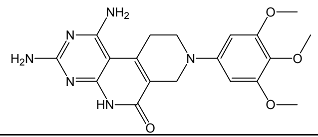
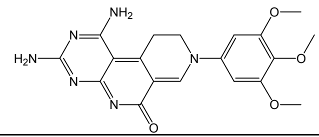
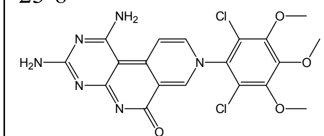
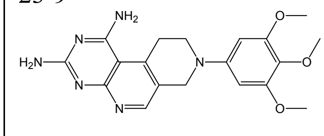
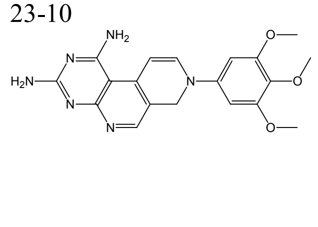
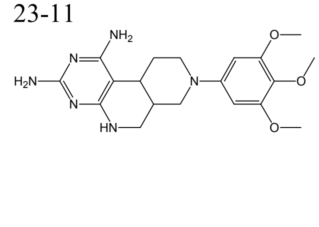
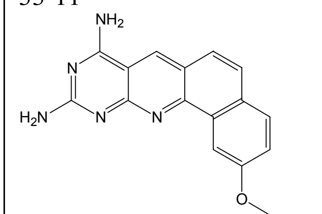
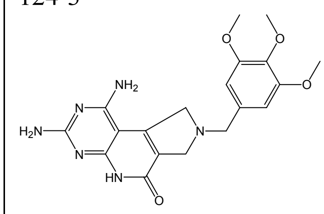
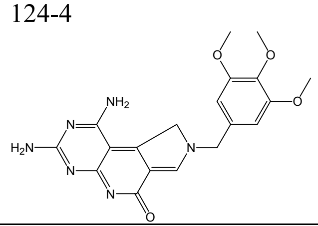
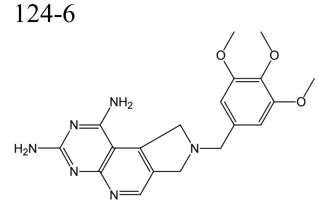
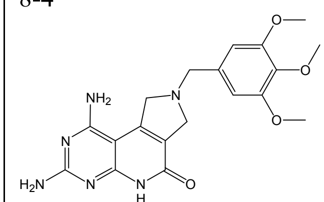
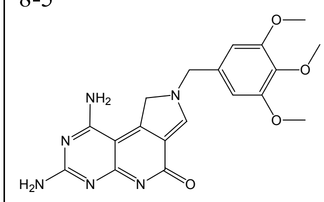
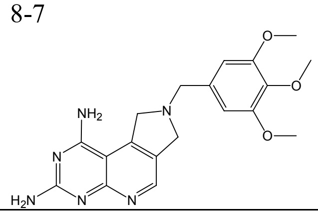
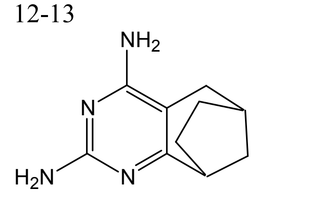
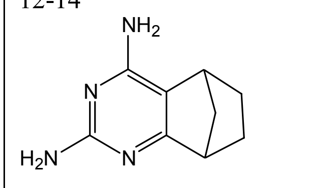
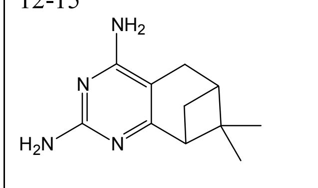
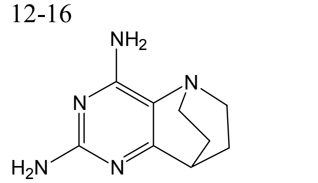
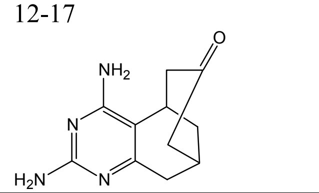
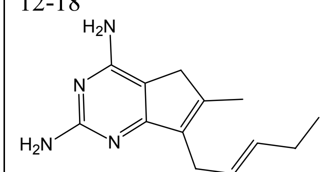
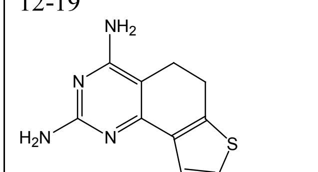
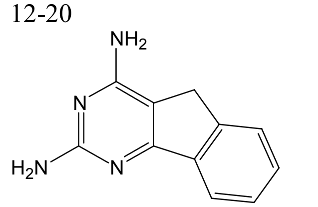
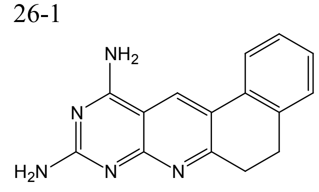
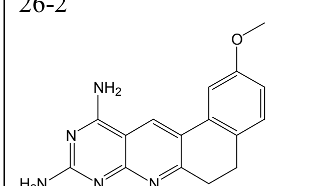
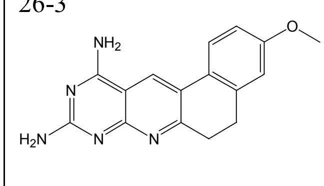
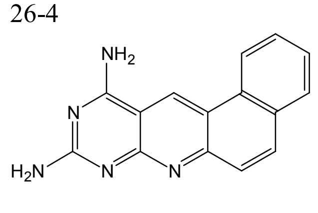
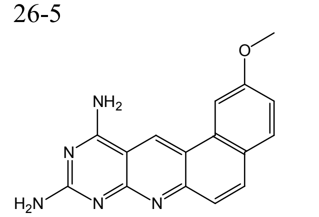
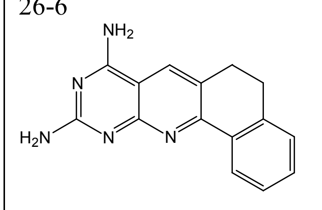
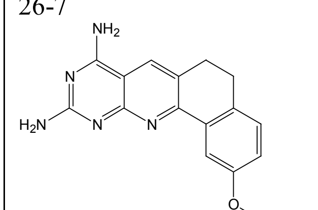
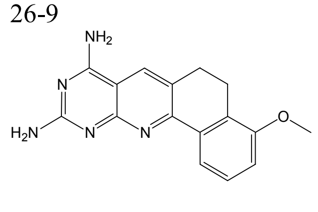
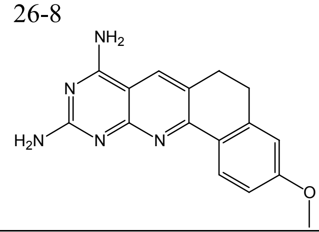
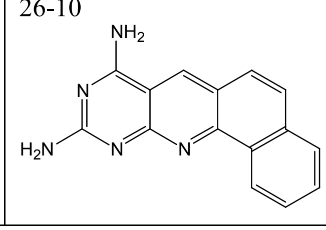
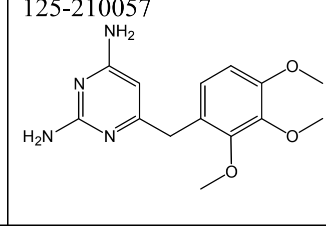
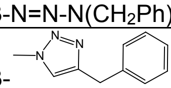
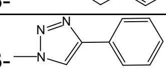
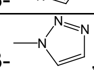
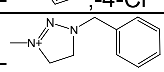
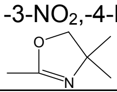
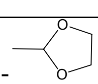
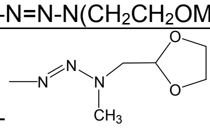
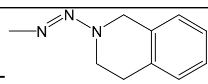
23-5 	23-6 	23-8 	23-9 
23-10 	23-11 	33-11 	124-3 
124-4 	124-6 	8-4 	8-5 
8-7 	12-13 	12-14 	12-15 
12-16 	12-17 	12-18 	12-19 
12-20 	26-1 	26-2 	26-3 
26-4 	26-5 	26-6 	26-7 
26-9 	26-8 	26-10 	125-210057 

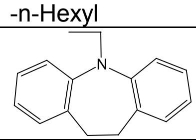
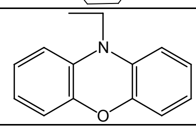
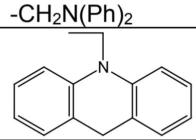
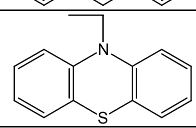
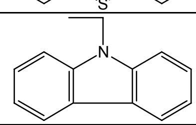
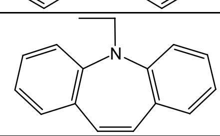
Table 3.2. DHFR inhibitors

Name[a]	R_1	R_2	R_3	Fa-mily	IC_{50} (μM)[b]			set[c]
					PC	TG	RL	
1-pyri-methamine	-4-Cl	-Et	$-NH_2$	A	3.7	0.39	2.3	1
1-3062	-3,4-Cl	-Et	$-NH_2$	A	1.08	0.094	0.19	0
1-7364	-3,4-Cl	-Me	$-NH_2$	A	1.68	3.7	0.32	1
1-115194	-3,4-Cl	$-CH_2OEt$	$-NH_2$	A	12.7	0.08	0.093	1
1-118203	-4-Cl	$-CH_2O$-(pNO_2-Ph)	$-NH_2$	A	85.1	1.37	42.5	2
1-125357	-3,4-Cl	$-CH=CHCH_2$-Ph	$-NH_2$	A	0.5	0.056	0.088	1
1-125358	-3,4-Cl	$-CH_2O$-Ph	$-NH_2$	A	13.5	0.11	1.28	1
1-125359	-3,4-Cl	$-CH_2O$-1-naphthyl	$-NH_2$	A	3.9	0.054	0.113	0
1-125850	-3,4-Cl	-CH=CH-2-naphthyl	$-NH_2$	A	12.8	0.86	2.06	2
1-211797	-4-Cl	$-CH_2CN$	$-NH_2$	A	490	>100	188	1
1-211804	-4-Cl	$-CF_3$	$-NH_2$	A	>42	>42	>42	2
1-212329	-4-Cl	$-CH_2NH$-(pNO_2-Ph)	$-NH_2$	A	17	1.49	1.06	1
1-302325	-3-NO_2,-4-Cl	-Et	$-NH_2$	A	0.85	0.013	0.015	1
1-319947	-3-N_3,-4-Cl	-Et	$-NH_2$	A	1.33	0.69	0.33	2
1-319949	-3-NO_2,-4-piperidine	-Et	$-NH_2$	A	5.05	0.43	0.9	1
1-330463	-3-$N^+\equiv N$,-4-Cl	-Et	$-NH_2$	A	0.19	0.023	0.032	1
1-330465	-3-N=N-N$(Me)_2$,-4-Cl	-Et	$-NH_2$	A	2.8	0.31	18.9	0
1-372950	-3-N=CHN$(Me)_2$,-4-Cl	-Et	$-NH_2$	A	10.6	1.6	3.5	1
1-372955	-4-piperidine	-Et	$-NH_2$	A	12	67.6	0.94	2
1-382035	-3-N(Me)CH_2-Ph,-4-NO_2	-Et	$-NH_2$	A	1.6	0.091	0.003	1
1-382036	-3-N(Me)CH_2-Ph,-4-NH_2	-Et	$-NH_2$	A	3.6	0.015	0.022	2
1-382042	-3-N(Et)CH_2-Ph,-4-NO_2	-Et	$-NH_2$	A	5.7	0.17	0.079	0
1-382046	-3-CH_2-Ph,-4-NO_2	-Et	$-NH_2$	A	1.03	0.019	0.025	2
1-400654	-4-Cl	-Et	$NHSO_2$ Me	A	1.4		0.28	1
2-4	-3-N=N-N(CH_2Ph)$(CH_2)_2$OAc,-4-Cl	-Et	$-NH_2$	A	0.17	0.69	19.4	1
2-11a	-3-  ,-4-Cl	-Et	$-NH_2$	A	5.18	0.52	1.38	0
2-11c	-3-  ,-4-Cl	-Et	$-NH_2$	A	3.53	0.37	0.15	0
2-11g	-3-  ,-4-Cl	-Et	$-NH_2$	A	24.8	0.28	0.38	1
2-21c	-3-  ,-4-Cl	-Et	$-NH_2$	A	13.5	1.38	3.3	2
2-23a	-3-NHCO-Ph,-4-Cl	-Et	$-NH_2$	A	10.8	0.51	4.1	2
2-23b	-3-NHCOC_6H_2-3,4,5-OMe,-4-Cl	-Et	$-NH_2$	A	1.3	0.08	0.99	1
2-24a	-3-NHCH_2-Ph,-4-Cl	-Et	$-NH_2$	A	0.12	0.07	0.63	1
2-24b	-3-N$(CH_3)CH_2C_6H_2$-3,4,5-OMe,-4-Cl	-Et	$-NH_2$	A	1.07	0.02	0.09	2
3-5-benzoprim	-3-NO_2,-4-NHCH_2-Ph	-Et	$-NH_2$	A	1.03	0.019	0.025	0
3-6-MBP	-3-NO_2,-4-N(Me)CH_2-Ph	-Et	$-NH_2$	A	1.6	0.091	0.003	0
3-8	-3-NO_2,-4-Cl	-Et	$-NH_2$	A	0.85	0.013	0.015	0
3-9	-3-NO_2,-4-NH$CH_2C_6H_4$-4-CO_2H	-Et	$-NH_2$	A	0.27	0.002	0.003	1
3-10	-3-NO_2,-4-N(Me)$CH_2C_6H_4$-4-CO_2H	-Et	$-NH_2$	A	0.27	0.003	0.004	0
3-14	-3-NO_2,-4-N(Me)$CH_2C_6H_4$-4-CONHMe	-Et	$-NH_2$	A	0.46	0.007	0.007	0
3-17	-3-NO_2,-4-N(Me)$CH_2C_6H_4$-4- 	-Et	$-NH_2$	A	0.95	0.014	0.015	2
3-22	-3-NO_2,-4-NH$CH_2C_6H_4$-4- 	-Et	$-NH_2$	A	0.93	0.015	0.016	1
3-25	-3-NO_2,-4-NH$CH_2C_6H_4$-4-$CONH_2$	-Et	$-NH_2$	A	0.53	0.004	0.008	0
3-26	-3-NO_2,-4-NH$CH_2C_6H_4$-4-CONHMe	-Et	$-NH_2$	A	0.58	0.004	0.006	0

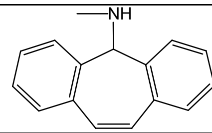
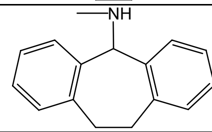
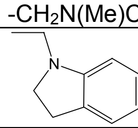
3-28	-3-NO_2,-4-$NHCH_2C_6H_4$-4-piperidine	-Et	-NH_2	A	2.2	0.14	0.27	2
3-27	-3-NO_2,-4-$NHCH_2C_6H_4$-4-CON$(Me)_2$	-Et	-NH_2	A	1.3	0.008	0.009	0
4-4a	-3-N=N-N$(Me)_2$,-4-Cl	-Et	-NH_2	A	2.8	0.31	18.9	0
4-4b	-3-N=CHN$(Me)_2$,-4-Cl	-Et	-NH_2	A	10.6		3.5	0
4-4c	-3-N=N-NHMe,-4-Cl	-Et	-NH_2	A	4.6		2.3	1
4-10a	-3-N=N-N(Me)Et,-4-Cl	-Et	-NH_2	A	14.1		202	1
4-10b	-3-N=N-N(Me)Pr,-4-Cl	-Et	-NH_2	A	10.2		>4	0
4-10c	-3-N=N-N(Me)CH_2Ph,-4-Cl	-Et	-NH_2	A	2.8		1.8	0
4-10d	-3-N=N-N(Me)CH_2CH_2N$(Et)_2$,-4-Cl	-Et	-NH_2	A	18.2		17.3	1
4-10e	-3-N=N-N$(Et)_2$,-4-Cl	-Et	-NH_2	A	29.3		17.7	0
4-10f	-3-N=N-N$(t\text{-}Bu)_2$,-4-Cl	-Et	-NH_2	A	>11		>11	0
4-10g	-3-N=N-1-pyrrolidyl,-4-Cl	-Et	-NH_2	A	>11		>11	1
4-10h	-3-N=N-1-piperidyl,-4-Cl	-Et	-NH_2	A	40.4		66.8	0
4-10i	-3-N=N-1-(4-Me-piperazine),-4-Cl	-Et	-NH_2	A	8.5		16.2	1
4-10j	-3-N=N-1-(4-OH-piperidine),-4-Cl	-Et	-NH_2	A	3.3		7.2	0
4-10k	-3-N=N-4-morpholine,-4-Cl	-Et	-NH_2	A	1.7		26.3	1
4-10l	-3-N=N-N(Me)CH_2CH_2OH,-4-Cl	-Et	-NH_2	A	3.5		27.9	0
4-10m	-3-N=N-N(Et)CH_2CH_2OH,-4-Cl	-Et	-NH_2	A	11.5		27	1
4-10n	-3-N=N-N(Pr)CH_2CH_2OH,-4-Cl	-Et	-NH_2	A	2.5		0.44	0
4-10o	-3-N=N-N(t-Bu)CH_2CH_2OH,-4-Cl	-Et	-NH_2	A	4.9		10.3	0
4-10p	-3-N=N-N(CH_2Ph)$(CH_2)_2OH$,-4-Cl	-Et	-NH_2	A	0.26		7	0
4-10q	-3-N=N-N$(CH_2CH_2OH)_2$,-4-Cl	-Et	-NH_2	A	0.44		5	0
4-10r	-3-N=N-N$(CH_2CH_2OMe)_2$,-4-Cl	-Et	-NH_2	A	0.91		26.1	0
4-10s	-3-  ,-4-Cl	-Et	-NH_2	A	0.57		0.5	2
4-11	-3-  ,-4-Cl	-Et	-NH_2	A	5		1.7	2
4-14a	-3-N=N-N(CH_2Ph)CH_2CH_2OCOMe,-4-Cl	-Et	-NH_2	A	0.17	0.69	19.4	0
5-10e	-3-N=N-N($CH_2C_6H_3$-3,4-OMe)CH_2CH_2OCOMe,-4-Cl	-Et	-NH_2	A	1.4	0.08	0.39	0
5-10f	-3-N=N-N($CH_2C_6H_3$-3,5-OMe)CH_2CH_2OCOMe,-4-Cl	-Et	-NH_2	A	0.86	0.378	0.845	0
5-10g	-3-N=N-N($CH_2C_6H_3$-3,4,5-OMe)CH_2CH_2OCOMe,-4-Cl	-Et	-NH_2	A	1.5	0.07	0.45	2
5-10h	-3-N=N-N($CH_2C_6H_4$-2-F)CH_2CH_2OCOMe,-4-Cl	-Et	-NH_2	A	92	7.8	14.9	0
5-10i	-3-N=N-N($CH_2C_6H_4$-3-F)CH_2CH_2OCOMe,-4-Cl	-Et	-NH_2	A	2.22	1.25	3.87	0
5-10j	-3-N=N-N($CH_2C_6H_4$-4-F)CH_2CH_2OCOMe,-4-Cl	-Et	-NH_2	A	1.91	4.4	3.83	0
5-10k	-3-N=N-N($CH_2C_6H_4$-2-Cl)CH_2CH_2OCOMe,-4-Cl	-Et	-NH_2	A	4.7	6.1	8.4	0
5-10l	-3-N=N-N($CH_2C_6H_4$-3-Cl)CH_2CH_2OCOMe,-4-Cl	-Et	-NH_2	A	0.58	1.16	1.29	0
5-10m	-3-N=N-N($CH_2C_6H_4$-4-Cl)CH_2CH_2OCOMe,-4-Cl	-Et	-NH_2	A	0.94	0.8	1.8	0
5-10n	-3-N=N-N($CH_2C_6H_4$-2-Me)CH_2CH_2OCOMe,-4-Cl	-Et	-NH_2	A	11.08	0.83	11.23	0
5-10o	-3-N=N-N($CH_2C_6H_4$-3-Me)CH_2CH_2OCOMe,-4-Cl	-Et	-NH_2	A	5.29	5.39	5.8	0
5-10p	-3-N=N-N($CH_2C_6H_4$-4-Me)CH_2CH_2OCOMe,-4-Cl	-Et	-NH_2	A	1.7	0.63	3.85	0
5-10q	-3-N=N-N($CH_2C_6H_4$-4-CF_3)CH_2CH_2OCOMe,-4-Cl	-Et	-NH_2	A	1.35	2.54	6.62	0
5-10r	-3-N=N-N($CH_2C_6H_4$-3-NO_2)CH_2CH_2OCOMe,-4-Cl	-Et	-NH_2	A	1.3	0.83	1.61	0
5-10t	-3-N=N-N(CH_2-1-naphthyl)CH_2CH_2OCOMe,-4-Cl	-Et	-NH_2	A	0.053	0.2	0.28	0

5-10u	-3-N=N-N(CH_2-2-naphthyl)CH_2CH_2OCOMe,-4-Cl	-Et	-NH_2	A	0.59	1.05	0.62	0
5-10v	-3-N=N-N(CH_2-4-pyridinyl)CH_2CH_2OCOMe,-4-Cl	-Et	-NH_2	A	3.46	0.64	6.19	0
5-10s	-3-N=N-N($CH_2C_6H_4$-4-NO_2)CH_2CH_2OCOMe,-4-Cl	-Et	-NH_2	A	1.21	85	3.33	0
5-12b	-3-N=N-N(CH_2Ph)(CH2)$_3$OH,-4-Cl	-Et	-NH_2	A	0.68	2.01	7.35	1
5-12c	-3-N=N-N(CH_2CH_2Ph)CH_2CH_2OH,-4-Cl	-Et	-NH_2	A	0.48	1.79	4.67	0
5-12d	-3-N=N-N(CH(Me)Ph)CH_2CH_2OH,-4-Cl	-Et	-NH_2	A	1.09	1.47	6.44	0
5-10a	-3-N=N-N(CH_2Ph)CH_2CH_2OCOEt,-4-Cl	-Et	-NH_2	A	1.3	0.7	3.3	0
5-10b	-3-N=N-N(CH_2Ph)CH_2CH_2OCO-iPr,-4-Cl	-Et	-NH_2	A	1.2	0.66	3.7	0
5-10c	-3-N=N-N(CH_2Ph)CH_2CH_2OCOPh,-4-Cl	-Et	-NH_2	A	0.85	0.22	0.69	0
5-10d	-3-N=N-N($CH_2C_6H_4$-4-OMe)CH_2CH_2OCOMe,-4-Cl	-Et	-NH_2	A	2.1	0.44	2.1	0
1-3075	-C_6H_4-4-CO_2H			B	>52	>52	>52	2
1-3077	-C_6H_4-3,4-Cl			B	0.154	0.002	0.029	2
1-5205	-C_6H_4-2,6-Me			B	>20		>20	1
1-96642	-naphthyl			B	0.72		0.16	1
1-104124	-C_6H_4-3-(CH_2)$_4$-C_6H_3-2,4-Cl			B	0.36	0.016	0.018	1
1-104129	-C_6H_4-4-(CH_2)$_4$-C_6H_3-2,4-Cl			B	2.23	0.034	0.043	0
1-109832	-C_6H_4-3-O(CH_2)$_3$O-C_6H_4-4-NH_2			B	0.17	0.03	0.16	1
1-109834	-C_6H_4-3-OCH_2-C_6H_4-3-NH_2			B	1.4	0.27	0.25	2
1-112529	-C_6H_4-3-O(CH_2)$_2$O-C_6H_4-4-NH_2			B	0.47	0.011	0.041	1
1-113220	-C_6H_4-4-(CH_2)$_2$$CONHC_6H_4$-4-$SO_2F$			B	0.59	0.03	0.003	2
1-115928	-C_6H_4-3-CN			B	1.6	0.17	0.22	1
1-118213	-C_6H_4-4-$OCH_2CONHC_6H_4$-3-SO_2F			B	1	0.026	0.074	1
1-122059	-C_6H_3-(3-Cl)-4-$OCH_2CH_2NHSO_2C_6H_4$-3-SO_2F			B	0.34		0.027	2
1-127153	-C_6H_4-4-(CH_2)$_4$-Ph			B	0.31	0.026	0.02	1
1-127752	-C_6H_4-3-(CH_2)$_2$-Ph			B	0.43	0.01	0.041	2
1-127753	-C_6H_4-4-(CH_2)$_2$-Ph			B	0.57	0.012	0.029	1
1-127754	-C_6H_4-3-(CH_2)$_4$-O-Ph			B	0.27	0.005	0.005	1
1-128189	-C_6H_4-4-NH_2			B	70.9	1.44	26	1
1-132275	-C_6H_3-(3-Cl)-4-$OCH_2CONHPh$			B	0.3		0.004	1
1-137765	-C_6H_4-3-$CH_2CONHPh$			B	0.44	0.084	0.28	1
1-158695	-C_6H_2-3,4,5-OMe			B	>26		>26	2
1-211327	-C_6H_4-4-CH(CN)(CH_2)$_3$-Ph			B	6.1	0.031	0.067	2
1-368890	-C_6H_3-3,5-CF3			B	66.9	0.58	0.57	1
6-55	-C_6H_3-2,5-OMe			B	>60	>60	>60	2
6-56	-C_6H_2-2,3,4-OMe			B	20.4	1.2	6.1	0
6-57	-C_6H_3-2,4-OMe			B	>90	>90	>90	1
6-58	-C_6H_3-3,5-OMe			B	12.9	16.4	11.4	1
6-59	-C_6H_2-3,4,5-OMe			B	>100	84	>100	0
6-61	-2-naphthyl-6-Cl			B	4.4	1.3	9.1	0
6-62	-2-naphthyl-5,6-Cl			B	3	1.4	3.1	0
6-63	-2-naphthyl-5,7-Cl			B	0.84	0.41	1.2	0
1-137187	-8-OMe	-CH_2CH_2-		C	6.2	2.5	1.25	1
1-137188	-7,8-Me	-CH_2CH_2-		C	0.59	0.038	0.036	1
1-137189	-7,9-Cl	-CH_2CH_2-		C	1.07	0.057	0.022	2
1-137191	-7-Cl	-CH=CH-		C	0.12	0.011	0.016	0
1-137192	-8-Me	-CH=CH-		C	0.46	0.052	0.06	2
6-50	-Me			D	1		0.062	1
6-51	-Et			D	>30		0.18	0
6-52	-SMe			D	>30		0.18	1
6-53	-Ph			D	6.8		0.45	2
6-49	-H			D	>30		2.8	1
7-1	-H			E	>9	1.4	15	2
7-2	-3,4,5-OMe			E	>35	>35	>35	2
7-3	-3,5-OMe			E	>22.8	>20.4	22.8	1

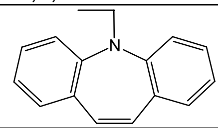
7-4	-2,4-Cl			E	22.6	13.1	50.9	1
7-5	-3,4-Cl			E	24.1	22.3	20.6	1
7-6	-2,6-Cl			E	40.5	31.7	81.2	0
7-7	-4-CONHCH(CO$_2$H)CH$_2$CH$_2$CO$_2$H			E	10.9	21.5	85.8	2
1-Metho-trexate	-4-CONHCH(CO$_2$H)CH$_2$CH$_2$CO$_2$H	-CH$_2$N(Me)-		F.1	0.001	0.014	0.003	1
1-29630	-2-Cl,-4-CONHCH(CO$_2$H) CH$_2$CH$_2$CO$_2$H,-6-Cl	-CH$_2$N(Me)-		F.1	0.000035	0.001	0.00033	0
1-98535	-4-CONHCH(CO$_2$H)CH$_2$CO$_2$H	-CH$_2$NH-		F.1	0.028		0.017	0
1-98579	-2-Cl,-4-CONHCH(CO$_2$H) CH$_2$CH$_2$CO$_2$H,-6-Br	-CH$_2$N(Me)-		F.1	0.00021	0.00093	0.00145	0
1-98580	-2-Br,-4-CONHCH(CO$_2$H)CH$_2$CH$_2$CO$_2$H	-CH$_2$N(Me)-		F.1	0.00109	0.00095	0.00055	1
1-107146	-2-F,-4-CONHCH(CO$_2$H)CH$_2$CH$_2$CO$_2$H	-CH$_2$N(Me)-		F.1	0.00072	0.0119	0.00036	0
1-117356	-4-CONHCH(CO$_2$H)CH$_2$CH$_2$CO$_2$H	-CH$_2$N(Me)-		F.1	0.00088		0.0062	0
1-127977	-4-CONHCH(CO$_2$H)(CH$_2$)$_4$NH$_2$	-CH$_2$N(Me)-		F.1	>8	>8	>8	0
1-131463	-4-CO$_2$H	-CH$_2$N(Me)-		F.1	0.94	2.27	0.11	1
1-136735	-2-I,-4-CONHCH(CO$_2$H)CH$_2$CH$_2$CO$_2$H	-CH$_2$N(Me)-		F.1	0.000104	0.0016	0.00165	0
1-136736	-2-F,-4-CONHCH(CO$_2$H) CH$_2$CH$_2$CO$_2$H,-6-F	-CH$_2$N(Me)-		F.1	0.00051	0.099	0.0048	0
1-137545	-2-Me,-4-CO$_2$H,-6-Me	-CH$_2$N(Me)-		F.1	0.19	0.28	0.038	2
1-144698	-2-CF$_3$,-4-CONHCH(CO$_2$H)CH$_2$CH$_2$CO$_2$H	-CH$_2$N(Me)-		F.1	0.00031	0.0067	0.00155	1
1-152737	-2-OMe,-4-CONHCH(CO$_2$H)CH$_2$CH$_2$CO$_2$H	-CH$_2$N(Me)-		F.1	0.00049		0.0011	2
1-169531	-4-CONHCH(CO$_2$H)CH$_2$CH$_2$CO$_2$H	-CH$_2$N(Et)-		F.1	0.00035		0.0014	0
1-233903	-4-CONH$_2$	-CH$_2$NH-		F.1	>1000		220	0
1-233904	-H	-CH$_2$N(Me)-		F.1	10	16.9	0.83	1
1-233910	-4-NHCOMe	-CH$_2$NH-		F.1	7.1		0.17	1
1-235791	-4-COMe	-CH$_2$NH-		F.1	0.72		0.075	1
1-236642	-4-CON(Me)$_2$	-CH$_2$NH-		F.1	>5		>10	0
1-241522	-4-CONH-Pr	-CH$_2$NH-		F.1	3.1		0.48	2
8-1	-H	-CH$_2$S-		F.1	9.5	0.77	246	0
9-12a	-2,3-(CH)$_4$-	-CH$_2$NH-		F.1	0.13	0.076	1.26	0
9-12c	-2,3-(CH)$_4$-	-CH$_2$S-		F.1	4.2	7	8.2	1
9-12d	-3-Me	-CH$_2$S-		F.1	21.2	1.8	8.48	0
9-12e	-4-Me	-CH$_2$S-		F.1	30	15	26	0
9-12g	-4-OMe	-CH$_2$S-		F.1	17.2	14	10.2	0
9-12h	-3,4-OMe	-CH$_2$S-		F.1	58.2	23.2	36.3	2
9-12i	-3-Cl	-CH$_2$S-		F.1	42.2	5.7	26.6	1
9-12j	-4-Cl	-CH$_2$S-		F.1	42.1	16.8	14.3	0
9-12k	-4-Cl	-CH$_2$NH-		F.1	1.5	0.59	0.3	2
9-12l	-2-Me,-5-OMe	-CH$_2$NH-		F.1	>3	0.44	>3	1
9-12m	-2-Me,-6-OMe	-CH$_2$NH-		F.1	15.8	1.85	5.7	1
9-12n	-2,5-OMe	-CH$_2$NH-		F.1	6.2	6.9	22.9	0
9-12o	-2,5-OMe	-CH$_2$N(Me)-		F.1	3.9	0.21	0.47	1
9-12p	-3,5-OMe	-CH$_2$NH-		F.1	0.96	0.11	0.88	1
9-12q	-3,4,5-OMe	-CH$_2$NH-		F.1	7	1	1.9	2
9-12r	-2-OMe,-5-CF$_3$	-CH$_2$NH-		F.1	21	10.6	21	1
9-12s	-3-OMe,-5-CF$_3$	-CH$_2$NH-		F.1	0.68	0.89	1.9	1
9-12t	-2,3-(CH)$_4$-	-CH$_2$CH$_2$-		F.1	97	0.82	60	2
9-12u	-2,5-OMe	-CH$_2$CH$_2$-		F.1	11.1	5.4	23.2	2
10-2d	-H	-CH$_2$CO$_2$-		F.1	9.8	5.8	7	2
9-12f	-3-OMe	-CH$_2$S-		F.1	9.9	1.6	7.7	1
10-9a	-H	-CH$_2$CO$_2$CH$_2$-		F.1	11.2	1.9	2.3	1
10-10a	-H	-CH=CHCO$_2$-		F.1	2.1	0.189	1.29	2
1-79658	-Me	-H		F.2	>20	>20	>20	2
1-174121	-CH$_2$N(Me)-1-naphthyl-4-CONHCH(CO$_2$H)CH$_2$CH$_2$CO$_2$H	-H		F.2	0.00019	0.00228	0.0025	0
1-232965	-CH$_2$-S-Ph	-H		F.2	9.5	0.77	246	0

1-235776	-CH_2-NH-Ph	-H		F.2	>1.9	2.7	>1.9	0
1-233912	-CH_2-O-Ph	-H		F.2	136	2.7	13	2
1-235777	-CH_2NH-1-naphthyl	-H		F.2	0.13	0.076	1.26	0
6-40	-$CH_2C_6H_4$-4-n-But	-R_1		F.2	3.9	14	44	0
6-41	-$CH_2C_6H_4$-3-CF_3	-R_1		F.2	26	39	64	1
6-42	-$CH_2C_6H_4$-4-Cl	-R_1		F.2	5.9	2.5	8.3	0
6-43	-$CH_2C_6H_4$-3,4-Cl	-R_1		F.2	4.4	>28	18	1
6-44	-CH_2CH_2Ph	-R_1		F.2	4.9	4.8	4.5	1
6-45	-$CH_2CH_2CH_2$Ph	-R_1		F.2	1.1	1	2.7	1
6-46	-$CH_2C_6H_{11}$	-R_1		F.2	2.6	0.46	27	2
6-47	-CH_2-2-naphthyl	-R_1		F.2	4	>31	>31	1
6-48	1,2-naphthyl	R_1		F.2	18.4	4.9	1.3	2
11- GR92754	-i-Butyl	-R_1		F.2	0.082	0.028	0.32	1
11-AH10639	-i-Pr	-R_1		F.2	0.2	0.033	1.1	0
11-AH2503	-Et	-R_1		F.2	0.62	1	2.2	1
11-AH2501	-n-Pentyl	-R_1		F.2	1.5	0.17	0.61	2
11-AH2504	-C_6H_4-2-OMe	-Et		F.2	2	0.018	0.41	2
11-AH2497	-$CH_2C_6H_{11}$	-R_1		F.2	>5	0.46	>5	0
11-AH2498	-n-Hexyl	-R_1		F.2	>5	46	>5	0
12-4b		-H		F.2	1.4	0.91	5.1	0
12-4d		-H		F.2	3.4	2.2	13	2
12-4g	-$CH_2N(Ph)_2$	-H		F.2	4.9	1.3	4	0
12-4c		-H		F.2	0.042	0.029	0.027	1
12-4e		-H		F.2	0.12	0.11	0.2	0
12-4f		-H		F.2	0.1	0.012	0.055	1
12-4a		-H		F.2	0.21	0.043	4.4	0
10-18d	-CH_2OH	-H		F.2	>51	>51	>51	1
1-122870	-4-CONHCH(CO_2H)CH_2CO_2H	-CH_2NH-	-Me	G.1	0.000 42	0.001 33	0.000 156	2
1- Trimetrexate	-3,4,5-OMe	-CH_2NH-	-Me	G.1	0.042	10	0.003	1
1-129516	-4-CONHCH($CH_2CO_2H)_2$	-CH_2NH-	-Me	G.1	0.000 22	0.000 294	0.000 266	0
1-132483	-3-CONHCH(CO_2H)CH_2CO_2H,-4- Cl	-CH_2NH-	-Me	G.1	0.005	0.010	0.015	1
1-184692	-4-CONHCH(CO_2H)CH_2CO_2H	-CH_2NH-	-Et	G.1	0.010	0.001	0.004	0
1-351521	-3,5-OMe	-CH_2-	-Me	G.1	0.031		0.002	2
13-8	-2,5-OMe	-CH_2NH-	-Cl	G.1	0.051	0.03	0.044	1
13-9	-3,4,5-OMe	-CH_2NH-	-Cl	G.1	0.033	0.005	0.006	0
13-10	-3,4,5-OMe	-CH_2N(Me)-	-Cl	G.1	0.012	0.006	0.012	1
13-11	-2,5-OMe	-$NHCH_2$-	-Cl	G.1	0.053	0.017	0.028	2
13-12	-2,5-OMe	-NH$(CH_2)_2$-	-Cl	G.1	12	0.99	2.1	1
13-13	-2,5-OMe	-NH(CH_2)3-	-Cl	G.1	26	1	1.9	1
13-14	-3,4,5-OMe	-$NHCH_2$-	-Cl	G.1	0.033	0.007	0.006	1

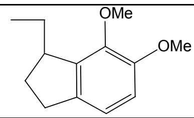
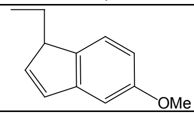
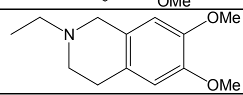
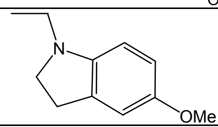
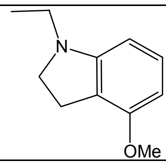
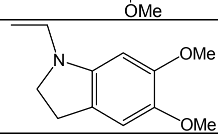
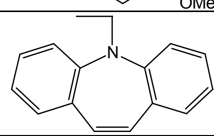
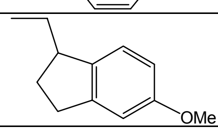
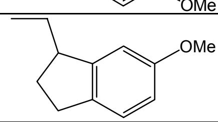
13-15	-3,4,5-OMe	-NH(CH_2)$_2$-	-Cl	G.1	43	2.6	5.4	0
13-16	-3,4,5-OMe	-NH(CH_2)3-	-Cl	G.1	95	14	19	2
13-17	-3,4,5-OMe	-N(Me)CH_2-	-Cl	G.1	0.17	0.016	0.038	0
14-28	-2,5-OMe	-CH_2NH-	-Cl	G.1	0.051	0.03	0.044	0
14-29	-3,4,5-OMe	-CH_2NH-	-Cl	G.1	0.033	0.005	0.006	0
14-30	-3,4,5-OMe	-CH_2N(Me)-	-Cl	G.1	0.012	0.006	0.012	0
9-16a	-2,3-(CH)$_4$-	-CH_2NH-	-H	G.1	0.21	0.027	0.055	1
9-16b	-4-Cl	-CH_2NH-	-H	G.1	0.6	0.075	0.073	0
9-16c	-2-Me,-4-Cl	-CH_2NH-	-H	G.1	0.33	0.033	0.023	2
15-3	-2,5-OMe	-NHCH_2-	-H	G.1	4.6	0.16	1.1	0
15-4	-3,5-OMe	-NHCH_2-	-H	G.1	2.2	0.12	0.84	0
15-5	-2,4-OMe	-NHCH_2-	-H	G.1	4.4	0.17	1.2	1
15-2	-H	-NHCH_2-	-H	G.1	8.7	0.32	0.66	0
15-6	-3,4,5-OMe	-NHCH_2-	-H	G.1	6.8	0.084	0.9	0
15-7	-2,3,4-OMe	-NHCH_2-	-H	G.1	4.9	0.19	1.3	0
15-8	-2,4,5-OMe	-NHCH_2-	-H	G.1	5.4	0.124	1.6	0
15-9	-2,4,6-OMe	-NHCH_2-	-H	G.1	116	0.95	22.7	0
15-10	-2,3-(CH)$_4$-	-NHCH_2-	-H	G.1	0.72	0.099	0.19	0
10-1a	-H	-CO_2CH_2-	-H	G.1	2.5	0.36	0.42	1
10-2a	-H	-CH_2CO_2-	-H	G.1	0.6	0.099	0.39	0
10-3a	-3,4,5-OMe	-CH_2CO_2-	-H	G.1	0.56	0.05	0.026	2
10-4a	-2,5-OMe	-CH_2CO_2-	-H	G.1	2.4	0.22	0.42	1
10-5a	-2,3,4-OMe	-CH_2CO_2-	-H	G.1	1.4	0.14	0.5	0
10-6a	-3,5-OMe	-CH_2CO_2-	-H	G.1	0.46	0.15	0.35	1
16-9a	-2-Br,-3,4,5-OMe	-CH_2NH-	-Cl	G.1	0.028	0.008	0.010	2
16-11	-2-Br,-3,4,5-OMe	-CH_2NH-	-H	G.1	0.55	0.039	0.33	0
15-11	-2,5-OMe	-N(Me)CH_2-	-H	G.1	0.087	0.03	0.026	0
15-12	-3,5-OMe	-N(Me)CH_2-	-H	G.1	0.024	0.009	0.008	2
15-13	-2,4-OMe	-N(Me)CH_2-	-H	G.1	0.1	0.039	0.043	0
15-14	-2,3,4-OMe	-N(Me)CH_2-	-H	G.1	0.052	0.017	0.019	1
15-15	-2,3-(CH)$_4$-	-N(Me)CH_2-	-H	G.1	0.017	0.021	0.017	0
10-7a	-3,4-OMe	-CH_2CO_2-	-H	G.1	0.51	0.066	0.11	0
10-8a	-4-CO_2H	-CH_2CO_2-	-H	G.1	1	1	1.1	0
10-11a	-H	-CH_2NH-	-H	G.1	0.98	0.1	0.44	0
10-12a	-H	-CH_2N(Me)-	-H	G.1	0.096	0.011	0.069	2
6-1	-Me	-H	-H	G.2	1.5	4.98	7.4	0
6-2	-CF_3	-H	-H	G.2	82		215	1
6-3	-F	-H	-H	G.2	0.41		5	1
6-4	-Cl	-H	-H	G.2	0.42		3.9	0
6-5	-Br	-H	-H	G.2	0.28		2.3	0
6-6	-I	-H	-H	G.2	0.52		2.4	0
6-7	-OMe	-H	-H	G.2	4.5	0.48	1.2	0
6-8	-OEt	-H	-H	G.2	4.7	8.7	11	1
6-9	-OPr	-H	-H	G.2	0.96	0.15	0.12	1
6-10	-OCH_2CF_3	-H	-H	G.2	13		0.14	1
6-11	-$OCH_2C_2F_5$	-H	-H	G.2	54		0.41	2
6-12	-$OCH_2C_3F_7$	-H	-H	G.2	71		6.1	0
6-13	-SMe	-H	-H	G.2	1.1		0.54	2
6-14	-H	-Me	-H	G.2	20		55	1
6-15	-H	-CF_3	-H	G.2	2.7	3.1	4.1	0
6-16	-H	-F	-H	G.2	119	104	452	1
6-17	-H	-Cl	-H	G.2	3.6	14	29	0
6-18	-H	-Br	-H	G.2	1.6		14	0
6-19	-H	-I	-H	G.2	1.6	1.6	5.9	0
6-20	-H	-OMe	-H	G.2	58	22	46	1
6-21	-H	-H	-Me	G.2	65	81	42	0
6-22	-H	-H	-CF_3	G.2	31		63	0
6-23	-H	-H	-F	G.2	41		189	0
6-24	-H	-H	-Cl	G.2	45		53	0
6-25	-H	-H	-Br	G.2	12		19	0
6-26	-Cl	-Cl	-H	G.2	9.7		3.5	2
6-27	-Cl	-Br	-H	G.2	0.14		0.8	0
6-28	-H	-Me	-Me	G.2	5.3	7.7	7.9	0

17-16	-C≡C-Ph	-H	-H	G.2	>30	>30	>30	1
17-17	-CH=CH-Ph	-H	-H	G.2	10	1.2	1.5	1
17-18	-CH_2CH_2-Ph	-H	-H	G.2	3.2	0.37	0.51	2
17-6	-$CH_2NHC_6H_3$-2,5-OMe	-H	-H	G.2	>5	>5	>5	2
17-8	-$CH_2N(Me)C_6H_3$-2,5-OMe	-H	-H	G.2	>40	>40	84	0
17-10	-$CH_2CH_2C_6H_3$-2,5-OMe	-H	-H	G.2	19	5.9	4.1	0
17-19	-$CH_2CH_2C_6H_3$-2,5-OMe	-H	-H	G.2	>18	>18	>18	2
17-7	-$CH_2NHC_6H_2$-3,4,5-OMe	-H	-H	G.2	>20	>20	>20	0
17-9	-$CH_2N(Me)C_6H_2$-3,4,5-OMe	-H	-H	G.2	>20	>20	53	1
17-11	-$CH_2CH_2C_6H_2$-3,4,5-OMe	-H	-H	G.2	14	0.93	3.7	2
18-5	-H		-H	G.2	13	5.8	1.5	1
18-6	-H		-H	G.2	0.51	0.13	0.019	2
10-17a	-H	-CO_2H	-H	G.2	>32	>32	>32	2
10-18a	-H	-CH_2OH	-H	G.2	51.7	44.9	40.3	1
19-1	-3,4,5-OMe	-CH_2NH-		H.1	4.6	0.054	0.29	2
19-2	-3,4,5-OMe	-$CH_2N(Me)$-		H.1	0.095	0.007	0.038	0
19-3	-3,4,5-OMe	-$CH_2N(CH_2C{\equiv}CH)$-		H.1	0.119	0.012	0.074	1
19-4	-2,5-OMe	-CH_2NH-		H.1	1.67	0.181	0.56	0
19-5	-2,5-OMe	-$CH_2N(Me)$-		H.1	0.3	0.015	0.26	2
19-6	-2,5-OMe	-$CH_2N(Et)$-		H.1	0.114	0.017	0.071	0
19-7	-2,5-OEt	-CH_2NH-		H.1	1.57	0.14	1.47	0
19-8	-2,5-OEt	-$CH_2N(Me)$-		H.1	0.319	0.017	0.116	0
19-9	-3,4-Cl	-CH_2NH-		H.1	6.8	0.11	0.15	2
19-10	-3,4-Cl	-$CH_2N(Me)$-		H.1	0.246	0.021	0.034	0
19-11	-2,5-Cl	-CH_2NH-		H.1	0.41	0.097	0.24	0
19-12	-2,6-Cl	-CH_2NH-		H.1	0.502	0.010	0.109	1
19-13	-4-Cl	-CH_2NH-		H.1	0.94	0.078	0.128	0
19-14	-4-Cl	-$CH_2N(Me)$-		H.1	0.171	0.022	0.067	0
19-15	-3-Br	-CH_2NH-		H.1	0.33	0.03	0.227	0
19-16	-3,4,5-OMe	-CH_2NHCH_2-		H.1	18.5		12.7	0
19-17	-3,4,5-OMe	-$CH_2N(Me)CH_2$-		H.1	3.1	0.33	1.63	2
19-18	-H			H.1	0.21	0.027	0.16	2
19-19	-2,3-$(CH)_4$-	-CH_2NH-		H.1	0.517	0.036	0.139	0
19-20	-2,3-$(CH)_4$-	-$CH_2N(Me)$-		H.1	0.1	0.023	0.047	2
20-5a	-H	-CH_2-		H.1	0.29	0.032	0.18	1
20-5b	-2-Me	-CH_2-		H.1	0.25	0.023	0.11	2
20-5c	-3-Me	-CH_2-		H.1	0.34	0.036	0.11	0
20-5d	-2-OMe	-CH_2-		H.1	0.45	0.014	0.12	1
20-5e	-3-OMe	-CH_2-		H.1	0.27	0.021	0.11	2
20-5f	-4-OMe	-CH_2-		H.1	0.44	0.05	0.077	0
20-5g	-3-CF_3	-CH_2-		H.1	0.4	0.14	0.19	0
20-5h	-3-OCF_3	-CH_2-		H.1	0.1	0.014	0.079	0
20-5i	-4-OCF_3	-CH_2-		H.1	0.58	0.073	0.17	1
20-5j	-2,5-OMe	-CH_2-		H.1	0.057	0.021	0.034	0
20-5k	-3,4-OMe	-CH_2-		H.1	0.1	0.023	0.063	1
20-5l	-3,4,5-OMe	-CH_2-		H.1	0.091	0.024	0.038	0
20-5m	-3,4-Cl	-CH_2-		H.1	15	2.6	7.3	1
20-9a	-Et			H.2	6.9	1.1	3	1
20-9b	-t-Bu			H.2	0.18	0.018	0.065	0
20-9c	-Ph			H.2	2.2	1.3	1.9	2
20-10	-CH_2-2-thienyl			H.2	0.21	0.062	0.08	0
20-11	-CH_2-3-thienyl			H.2	1.7	0.049	0.13	1
20-12	-CH_2-2-tetrahydrofuran			H.2	0.88	0.04	0.27	1
8-2a	-H	-CH_2S-		I.1	2	0.13	0.52	2

8-2b	-2,3-$(CH)_4$-	-CH_2S-		I.1	0.47	0.049	0.16	0
8-2c	-3,4-$(CH)_4$-	-CH_2S-		I.1	0.38	0.048	0.086	0
14-5	-2,5-OMe	-CH_2NH-		I.1	1.4	0.1	0.43	0
14-6	-3,4,5-OMe	-CH_2NH-		I.1	2.6	0.082	0.3	1
14-7	-3,4,5-OMe	-CH_2N(Me)-		I.1	0.13	0.005	0.026	0
14-8	-4-Cl	-CH_2N(Me)-		I.1	0.062	0.015	0.022	2
14-9	-3-Cl	-CH_2N(Me)-		I.1	2.1	0.02	0.067	0
14-10	-3,4-Cl	-CH_2N(Me)-		I.1	0.022	0.098	0.032	0
21-3	-H	-CH_2S-		I.1	2	0.13	0.52	0
21-4	-H	-CH_2NH-		I.1	1.7	0.085	0.26	0
21-5	-H	-CH_2N(Me)-		I.1	0.29	0.008	0.024	0
21-6	-2-OMe	-CH_2NH-		I.1	2.7	0.12	0.42	0
21-7	-2-OMe	-CH_2N(Me)-		I.1	0.51	0.026	0.12	2
21-8	-3-OMe	-CH_2NH-		I.1	1.7	0.1	0.2	2
21-9	-3-OMe	-CH_2N(Me)-		I.1	0.097	0.015	0.035	0
21-10	-4-OMe	-CH_2NH-		I.1	0.85	0.054	0.073	0
21-11	-4-OMe	-CH_2N(Me)-		I.1	0.25	0.016	0.018	1
21-12	-2-Cl	-CH_2NH-		I.1	0.53	0.11	0.14	0
21-13	-2-Cl	-CH_2N(Me)-		I.1	0.21	0.015	0.12	0
21-14	-3-Cl	-CH_2NH-		I.1	2	0.13	0.14	1
21-15	-2,4-OMe	-CH_2NH-		I.1	5.5	0.14	0.32	1
21-16	-2,4-OMe	-CH_2N(Me)-		I.1	0.16	0.014	0.016	0
21-17	-2,5-OMe	-CH_2NH-		I.1	4.4	0.12	0.28	0
21-18	-2,5-OMe	-CH_2N(Me)-		I.1	0.21	0.025	0.05	0
21-19	-3,4-OMe	-CH_2NH-		I.1	0.9	0.09	0.06	0
21-20	-3,4-OMe	-CH_2N(Me)-		I.1	0.091	0.010	0.003	0
21-21	-2,4-Cl	-CH_2NH-		I.1	0.73	0.05	0.088	1
21-22	-2,4-Cl	-CH_2N(Me)-		I.1	0.5	0.05	0.058	0
21-23	-2,5-Cl	-CH_2NH-		I.1	1.6	0.091	0.2	0
21-24	-2,5-Cl	-CH_2N(Me)-		I.1	0.15	0.025	0.047	0
21-25	-2,6-Cl	-CH_2NH-		I.1	1	0.028	0.082	0
21-26	-2,6-Cl	-CH_2N(Me)-		I.1	0.17	0.03	0.048	1
21-27	-3,4-Cl	-CH_2NH-		I.1	0.41	0.057	0.054	0
21-28	-3,4-Cl	-CH_2N(Me)-		I.1	0.038	0.027	0.017	0
21-29	-3,4,5-OMe	-CH_2NH-		I.1	2	0.04	0.2	0
21-30	-3,4,5-Cl	-CH_2NH-		I.1	0.66	0.087	0.044	0
21-31	-3,4,5-Cl	-CH_2N(Me)-		I.1	0.25	0.038	0.087	0
21-32	-2,4,6-Cl	-CH_2NH-		I.1	2	0.046	0.57	0
21-33	-2,4,6-Cl	-CH_2N(Me)-		I.1	0.12	0.044	0.052	0
21-34	-2,3-$(CH)_4$-	-CH_2S-		I.1	0.47	0.049	0.16	0
21-35	-3,4-$(CH)_4$-	-CH_2S-		I.1	0.38	0.048	0.086	0
21-36	-2,3-$(CH)_4$-	-CH_2NH-		I.1	0.23	0.026	0.04	1
21-37	-3,4-$(CH)_4$-	-CH_2NH-		I.1	1.6	0.16	0.21	0
21-38	-2,3-$(CH)_4$-	-CH_2N(Me)-		I.1	0.04	0.018	0.007	0
21-39	-3,4-$(CH)_4$-	-CH_2N(Me)-		I.1	0.052	0.016	0.007	0
21-40	-4-COMe	-CH_2NH-		I.1	0.41	0.027	0.003	0
21-41	-4-COMe	-CH_2N(Me)-		I.1	0.13	0.015	0.005	1
21-42	-4-COMe	-CH_2N(C≡CH)-		I.1	0.22	0.02	0.015	1
21-43	-4-$COCF_3$	-CH_2N(Me)-		I.1	0.25	0.046	0.032	0
21-44	-4-$COCF_3$	-CH_2N(C≡CH)-		I.1	0.12	0.054	0.008	0
22-4	-2-OMe	-S-		I.1	2.2	0.058	0.23	1
22-5	-4-OMe	-S-		I.1	0.7	0.045	0.075	1
22-6	-3,4-OMe	-S-		I.1	0.086	0.019	0.018	0
22-7	-2-OMe	-SO_2-		I.1	3.2	0.21	1.4	2
22-8	-4-OMe	-SO_2-		I.1	10.5	1	2	1
22-9	-3,4-OMe	-SO_2-		I.1	2.7	0.94	0.88	1
22-11	-2-OMe	-NH-		I.1	8.7	0.46	0.26	0
22-12	-4-OMe	-NH-		I.1	90.4	2.8	3.8	2
22-15	-3,4-OMe	-NH-		I.1	40.4	0.68	1.1	0
22-19	-4-OMe	-N(Me)-		I.1	0.22	0.009	0.007	0
22-21	-3,4-OMe	-N(Me)-		I.1	0.0023	0.00088	0.0004	2
22-14	-2,5-OMe	-NH-		I.1	16.1	0.73	3.6	2

22-20	-2,5-OMe	-N(Me)-		I.1	0.034	0.041	0.004	0
22-16	-3,4,5-OMe	-NH-		I.1	25.9	2.4	3.2	1
22-22	-3,4,5-OMe	-N(Me)-		I.1	0.021	0.007	0.004	0
22-23	-2,3-$(CH)_4$-	-N(Me)-		I.1	5.1	2.1	3.3	0
22-17	-3,4-$(CH)_4$-	-NH-		I.1	15	1.1	2	1
22-24	-3,4-$(CH)_4$-	-N(Me)-		I.1	0.032	0.015	0.010	0
22-10	-H	-NH-		I.1	8.3	0.3	0.43	0
22-18	-H	-N(Me)-		I.1	0.010	0.002	0.001	2
22-13	-4-Cl	-NH-		I.1	14.6	0.83	0.82	2
22-25	-3,4,5-OMe	-$NHCH_2$-		I.1	29.4	0.49	1.4	2
18-4				I.2	0.037	0.034	0.053	2
8-3	-H	-CH_2S-	-H	J.1	1.3	0.47	1.9	1
8-4	-2,3-$(CH)_4$-	-CH_2S-	-H	J.1	0.95	0.39	3.4	0
8-5	-3,4-$(CH)_4$-	-CH_2S-	-H	J.1	0.17	0.09	0.22	0
8-6	-2,5-Cl	-CH_2S-	-H	J.1	5.9	2	2.5	2
8-7	-3,5-Cl	-CH_2S-	-H	J.1	11	6.2	38	1
8-8	-3,4-OMe	-CH_2S-	-H	J.1	2.2	1.1	4	1
23-4a	-3,4,5-OMe	-CH_2NH-	-Me	J.1	0.086	0.007	0.002	1
23-5a	-3,4,5-OMe	-CH_2N(Me)-	-Me	J.1	0.013	0.001	0.007	0
23-6a	-3,4,5-OMe	-CH_2N(CHO)-	-Me	J.1	0.55	0.013	0.11	2
23-4b	-3,4-Cl	-CH_2NH-	-Me	J.1	0.32	0.028	0.053	0
23-5b	-3,4-Cl	-CH_2N(Me)-	-Me	J.1	0.1	0.027	0.042	1
23-6b	-3,4-Cl	-CH_2N(CHO)-	-Me	J.1	0.51	0.083	0.14	2
23-4c	-3,4,5-Cl	-CH_2NH-	-Me	J.1	0.063	0.012	0.033	0
23-5c	-3,4,5-Cl	-CH_2(Me)-	-Me	J.1	0.105	0.038	0.036	0
23-6c	-3,4,5-Cl	-CH_2CHO-	-Me	J.1	0.52	0.094	0.25	0
24-2a	-2,5-OMe	-CH_2NH-	-Me	J.1	0.046	0.016	0.128	0
24-2b	-3,5-OMe	-CH_2NH-	-Me	J.1	0.023	0.005	0.043	0
24-2c	-2,4-OMe	-CH_2NH-	-Me	J.1	0.316	0.057	0.214	1
24-2d	-3,4-OMe	-CH_2NH-	-Me	J.1	0.044	0.009	0.008	0
24-2e	-2,5-OEt	-CH_2NH-	-Me	J.1	0.077	0.017	0.017	1
24-3a	-2,5-OMe	-CH_2N(Me)-	-Me	J.1	0.216	0.030	0.407	2
24-3b	-3,5-OMe	-CH_2N(Me)-	-Me	J.1	0.13	0.058	0.17	1
24-3c	-2,4-OMe	-CH_2N(Me)-	-Me	J.1	0.32	0.029	0.044	0
24-3e	-2,5-OEt	-CH_2N(Me)-	-Me	J.1	3.1	0.1	3000	0
24-4a	-3,4,5-OMe	-$CH_2N(CH_2C{\equiv}CH)$-	-Me	J.1	0.054	0.008	0.012	0
24-4b	-3,4,5-OMe	-CH_2N(i-Pr)-	-Me	J.1	0.013	0.007	0.018	1
24-4c	-3,4,5-OMe	-CH_2N(Et)-	-Me	J.1	0.050	0.003	0.011	0
24-5a	-2,3-$(CH)_4$-	-CH_2NH-	-Me	J.1	0.573	0.015	0.030	0
24-5b	-2,3-$(CH)_4$-,-4-OMe	-CH_2NH-	-Me	J.1	0.041	0.023	0.054	0
14-20	-3,4,5-OMe	-CH_2NH-	-Me	J.1	0.086	0.007	0.002	0
14-21	-3,4,5-OMe	-CH_2N(Me)-	-Me	J.1	0.013	0.001	0.008	0
14-22	-3,4-Cl	-CH_2N(Me)-	-Me	J.1	0.1	0.027	0.042	0
9-13a	-2,3-$(CH)_4$-	-CH_2NH-	-Me	J.1	0.22	0.24	0.11	0
9-13b	-H	-CH_2S-	-Me	J.1	0.44	0.034	0.43	1
9-13c	-2,3-$(CH)_4$-	-CH_2S-	-Me	J.1	0.29	0.03	0.55	0
9-13d	-3-Me	-CH_2S-	-Me	J.1	0.17	0.065	0.33	0
9-13e	-4-Me	-CH_2S-	-Me	J.1	0.53	0.057	0.5	0
9-13f	-3-OMe	-CH_2S-	-Me	J.1	0.34	0.11	0.6	0
9-13g	-4-OMe	-CH_2S-	-Me	J.1	0.56	0.063	0.52	0
9-13h	-3,4-OMe	-CH_2S-	-Me	J.1	0.15	0.03	0.18	0
9-13i	-3-Cl	-CH_2S-	-Me	J.1	0.068	0.09	0.19	0
9-13j	-4-Cl	-CH_2S-	-Me	J.1	0.36	0.09	0.37	0
9-13k	-4-Cl	-CH_2NH-	-Me	J.1	0.11	0.032	0.072	1
9-13l	-2-Me,5-OMe	-CH_2NH-	-Me	J.1	0.038	0.023	0.15	1
9-13m	-2-Me,-6-OMe	-CH_2NH-	-Me	J.1	1	0.1	0.32	0
9-13n	-2,5-OMe	-CH_2NH-	-Me	J.1	0.011	0.014	0.01	0
9-13o	-2,5-OMe	-CH_2N(Me)-	-Me	J.1	0.7	0.028	0.46	0
9-13p	-3,5-OMe	-CH_2NH-	-Me	J.1	0.049	0.004	0.031	0
9-13q	-3,4,5-OMe	-CH_2NH-	-Me	J.1	0.013	0.003	0.005	0

9-13r	-2-OMe,-5-CF_3	-CH_2NH-	-Me	J.1	0.044	0.022	0.02	1
9-13s	-3-OMe,-5-CF_3	-CH_2NH-	-Me	J.1	0.02	0.018	0.017	1
9-13t	-2,3-$(CH)_4$-	-CH_2CH_2-	-Me	J.1	0.064	0.026	0.135	2
9-13u	-2,5-OMe	-CH_2CH_2-	-Me	J.1	0.34	0.008	0.77	1
9-14a	-2,5-Me	-CH_2NH-	-Me	J.1	0.03	0.016	0.12	1
9-14b	-2-Me,-4-OMe	-CH_2NH-	-Me	J.1	0.17	0.007	0.029	1
9-14c	-2-OMe,-5-Me	-CH_2NH-	-Me	J.1	0.068	0.015	0.16	0
9-14d	-2,4-OMe	-CH_2NH-	-Me	J.1	0.091	0.01	0.061	0
9-14e	-2,4-OMe	-CH_2N(Me)-	-Me	J.1	0.35	0.053	0.23	0
9-14f	-3,5-OMe	-CH_2N(Me)-	-Me	J.1	0.05	0.004	0.013	0
9-14g	-3,4-OMe	-CH_2NH-	-Me	J.1	0.051	0.003	0.007	0
9-14h	-3,4-OMe	-CH_2N(Me)-	-Me	J.1	0.32	0.003	0.004	0
9-14i	-2,5-OEt	-CH_2NH-	-Me	J.1	0.125	0.022	0.37	0
9-14j	-2,5-OEt	-CH_2N(Me)-	-Me	J.1	0.167	0.029	0.53	0
9-14k	-2-OMe,-5-CF_3	-CH_2N(Me)-	-Me	J.1	0.093	0.038	0.23	0
9-14l	-2,3-$(CH)_4$-	-CH_2NH-	-Me	J.1	0.1	0.026	0.13	0
9-14m	-2,3-$(CH)_4$-	-CH_2N(Me)-	-Me	J.1	0.15	0.016	0.14	1
9-15a	-2,3-$(CH)_4$-	-CH_2NH-	-H	J.1	0.26	0.15	0.23	1
9-15b	-4-Cl	-CH_2NH-	-H	J.1	0.97	0.3	0.72	0
9-15c	-3,4,5-OMe	-CH_2NH-	-H	J.1	2	0.13	0.81	0
9-16d	-2,5-OMe	-CH_2NH-	-H	J.1	0.75	0.14	0.46	0
25-9	-H	-NHCH_2-	-H	J.1	2.7	0.52	2.1	1
25-10	-H	-N(Me)CH_2-	-H	J.1	0.068	0.032	0.14	0
25-11	-3,4,5-OMe	-NHCH_2-	-H	J.1	14.1	0.35	3.3	0
25-12	-3,4,5-OMe	-N(Me)CH_2-	-H	J.1	0.061	0.014	0.033	2
25-13	-3,4,5-OMe	-NHCH(Me)-	-H	J.1	9.2	0.194	1.27	1
25-14	-2,3,4-OMe	-NHCH_2-	-H	J.1	15.3	0.67	3.24	0
25-15	-2,3,4-OMe	-N(Me)CH_2-	-H	J.1	0.079	0.026	0.03	0
25-16	-2,4,6-OMe	-NHCH_2-	-H	J.1	20.7	0.23	1.2	1
25-17	-2,4,5-OMe	-NHCH_2-	-H	J.1	5.5	0.48	1.1	0
25-18	-3,4-OMe	-NHCH_2-	-H	J.1	4.8	0.73	1.5	0
25-19	-3,5-OMe	-NHCH_2-	-H	J.1	5.7	1.2	3.4	1
25-20	-3,5-OMe	-N(Me)CH_2-	-H	J.1	0.076	0.031	0.072	0
25-21	-2,5-OMe	-NHCH_2-	-H	J.1	3.8	0.31	0.35	1
25-22	-2,5-OMe	-N(Me)CH_2-	-H	J.1	0.084	0.006	0.057	0
25-23	-2,3-$(CH)_4$-	-NHCH_2-	-H	J.1	3.9	0.98	0.24	0
25-24	-2,3-$(CH)_4$-,-4-OMe	-NHCH_2-	-H	J.1	8.2	0.38	0.43	2
25-25	-2,3-$(CH)_4$-,-6-OMe	-NHCH_2-	-H	J.1	15.4	0.71	0.37	0
25-26	-4-O-Ph	-NHCH_2-	-H	J.1	24.3	3.7	2.9	0
25-27	-H	-NH$(CH_2)_2$-	-H	J.1	1.94	4.45	0.28	0
25-28	-H	-N(Me)CH_2CH_2-	-H	J.1	1.8	0.92	1.4	2
26-7	-2,5-OMe,-4-pyrrolo	-CH_2NH-	-Me	J.1	0.35	0.033	0.23	1
26-8	-2-pyrrolo,-4,5-OMe	-CH_2NH-	-Me	J.1	1.8	0.6	3.5	1
26-9	-2,3,5,6-OMe,-4-pyrrolo	-CH_2NH-	-Me	J.1	0.62	0.075	0.17	2
26-10	-2-OMe,-5-Ph	-CH_2NH-	-Me	J.1	0.64	0.068	0.44	0
27-5	-H	-CH_2NH-	-Me	J.1	0.08	0.017	0.17	0
27-6	-2-OMe	-CH_2NH-	-Me	J.1	0.117	0.023	0.169	0
27-7	-3-OMe	-CH_2NH-	-Me	J.1	0.069	0.007	0.080	0
27-8	-4-OMe	-CH_2NH-	-Me	J.1	0.095	0.012	0.056	0
27-9	-2-Cl	-CH_2NH-	-Me	J.1	0.047	0.007	0.088	0
27-10	-3-Cl	-CH_2NH-	-Me	J.1	0.023	0.011	0.037	0
27-11	-4-Cl	-CH_2NH-	-Me	J.1	0.055	0.019	0.051	0
27-12	-4-Br	-CH_2NH-	-Me	J.1	0.081	0.010	0.035	0
27-13	-3-OMe	-CH_2(Me)-	-Me	J.1	0.03	0.006	0.018	0
27-14	-4-OMe	-CH_2(Me)-	-Me	J.1	0.035	0.007	0.013	0
27-15	-2-Cl	-CH_2(Me)-	-Me	J.1	0.084	0.018	0.1	0
27-16	-4-Cl	-CH_2(Me)-	-Me	J.1	0.029	0.005	0.026	0
27-17	-4-Br	-CH_2(Me)-	-Me	J.1	0.037	0.03	0.036	0
10-2b	-H	-CH_2CO_2-	-H	J.1	5.45	1.9	3.9	2
10-2c	-H	-CH_2CO_2-	-Me	J.1	0.25	0.056	0.23	1
28-3	-3,4,5-OMe	-CH_2N(Me)-	-H	J.1	0.24	0.009	0.28	1
28-4	-3,4,5-OMe	-CH_2N(Et)-	-H	J.1	0.19	0.049	0.12	0
28-5	-3,4,5-OMe	-CH_2N(CHO)-	-H	J.1	18.5	1.1	7.4	0

28-6	-3,4,5-OMe	-CH=CH-	-H	J.1	>5.0	1.4	12.9	0
28-7	-3,4-OMe	-CH=CH-	-H	J.1	2.6	1.4	2.1	1
28-8	-4-OMe	-CH=CH-	-H	J.1	5.3	1.5	11.8	2
28-9	-3,4,5-OMe	$-CH_2CH_2-$	-H	J.1	5	0.2	1.14	2
28-10	-3,4-OMe	$-CH_2CH_2-$	-H	J.1	1.4	0.2	0.61	0
28-11	-4-OMe	$-CH_2CH_2-$	-H	J.1	0.29	0.25	0.26	1
29-2-piritrexim	-2,5-OMe	$-CH_2-$	-Me	J.1	0.031	0.017	0.002	2
26-4		-Me		J.2	0.41	0.049	0.23	1
26-5		-Me		J.2	0.57	0.077	0.47	2
26-6		-Me		J.2	0.85	0.11	0.13	2
30-3		-Me		J.2	0.25	0.057	0.17	1
30-2		-Me		J.2	0.29	0.048	0.15	0
30-4		-Me		J.2	0.41	0.049	0.23	0
18-3		-H		J.2	0.043	0.04	0.19	2
26-2		-Me		J.2	0.29	0.048	0.15	1
26-3		-Me		J.2	0.25	0.057	0.17	0
10-18b	$-CH_2OH$	-H		J.2	>104	>104	>104	1
10-18c	$-CH_2OH$	-Me		J.2	10.5	3.2	8.5	2
31-4	$-CH_2Ph$			J.3	16	1.4		1
31-3	$-CH_2C_6H_4-CONHCH(CO_2H)CH_2CH_2CO_2H$			J.3	1.8	0.9		1
31-5	$-CH_2C_6H_4$-4-OMe			J.3	>15	2.9		2
31-6	$-CH_2C_6H_3$-3,5-OMe			J.3	14	2.7		1
31-7	$-CH_2C_6H_3$-2,4-Cl			J.3	4	2.1		2
31-8	$-CH_2C_6H_3$-3,4-Cl			J.3	5	5.1		1
31-9	$-CO_2CH_2$-Ph			J.3	>15	22		2
23-7	-3,4,5-OMe	$-CH_2CHO-$	-Me	K	>2.6		>2.6	1
23-8b	-3,4,5-OMe	$-CH_2CHO-$	-Me	L	>1.2		>1.2	2
23-8	-3,4,5-OMe	$-CH_2NH-$	-Me	L	>3700		3700	1
28-12	-3,4,5-OMe	$-CH_2CH_2-$	-H	L	61.7	0.47	6.1	2
28-13	-3,4-OMe	$-CH_2CH_2-$	-H	L	7.7	1.1	2.1	0
32-8	-2,5-OMe	$-CH_2-$		M	3.3	0.3	1.4	2
32-9	-3,4,5-OMe	$-CH_2-$		M	6.9	0.2	2.2	1
32-10	-2-Br,-3,4,5-OMe	$-CH_2-$		M	0.51	0.09	0.35	1
32-11	-3,4,5-OMe	$-CH_2CH_2-$		M	30	3.2	9.5	1

32-12	-3,4,5-OMe	-$(CH_2)_3$-		M	9.3	1.7	0.94	1
33-1	-4-$CONHCH(CO_2H)CH_2CH_2CO_2H$	-CH_2NH-	O	N.1	0.9	0.7	1.3	2
33-2	-4-$CONHCH(CO_2H)CH_2CH_2CO_2H$	-CH_2N(Me)-	O	N.1	0.035	19.8	0.43	0
33-3	-3,4,5-OMe	-CH_2NH-	O	N.1	>4	>4	>37.0	1
33-4	-3,4,5-Cl	-CH_2NH-	O	N.1	8.3	>3.9	25.6	0
33-5	-3,4-Cl	-CH_2NH-	O	N.1	>35	89.3	35.2	0
33-6	-2,5-OMe	-CH_2NH-	O	N.1	>21	>21	>21	1
34-1	-3,4,5-OMe	-CH_2NH-	NH	N.1	>23	8.1	56.3	2
34-2	-3,4-OMe	-CH_2NH-	NH	N.1	119	4.3	116	0
34-3	-4-OMe	-CH_2NH-	NH	N.1	279	6	63	1
34-4	-2,5-OMe	-CH_2NH-	NH	N.1	45.7	1.7	156	1
34-5	-2,5-OEt	-CH_2NH-	NH	N.1	>21	5.3	70	0
34-6	-3,4-Cl	-CH_2NH-	NH	N.1	35.3	1.4	14.4	0
34-7	-2,3-$(CH)_4$-	-CH_2NH-	NH	N.1	307	1.1	59.3	2
34-8	-H	-CH_2NH-	NH	N.1	252	3.9	>252	0
34-9	-4-$CONHCH(CO_2H)CH_2CH_2CO_2H$	-CH_2NH-	NH	N.1	0.038	0.21	0.044	0
35-2	-2,5-OMe	-CH_2N(Me)-	NH	N.1	>12	3.4	>12	0
35-3	-3,4-Cl	-CH_2N(Me)-	NH	N.1	28.3	1	3	2
35-4	-2,3-$(CH)_4$-	-CH_2N(Me)-	NH	N.1	209	0.87	8.2	0
35-5	-3,4-OMe	-CH_2S-	NH	N.1	11.1	2.6	16.7	2
35-6	-3,4-Cl	-CH_2S-	NH	N.1	58.5	11.6	5.3	1
35-7	-2,3-$(CH)_4$-	-CH_2S-	NH	N.1	10.6	0.81	3	2
35-8	-3,4-$(CH)_4$-	-CH_2S-	NH	N.1	929	9.2	82.9	0
35-10	-4-$CONHCH(CO_2H)CH_2CH_2CO_2H$	-CH_2N(Me)-	NH	N.1	0.044	0.15	0.06	1
36-7a	-3,4,5-OMe	-CH_2NH-	S	N.1	>10	>10	>10	0
36-7b	-2,5-OMe	-CH_2NH-	S	N.1	>10	>10	>10	0
36-7c	-3,4,5-OMe	-CH_2N(Me)-	S	N.1	>10	>10	>10	2
36-7d	-2,5-OMe	-CH_2N(Me)-	S	N.1	>10	>10	>10	1
36-7e	-3,5-Cl,-4-(1-pyrrolo)	-CH_2NH-	S	N.1	>10	>10	>10	2
37-1e	-H	-CH_2-	NH	N.1	>189	>130	270	2
38-3	-H	-CH_2S-	O	N.1	>26	>26	252	0
38-4	-2,3-$(CH)_4$-	-CH_2S-	O	N.1	19	19	23	0
38-5	-3,4-$(CH)_4$-	-CH_2S-	O	N.1	0.65	11.6	12.3	0
38-6	-2,3-$(CH)_4$-	-CH_2NH-	O	N.1	13.5	37	12	0
38-7	-3,4-$(CH)_4$-	-CH_2NH-	O	N.1	41	38	36.5	0
38-8	-3,4-$(CH)_4$-	-CH2O-	O	N.1	14	>42	60.3	2
38-9	-2-O-Ph	-CH_2NH-	O	N.1	>12	>12	>12	2
38-10	-4-O-Ph	-CH_2NH-	O	N.1	8.1	32.4	16.2	1
38-11	-2-Ph	-CH_2NH-	O	N.1	7.7	45.4	137	1
38-12	-3,4-$(CH)_4$-	-CH_2N(Me)-	O	N.1	14.8	23.6	14.6	2
38-13	-2,5-Cl	-CH_2NH-	O	N.1	50.9	>47	71.9	1
38-14	-3,4-Cl	-CH_2(Me)-	O	N.1	44.8	>27	>27	1
38-15	-3,4,5-Cl	-CH_2(Me)-	O	N.1	284	21.5	34.3	0
38-16	-3-OMe	-CH_2NH-	O	N.1	>31.3	>31.3	>31.3	0
38-17	-2,5-OMe	-CH_2(Me)-	O	N.1	>27	>27	>27	0
39-3a	-4-$CONHCH(CO_2H)CH_2CH_2CO_2H$	-CH_2CH_2-	NH	N.1	1.2	1.2		1
39-3b	-4-$CONHCH(CO_2H)CH_2CH_2CO_2H$	-CH_2CH_2-	NCH_2 Ph	N.1	>19	>19		2
40-3	-Me	-C_6H_3-2,5-OMe	S	N.2	4.8	0.13	0.37	1
40-4	-Me	-$CH_2C_6H_3$-2,5-OMe	S	N.2	14	0.07	0.4	1
40-5	-Me	-$(CH_2)_2$C6H3-2,5-OMe	S	N.2	1.2	3.3	5.9	2
40-6	-Me	-C_6H_2-3,4,5-OMe	S	N.2	>8	0.32	1.8	2
40-7	-Me	-$CH_2C_6H_2$-3,4,5-OMe	S	N.2	>8	0.63	51	1
40-8	-Me	-$(CH_2)_2C_6H_2$-3,4,5-OMe	S	N.2	>8	18	25	1
40-9	-$CH_2CH_2C_6H_3$-2,5-OMe	-H	S	N.2	28	5.8	3.1	1
40-10	-$CH_2CH_2CH_2C_6H_3$-2,5-OMe	-H	S	N.2	55	24	26	0
40-table2-1	-Me	-Ph	S	N.2	26	3.9	57	1
40-table2-2	-Me	-C_6H_3-3,4-Cl	S	N.2	16	>28	4.6	2
40-table2-3	-Me	-CH_2Ph	S	N.2	35	6.2	14	2
40-table-2-4	-C_6H_4-4-Cl	-Me	S	N.2	>100	>70	>100	0

40-table-2-5	-CH_2Ph	-Me	S	N.2	>100	>100	>37	0
40-table-2-6	-$(CH_2)_4$-	R1	S	N.2	2.1	2.8	3.9	2
40-table-2-7	-$(CH_2)_5$-	R1	S	N.2	1.4	0.86	1.3	0
40-table-2-8	-$(CH_2)_6$-	R1	S	N.2	6.5		1.9	0
40-table-2-9	-$(CH_2)_{10}$-	R1	S	N.2	>50		9.3	0
40-table-2-10	-$CH_2CH_2N(CH_2$-Ph)CH_2-	R1	S	N.2	5.2	0.85	0.57	2
16-8a	-Me	-$CH_2C_6H_1$-2-Br,-3,4,5-OMe	S	N.2	>12	0.21	0.93	2
16-12	-$CH_2CH_2C_6H_1$-2-Br-3,4,5-OMe	-H	S	N.2	49	2.5	2.8	2
16-10	-$CH_2NHC_6H_1$-2-Br-3,4,5-OMe	-H	S	N.2	200	25	43	1
36-6a	-$CH_2NHC_6H_2$-3,4,5-OMe	-Br	S	N.2	13	34	17	1
36-6b	-$CH_2NHC_6H_3$-2,5-OMe	-Br	S	N.2	>100	>100	33	1
36-6c	-$CH_2N(Me)C_6H_2$-3,4,5-OMe	-Br	S	N.2	31	127	28	0
36-6d	-$CH_2N(Me)C_6H_3$-2,5-OMe	-Br	S	N.2	>10	>10	>10	0
36-6e	-$CH_2NC_6H_3$-3,5-Cl,-4-(1-pyrrolo)	-Br	S	N.2	7.5	26	10	0
37-1a	-Ph	-H	NH	N.2	>186	12	9.1	1
37-1b	-C_6H_4-4-Cl	-H	NH	N.2	>161	113	62	0
37-1c	-C_6H_3-3,4-Cl	-H	NH	N.2	33	16	23	2
37-1d	-C_6H_2-3,4,5-OMe	-H	NH	N.2	8.3	14	27	2
41-4a	-CH_2NH-1-anthracene	-H	O	N.2	51	15.4	>51	0
41-4b	-CH_2NH-2-anthracene	-H	O	N.2	>14	>14	>14	0
41-4c	-CH_2NH-1-fluorene	-H	O	N.2	>29	>29	>29	1
41-4d	-CH_2NH-2-fluorene	-H	O	N.2	36.2	27.7	>10	0
41-4e	-CH_2NH-3-(2-methoxy-dibenzofuran)	-H	O	N.2	10.3	>32	>32	0
41-4f	-CH_2NH-3-(N-ethyl-carbazole)	-H	O	N.2	16.2	4.5	12.6	1
41-4g	-CH_2NH-2-(9-hydroxy-fluorene)	-H	O	N.2	>13	63	>13	1
41-4h	-CH_2NH-2-(9-oxofluorene)	-H	O	N.2	>63	>63	>63	2
41-4i	-CH_2NH-4-(9-oxofluorene)	-H	O	N.2	19.3	>38	>38	0
41-5a	-CH_2N(Me)-3-(2-methoxy-dibenzofuran)	-H	O	N.2	>54	392	241	1
41-5b	-CH_2N(Me)-3-(N-ethyl-carbazole)	-H	O	N.2	18.7	22.6	24.7	0
41-7a	-CH_2S-2-biphenyl	-H	O	N.2	46	105	49	1
41-7b	-CH_2S-3-biphenyl	-H	O	N.2	22	23	31	1
41-7c	-CH_2S-4-biphenyl	-H	O	N.2	41	79	351	1
41-7e	-CH_2S-3-phenoxyPh	-H	O	N.2	33	19	39	1
41-7d	-CH_2S-2-phenoxyPh	-H	O	N.2	14	38	18	2
41-7f	-CH_2S-4-phenoxyPh	-H	O	N.2	25	259	65	0
42-8	-4-Cl	-S-		N.3	40	3.1	24.6	1
42-9	-3,4-Cl	-S-		N.3	>20	11.7	>20	0
42-10	-4-NO_2	-S-		N.3	>15	244	7470	2
42-11	-3,4-OMe	-S-		N.3	>21	20	>21	1
43-7	-2,5-OMe	-CH_2NH-		N.3	47	2.2	47	1
43-8	-3,5-OMe	-CH_2NH-		N.3	20	2.6	20	0
43-9	-2,4-OMe	-CH_2NH-		N.3	16	7.1	16	0
43-10	-3,4,5-OMe	-CH_2NH-		N.3	22	9.3	22	0
43-11	-2,5-Cl	-CH_2NH-		N.3	25	0.66	25	0
43-12	-3,5-Cl	-CH_2NH-		N.3	66.2	4.1	63	0
43-13	-2,4-Cl	-CH_2NH-		N.3	15	2.2	15	0
43-14	-3-Cl	-CH_2NH-		N.3	26	3.5	26	2
43-15	-2,5-OMe	-CH_2N(Me)-		N.3	56	10.3	56	0
43-16	-3,5-OMe	-CH_2N(Me)-		N.3	87	4.2	87	0
43-17	-3,4,5-OMe	-CH_2N(Me)-		N.3	45	37	45	2
44-2	-Ph			O	153	28	59.9	1
44-3	-C_6H_2-3,-4,5-OMe			O	59.4	2.2	13	1
44-4	-C_6H_2-2,-3,4-OMe			O	17	1.8	3.7	0
44-5	-C_6H_2-2,-4,6-OMe			O	31.5	0.84	1.88	1
44-6	-C_6H_2-2,-4,5-OMe			O	22	1.5	7	0
44-7	-C_6H_3-2,-5-OMe			O	37.4	1.7	3.5	1
44-8	-C_6H_3-3,-5-OMe			O	9	3	32.4	1
44-9	-C_6H_3-3,-4-OMe			O	12	14.2	52.3	0
44-10	-C_6H_3-2,-4-OMe			O	14	2.4	0.9	0

44-11	-C_6H_3-3,-4-Cl			O	19.5	6.7	252	2
44-12	-C_6H_3-2,-6-Cl			O	11	11	11	1
44-13	-C_6H_2-3,-5-OMe-4-OH			O	68	1	15.3	0
44-14	-C_6H_2-2-NO_2-4,5-OMe			O	25	16.8	25	2
44-15	-C_6H_4-4-Ph			O	605	32.9	30.1	0
44-16	-4-pyridine			O	18	20	18	1
44-17	-2-naphthyl			O	113	27	280	0
44-18	-1-naphthyl-4-OMe			O	13	13	13	2
44-19	-9-fluorenyl			O	35	30.5	29.8	2
44-20	-CH_2-Ph			O	9.8	0.5	1.6	2
44-21	-CH_2NH-C_6H_2-3,4,5-OMe			O	34	34	34	1
44-22	-CH_2NH-C_6H_3-2,5-OMe			O	29	49	29	2
44-23	-CH_2S-2-naphthyl			O	105	1.02	107	2
45-15a	-3,4,5-OMe	-CH_2-		P	>32	21	69	2
45-15b	-3,4,5-OMe	-CH_2CH_2-		P	1.8	0.14	0.51	1
45-15c	-3,4,5-OMe	-$(CH_2)_3$-		P	1.3	0.14	0.22	0
46-117496	-3,5-OMe,-4-pyrrolo			Q	12.8			0
46-125279	-3,5-OMe,-4-pyrrolidine			Q	50.2			2
46-134670	-3,5-Cl,-4-pyrrolo			Q	6.4			2
46-135705	-3,5-SMe,-4-pyrrolo			Q	5			1
46-118960	-3,4-OMe,-5-pyrrolo			Q	4.5			1
46-115736	-3,5-(2,5-dimethylpyrrolo),-4-Me			Q	2.8			1
46-119016	-3,5-pyrrolo,-4-OMe			Q	1.7			1
46-464561	-3-OMe,-4-pyrrolo,-5-OCH_2Ph			Q	1.3			0
46-118958	-3,5-OEt,-4-pyrrolo			Q	0.8			1
46-084484	-2-Br,-3,4,5-OMe			Q	0.95			2
46-182049	-2,3-(-CHCHCH_2O-),-4,5-OMe			Q	1.5			2
29-1-trimethoprim	-3,4,5-OMe			Q	12	2.7	130	1
29-3a	-2-OMe,-5-O-n-pentyl			Q	19	6.5	44	2
29-3b	-2-OMe,-5-O-n-hexyl			Q	14	9.2	29	0
29-3c	-2-OMe,-5-O-n-heptyl			Q	5.6	5	11	0
29-3d	-2-OMe,-5-O-n-octyl			Q	44	15	30	0
29-3e	-2-OMe,-5-O-n-nonyl			Q	51	32	86	0
29-3g	-2-OMe,-5-O$(CH_2)_4CO_2H$			Q	0.049	0.11	3.9	0
29-3i	-2-OMe,-5-O$(CH_2)_6CO_2H$			Q	2.6	0.18	12	0
29-3h	-2-OMe,-5-O$(CH_2)_5CO_2H$			Q	0.8	0.48	17	0
29-3j	-2-OMe,-5-O$(CH_2)_7CO_2H$			Q	7.1	0.36	12	0
29-3k	-2-OMe,-5-O$(CH_2)_8CO_2H$			Q	4.8	0.43	12	0
29-3f	-2-OMe,-5-O$(CH_2)_3CO_2H$			Q	0.25	0.18	2.6	1
46-168820	-3-OMe,-4-Br,-5-OCH_2CONH_2			Q	14			2
47-5				-	>5		>5	1
47-6				-	1.28	1.12	0.24	1
47-8				-	>1	2.3	0.5	2
47-9				-	>2.27		>2.7	1
47-10				-	4.6	0.76	0.47	2
47-11				-	2.2	0.29	0.16	0
14-11				-	>1.85	1.85	>18.5	0
48-3				-	94.5	9	42.9	1
48-4				-	3.1	0.62	20.3	2
48-6				-	12.6	5.3	11.7	2
49-4				-	94.5	9	42.9	0
49-5				-	3.1	0.62	20.3	0
49-7				-	12.6	5.3	11.7	0
50-13				-	6.1	5.1	59	1
50-14				-	117	5.6	43	0
50-15				-	77	39	112	1
50-16				-	>46	46	>46	2
50-17				-	260	34	237	1
50-18				-	176	30	101	2
50-19				-	>352	101	30	1
50-20				-	>205	20	29	1
51-1				-	0.65	0.16	0.032	1

51-2				-	0.22	0.091	0.023	2
51-3				-	7.7	4.7	1.6	0
51-4				-	0.17	0.3	0.055	1
51-5				-	0.7	0.34	0.092	2
51-6				-	15.8	0.18	0.64	0
51-7				-	38.4	0.57	3.2	0
51-9				-	8.6	0.15	0.6	0
51-8				-	6.7	0.11	0.24	0
51-10				-	>0.4	0.46	2.4	0
46-210057				-	>30			2

[a] names that begin with a number are formed by hyphenating the reference number and the label given to the compound in the reference.

[b] IC_{50} values in μM for *P. carinii* (PC), *T. gondii* (TG) and rat liver (RL). Only the PC activities are used in this work, as they are the most complete set.

[c] training set, test set and unused compounds are indicated as 1, 2, and 0, respectively.

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Table 4.1. ER ligands from NTP compilation (ER–tox set)

Name[a]	Substance	CASRN[b]	from NCI[c]	RBA[d]	set[e]
1	4,4'-(1,3-Adamantanediyl)diphenol		0	6.5	0
2	2-(1-Adamantyl)-4-methylphenol	41031-50-9	0	0	1
3	4-(1-Adamantyl)phenol	29799-07-3	1	1.3	1
4	Alachlor	15972-60-8	1	0	2
5	Aldosterone	52-39-1	1	0	2
6	Aldrin	309-00-2	1	0	2
7	δ-*trans* Allethrin	584-79-2	1	0	1
8	*p* -(7-Alloxyl)-11-ethyldibenzo-[*b,f*]thiepin-10-yl)phenol	85850-86-8	0	5.2	1
9	*p* -(3-(Alloxyl)-11-ethyl-6*H* - dibenzo[*b,f*]thiocin-12-yl)phenol hemihydrate	85850-88-0	0	9.2	2
10	*p* -(2-(Alloxyl)-6-ethyl-11,12-dihydrodibenzo[*a,e*]cyclooctene-5- yl)phenol	85850-87-9	0	15	1
11	3-(Alloxyl)-10-ethyl-11-(4- hydroxyphenyl)dibenz[*b,f*]oxepin	85850-85-7	0	0.2	0
12	3-(Alloxyl)-10-ethyl-11- phenyldibenzo[*b,f*]thiepin	85850-82-4	1	0.5	0
13	3-(Alloxyl)-11-ethyl-12-phenyl 6*H* - dibenzo[*b,f*]thiocin	85850-84-6	1	0.1	0
14	3-(Alloxyl)-10-ethyl-11- phenyldibenz[*b,f*]oxepin	83807-07-2	1	0.1	2
15	3-(Alloxyl)-11-ethyl-12-phenyl 5,6- dihydroxydibenz[*a,e*]cyclooctene	85850-83-5	1	0.4	0
16	Amaranth	915-67-3	1	0	1
17	2-Aminoestratriene-3,17β-diol	107900-30-1	0	12	1
18	4-Aminoestratriene-3,17β-diol	107900-31-2	0	16	0
19	2-Aminoestratrien-17β-ol	17522-06-4	0	4	0
20	4-Aminoestratrien-17β-ol	17522-04-2	0	0.2	0
21	4-Aminophenyl ether	101-80-4	1	0	2
22	4-*tert* -Amylphenol	80-46-6	1	0.0003	0
23	3β-Androstanediol	25126-76-5	1	3	2
24	5α-Androstane-3α,17β-diol	1852-53-5	1	0.0007	0
25	5α-Androstane-3β,17β-diol	571-20-0	1	0.2	0
26	5β-Androstane-3α,17β-diol	1851-23-6	1	0	0
27	5β-Androstanedione	5982-99-0	0	0	0
28	5α-Androstane-3,17-dione	846-46-8	1	0	2
29	5α-Androstane-3α-ol-17-one	53-41-8	1	0	0
30	4-Androstenediol	1156-92-9	1	0.5	1
31	5-Androstenediol	521-17-5	1	3	0
32	4-Androstenedione	63-05-8	1	0.002	1
33	Anthracene	120-12-7	1	0	1
34	Apigenin	520-36-5	1	0.2	0
37	Atrazine	1912-24-9	1	0.00006	0
38	Aurin	603-45-2	1	0.03	1
39	Baicalein	491-67-8	1	0.0009	0
40	Benomyl	17804-35-2	1	0	2
41	Benz[*a*]anthracene	56-55-3	1	0	0
42	Benzeneacetonitrile α-[bis(4- hydroxyphenyl) methylene]	66422-14-8	0	100	2
43	Benzo[*a*]carbazole	239-01-0	1	0	2
44	Benzo[*c*]carbazole		0	0	0
45	Benzo[*b*]fluoranthene	205-99-2	1	17	1
46	Benzo[*k*]fluoranthene	207-08-9	1	27	0
47	Benzo[*a*]fluorene	238-84-6	1	0	2
48	Benzo[*b*]fluorene	243-17-4	1	0	0
49	Benzo[*b*]naptho[2,1-d]thiophene	239-35-0	1	0	1
50	Benzo[*b*]naptho[2,3-d]thiophene	243-46-9	1	0	0
51	Benzo[*ghi*]perylene	191-24-2	1	0	0
52	Benzo[*c*]phenanthrene	195-19-7	1	0	0
53	Benzo[*a*]pyrene	50-32-8	1	14	0
54	Benzo[*e*]pyrene	192-97-2	1	57	0
55	Benzyl alcohol	100-51-6	1	0	2
56	4-Benzyloxyphenol	103-16-2	1	0.0004	1
57	Benzylparaben	94-18-8	1	0.003	0
58	Biochanin A	491-80-5	1	0.002	0
59	Bis(*m* -acetoxy)-1,1,2-triphenylbut-1- ene	100808-56-8	0	12	2

60	Bis(*p* -acetoxy)-1,1,2-triphenylbut-1- ene	100808-54-6	0	73	1
61	Bisdesoxyestradiol	1217-09-0	1	0.1	2
62	1,1-Bis-(4-hydroxyphenyl) ethane	2081-08-5	0	0.0009	1
63	4,4-Bis(4-hydroxyphenyl)heptane	7425-79-8	0	0.2	1
64	3,4-Bis(3-hydroxyphenyl)hexane	68266-24-0	1	10	2
65	3,3-Bis(4-hydroxyphenyl)pentane	3600-64-4	0	0.2	0
66	1,1-Bis(4-hydroxyphenyl)propane	1576-13-2	0	0.2	0
67	2,2-Bis(4-hydroxyphenyl)propanol	142648-65-5	0	0.008	2
68	2,2-Bis(*p*-hydroxyphenyl)-1,1,1- trichloroethane	2971-36-0	1	1.5	2
69	Bisphenol A	80-05-7	1	0.1	1
70	Bisphenol A bis(chloroformate)	2024-88-6	1	0.02	1
71	Bisphenol A diglycidyl ether	1675-54-3	1	0	1
72	Bisphenol A diglycidyl ether dimethacrylate	1565-94-2	1	0	2
73	Bisphenol A dimethacrylate	3253-39-2	1	0.01	1
74	Bisphenol A ethoxylate	68140-85-2	0	0	1
75	Bisphenol A ethoxylate diacrylate	64401-02-1	0	0.0005	0
76	Bisphenol A glucuronide		0	0	1
77	Bisphenol A propoxylate	37353-75-6	0	0	2
78	Bisphenol AF	1478-61-1	1	1	1
79	Bisphenol B	77-40-7	1	0.1	1
80	Bisphenol C	79-97-0	1	0.3	1
81	Bisphenol C 2	14868-03-2	1	2.3	0
82	Bisphenol E	6052-84-2	1	0.04	1
83	2,2'-Bisphenol F	2467-02-9	1	0	2
84	4,4'-Bisphenol F	620-92-8	1	0.08	0
85	Bisphenol S	80-09-1	1	0.0009	2
86	16α-Bromo-17β-estradiol	54982-79-5	1	65	1
87	1,3-Butanediol, 4-[4-(1,2,3,4-tetrahydro-6-hydroxy-2-phenyl-1-naphthalenyl)phenoxy]	107144-85-4	0	30	1
88	1,3-Butanediol, 4-[4-(1,2,3,4-tetrahydro-6-methoxy-2-phenyl-1- naphthenyl)phenoxy]	107163-56-4	0	0.1	0
89	Butolame	150748-23-5	0	0.1	0
90	Butyl 4-aminobenzoate	94-25-7	1	0	1
91	*n* -Butylbenzene	104-51-8	1	0	0
92	*sec* -Butylbenzene	135-98-8	1	0	1
93	Butyl benzyl phthalate	85-68-7	1	0.002	0
94	Butylparaben	94-26-8	1	0.001	2
95	2-*sec* -Butylphenol	89-72-5	1	0.0003	1
96	2-*tert* -Butylphenol	88-18-6	1	0.001	1
97	3-*tert* -Butylphenol	585-34-2	1	0.0007	2
98	4-*sec* -Butylphenol	99-71-8	1	0.0004	2
99	4-*tert* -Butylphenol	98-54-4	1	0.0003	2
100	Butyl phthalyl *n* -butyl glycolate	85-70-1	1	0	0
101	Caffeine	58-08-2	1	0	2
102	Carbaryl	63-25-2	1	0	2
103	Carbofuran	1563-66-2	1	0	2
105	(±)-Catechin	7295-85-4	1	0	0
106	Chalcone	94-41-7	1	0.002	0
107	Chlordane	57-74-9	1	0	0
108	α-Chlordane	5103-71-9	1	0	0
109	Chlormequat chloride	999-81-5	1	0	2
110	2-Chloro-4-amino-6-isopropylamino-1,3,5 triazine	6190-65-4	1	0.00002	0
111	2'-Chloro-4,4'-biphenyldiol	56858-70-9	1	1.1	2
112	2-Chloro-4-biphenylol	23719-22-4	1	0.02	0
113	4-Chloro-4'-biphenylol	28034-99-3	1	0.03	1
114	4-Chloro-*m* -cresol	59-50-7	1	0.0004	1
115	2-Chloro-4,6-diamino-*S* -triazine	3397-62-4	1	0.0003	1
116	2-Chloro-4-ethylamino-6-amino-1,3,5- triazine	1007-28-9	1	0	1
117	2-Chloro-4-ethylamino-6-(1- hydroxyisopropyl)amino-1,3,5-triazine	142179-80-4	0	0	2
118	2-Chloro-4-isopropylamino-6-(1- hydroxyisopropylamino)-1,3,5-triazine	142200-36-0	0	0	0
119	11β-Chloromethylestradiol	71794-60-0	1	100	1
120	2-Chloro-4-methylphenol	6640-27-3	1	0.0002	2
121	4-Chloro-2-methylphenol	1570-64-5	1	0.0002	0
122	2-Chlorophenol	95-57-8	1	0	1
123	4-Chlorophenol	106-48-9	1	0.004	2

124	Chlorotamoxifen	77588-46-6	1	0	1
125	Cholesterol	57-88-5	1	0	2
126	Chrysene	218-01-9	1	0	0
127	Chrysin	480-40-0	1	0	1
128	Cineole	470-82-6	1	0	2
129	Cinnamic acid	621-82-9	1	0	2
130	*cis* -Clomiphene	15690-55-8	1	0.1	1
131	*trans* -Clomiphene	911-45-5	1	13	0
132	Clomiphene citrate	50-41-9	1	0.7	0
133	Colchicine	64-86-8	1	0	1
134	Corticosterone	50-22-6	1	0	0
135	Cortisol	50-23-7	1	0	1
136	Coumestrol	479-13-0	1	20	2
137	*p* -Cumyl phenol	599-64-4	1	0.005	0
138	Cyclofenil diphenol	5189-40-2	1	5	2
139	Cycloprop[14*R* ,15α]estra-1,3,5(10)- triene-3,17β-diol, 3',15-dihydro-	73860-54-5	0	45	0
140	Cycloprop[14*S* ,15S]estra-1,3,5(10)- triene-3,17β-diol, 3',15-dihydro-	105455-76-3	1	100	0
141	Cypermethrin	52315-07-8	1	0	1
142	Daidzein	486-66-8	1	0.1	0
143	*m,p* '-DDD	4329-12-8	1	0	1
144	*o,p* '-DDD	53-19-0	1	0.002	0
147	(*R*)-4'-Deoxyindenestrol A	138515-00-1	0	0.2	0
148	(*R*)-5-Deoxyindenestrol A	138515-02-3	0	0.9	0
151	(*S*)-4'-Deoxyindenestrol A	138514-99-5	0	1.8	0
152	(*S*)-5-Deoxyindenestrol A	138515-01-2	0	5.6	0
153	1,3-Diacetoxy-17α-ethinyl-7α-methyl-1,3,5(10)-estratrien-17β-ol		0	20	2
154	1,3-Dibenzoyloxy-17α-ethinyl-7α- methyl-1,3,5(10)-estratrien-17β-ol		0	7.3	0
155	1,3-Dibenzyltetramethyldisiloxane		0	0	1
156	14-Dehydroestradiol-17β	58699-19-7	0	107	1
157	14-Dehydroestradiol-17β 3-methyl ether	35664-58-7	0	0.8	0
158	14-Dehydroestrone	2119-18-8	0	9	2
159	14-Dehydroestrone 3-methyl ether	17550-11-7	0	0	0
160	17-Desoxyestradiol	53-63-4	1	31	1
161	3-Deoxyestradiol	2529-64-8	1	4.3	0
162	3-Deoxyestrone	53-45-2	1	0.006	0
163	4,4'-Diaminostilbene dihydrochloride	66635-40-3	0	0	2
165	Dehydroepiandrosterone	53-43-0	1	0.01	2
166	Dexamethasone	50-02-2	1	0	1
167	Dibenz[*ah*]anthracene	53-70-3	1	0	0
168	Dibenzo-18-crown-6	14187-32-7	1	0	2
169	Dibutyl benzyl phthalate		0	0	2
170	*o,p* '-DDE	3424-82-6	1	0.0002	1
171	*o,p* '-DDT	789-02-6	1	0.07	2
172	*p,p* '-DDD	72-54-8	1	0.002	0
173	*p,p* '-DDE	72-55-9	1	0.00004	1
174	*p,p* '-DDT	50-29-3	1	0.006	1
175	2,6-Di-*tert* -butylphenol	128-39-2	1	0	2
176	Dibutyl phthalate	84-74-2	1	0.001	1
177	2,4'-Dichlorobiphenyl	34883-43-7	1	0.0002	1
178	2,5-Dichlorobiphenyl	34883-39-1	1	0	1
179	3,4-Dichlorobiphenyl	2974-92-7	1	0	0
180	3,5-Dichlorobiphenyl	34883-41-5	1	0	2
181	4,4'-Dichlorobiphenyl	2050-68-2	1	0	1
182	2,5-Dichloro-2'-biphenylol	53905-30-9	1	0	1
183	2,5-Dichloro-3'-biphenylol	53905-29-6	1	0.002	0
184	2',5'-Dichloro-4-biphenylol	53905-28-5	1	0.2	0
185	2,6-Dichloro-4'-biphenylol	79881-33-7	1	0.3	0
186	3,4-Dichloro-2'-biphenylol	209613-97-8	0	0	1
187	3,4-Dichloro-3'-biphenylol	14962-34-6	0	0	2
188	3,4-Dichloro-4'-biphenylol	53890-77-0	0	0.3	0
189	3,5-Dichloro-2'-biphenylol		0	0	1
190	3,5-Dichloro-4'-biphenylol		0	0	1
191	3,5-Dichloro 2-hydroxy-2-methylbut-3 enanalide	16776-82-1	0	0.0002	2

192	2,4-Dichlorophenoxyacetic acid	94-75-7	1	0	2
193	2-[[(3,5-Dichlorophenyl)amino]-carbamoyl]oxy]-2-methyl-3-butenoic acid	119209-27-7	1	0.0002	1
194	Dieldrin	60-57-1	1	0.0001	1
195	Dienestrol	84-17-3	1	97	1
197	β-Dienestrol	35495-11-5	1	0.4	0
198	1,3-Diethyl-6,4'-dihydroxy-2- phenylindene		0	79	0
199	Di-2-ethylhexyl adipate	103-23-1	1	0	1
200	Diethylhexyl phthalate	117-81-7	1	0	0
201	1,3-Diethyl-4-hydroxy-2- phenylindene		0	9.3	0
202	1,3-Diethyl-6-hydroxy 2-phenylindene		0	2.2	0
203	*meso-p* -(α,β-Diethyl-*p* - methylphenethyl)phenol	267408-76-4	0	4	0
204	Diethyl phthalate	84-66-2	1	0	2
205	(5*R* ,11*R*)-5,11-Diethyl-5,6,11,12- tetrahydrochrysene-2,8-diol		0	23	0
206	(5*S* ,11*S*)-5,11-Diethyl-5,6,11,12- tetrahydrochrysene-2,8-diol		0	0.9	0
208	2,2'-Dihydroxy-4- methoxybenzophenone	131-53-3	1	0	0
209	2,2'-Dihydroxybenzophenone	835-11-0	1	0	1
210	2,4-Dihydroxybenzophenone	131-56-6	1	0.002	1
211	3-(2,3 Dihydroxypropoxy)-10-ethyl-11-phenyldibenz[*b,f*]oxepin	85850-89-1	0	0.07	0
212	3,3'-Diethylstilbestrol	5959-71-7	1	3	2
213	3,3'-Dihydroxyhexestrol	79199-51-2	1	15	1
214	4,4'-Dihydroxybenzophenone	611-99-4	1	0.008	2
215	4,4'-Dihydroxybiphenyl	92-88-6	1	0	1
216	5,11-*trans* -Diethyl-5,6,11,12- tetrahydrochrysene-2,8-diol		0	221	0
217	5,6-Dihydro-8-[2-(dimethylamino)ethoxy]-12-ethyl-11-phenyl-dibenzo[a,e]cyclooctene,hydrate	85850-78-8	1	0.2	2
218	5α-Dihydrotestosterone	521-18-6	1	0.03	0
219	5β-Dihydrotestosterone	571-22-2	1	0	0
220	6,4'-Dihydroxyflavone	63046-09-3	0	0.2	1
221	Diethylstilbestrol	56-53-1	1	419	0
222	Diethylstilbestrol dimethyl ether	130-79-0	1	0.06	1
223	Diethylstilbestrol epoxide	6052-82-0	1	9.4	2
224	Diethylstilbestrol phenanthrene		0	0.3	1
225	Dihexyl phthalate	84-75-3	1	0	0
226	Dihydrogenistein	21554-71-2	0	0.1	1
228	Dihydroxydiethylstilbestrol	7507-01-9	1	0.5	1
229	Diisobutyl phthalate	84-69-5	1	0	0
230	Diisodecyl phthalate	26761-40-0	1	0	0
231	Diisoheptyl phthalate	41451-28-9	0	0	0
232	Diisononyl phthalate	28553-12-0	1	0	0
233	11β-[2-(*N,N* -Dimethylamino)ethoxy]estra-1,3,5(10) triene-3,17β-diol		0	1.6	2
234	3-[2-(Dimethylamino)ethoxy]-11-ethyl-12-phenyl)-6*H* - dibenzo[*b,f*]thioctin	85850-79-9	1	1.1	0
235	3-[2-(Dimethylamino)ethoxy]-10- ethyl-11-phenyldibenz[*b,f*]oxepin	85850-76-6	1	0.02	1
236	7-[2-(Dimethylamino)ethoxy]-11- ethyl-10-phenyldibenz[*b,f*]thiepin	85850-77-7	1	0.1	0
237	11β-[3-(*N,N* -Dimethylamino)-propoxy]estra-1,3,5 (10)-triene-3,17β- diol	130043-38-8	0	2.6	1
238	α,α-Dimethyl-β-ethylallenolic acid	15372-37-9	0	1	1
239	2,6-Dimethylhexestrol	334707-28-7	0	13	1
240	1,6-Dimethylnaphthalene	575-43-9	1	0	0
241	Dimethyl phthalate	131-11-3	1	0	1
242	α,α-Dimethylstilbestrol	552-80-7	1	72	2
243	Dimethyl sulfoxide	67-68-5	1	0	1
244	5,11-*trans* -Dimethyl-5,6,11,12- tetrahydrochrysene-2,8-diol		0	222	2
245	(5*R* ,11*R*)-5,11-Dimethyl-5,6,11,12- tetrahydrochrysene-2,8-diol		0	24	0
246	(5*S* ,11*S*)-5,11-Dimethyl-5,6,11,12- tetrahydrochrysene-2,8-diol		0	9.3	0
247	Di-*n* -octyl phthalate	117-84-0	1	0	0
248	Diphenolic acid	126-00-1	1	0.0007	1
249	*trans,trans* -1,4-Diphenyl-1,3-butadiene	886-65-7	1	0	1
250	4-[1,2-(Diphenyl-1-butenyl)]phenol acetate	100808-55-7	0	21	0
251	2,3-Diphenylindenone-1		0	0.01	1
252	4-[1- (Diphenylmethylene)propyl]phenol acetate	82333-68-4	0	2	0
253	1,3-Diphenyltetramethyldisiloxane	56-33-7	1	0.0007	2
254	5,11-*trans* -Dipropyl-5,6,11,12- tetrahydrochrysene-2,8-diol		0	34	1
255	(5*R* ,11*R*)-5,11-Dipropyl-5,6,11,12- tetrahydrochrysene-2,8-diol		0	5.2	0
256	(5*S* ,11*S*)-5,11-Dipropyl-5,6,11,12- tetrahydrochrysene-2,8-diol		0	1.6	0
257	4-Dodecylphenol	104-43-8	1	0.02	1

258	Doisynoestrol	15372-34-6	1	0.002	1
259	Dopamine	51-61-6	1	0	1
260	Droloxifene	82413-20-5	1	8.9	0
261	Empenthrin	54406-48-3	1	0	2
262	α–Endosulfan	959-98-8	1	0	1
263	α,β–Endosulfan	115-29-7	1	0.00009	0
264	β-Endosulfan	33213-65-9	1	0.00006	0
265	16-Epiestriol	547-81-9	1	44	0
266	17-Epiestriol	1228-72-4	1	29	0
267	Epitestosterone	481-30-1	1	0	0
268	Equilenin	517-09-9	1	8	1
269	Equilin	474-86-2	1	24	1
270	Equol	531-95-3	1	0.2	2
271	*Erythro* -MEA	20576-52-7	0	135	1
272	16α-Estradiol	1090-04-6	1	34	2
273	17α-Estradiol	57-91-0	1	28	0
274	17β-Estradiol	50-28-2	1	100	0
276	Estradiol 17-acetate		0	29	1
277	17β-Estradiol 3-acetate	4245-41-4	1	97	2
278	Estradiol benzoate	50-50-0	1	11	0
279	Estradiol diacetate	3434-88-6	1	11	0
280	17β-Estradiol 3-methyl ether	1035-77-4	1	1.9	0
281	9-dehydro-Estratetraene-3,17β-diol	791-69-5	0	80	0
282	Estra-1,3,5(10),6-tetraen-17-one, 3- hydroxy-		0	10	1
283	Estra-1,3,5(10)-triene-3,17β- diol,14α,15α–epoxy-	79581-12-7	0	5	2
284	Estra-1,3,5(10)-triene-3,17β- diol,14β,15β-epoxy-	79645-49-1	0	0	0
285	Estra-1,3,5(10)-triene-3,14,17β-triol	16288-09-8	1	2	1
286	Estratriene-3,6α,17β-triol	1229-24-9	0	0.7	1
287	Estriol	50-27-1	1	23	0
288	Estrone	53-16-7	1	38	1
289	Estrone 3-acetate	901-93-9	1	191	0
290	Estrone 3-methyl ether	1624-62-0	1	0	0
291	Estrone-3-sulfate	481-97-0	1	0	2
292	17α-Ethinyl estradiol	57-63-6	1	250	1
293	17β-Ethinyl estradiol	4717-38-8	1	190	0
294	Ethyl cinnamate	103-36-6	1	0	2
295	3-Ethyl-6,4'-dihydroxy-2- phenylindene		0	16	0
296	2-Ethylhexyl paraben	5153-25-3	1	0.02	0
297	4-Ethyl-7-hydroxy-3-(methoxyphenyl) 2*H* -1-benzopyran-2-one	5219-17-0	1	0.9	2
298	3-[(10-Ethyl-11-(*p*-hydroxyphenyl)dibenzo[*b,f*]oxepin-3- yl)oxy]-1,2-propanediol, hydrate (4:1)	85850-93-7	0	0.9	2
299	3-[(10-Ethyl-11-(*p*-hydroxyphenyl)dibenzo[*b,f*]thiepin-3- yl)oxy]-1,2-propanediol	85850-94-8	0	11	0
300	3-[(11-Ethyl-12-(*p* -hydroxyphenyl)-6*H* -dibenzo[*b,f*]thiocin-3-yl)oxy]-1,2-propanediol	85864-54-6	0	5	1
301	3-[(6-Ethyl-5-(*p* -hydroxyphenyl)-11,12-dihydrodibenzo[*a,e*]cycloocten2-yl)oxy]-1,2-propanediol	85850-95-9	0	9.1	2
302	3-Ethyl-4'-hydroxy-2-phenylindene		0	2.3	1
303	3-Ethyl-6-hydroxy 2-phenylindene		0	0.6	0
304	3-Ethyl-4'-hydroxy 2-phenylindenone-1		0	1.6	0
305	3-Ethyl-6-hydroxy 2-phenylindenone-1		0	1.2	2
306	3-Ethyl-4-(*p* -methoxyphenyl)-2-methyl-3-cyclohexene-1-carboxylic acid	1755-52-8	1	0.8	2
307	Ethyl paraben	120-47-8	1	0.0006	1
308	2-Ethylphenol	90-00-6	1	0	1
309	3-Ethylphenol	620-17-7	1	0.0001	1
310	4-Ethylphenol	123-07-9	1	0.00007	1
311	3-[(10-Ethyl-11-phenyldibenzo-[*b,f*]thiepin-3-yl)oxy]-1,2- propanediol, complexed with isopropyl	85850-90-4	0	0.7	1
312	3-[(11-Ethyl-12-phenyl-6*H* -dibenzo [*b,f*]thiocin-3-yl)oxy]-1,2-propanediol, hydrate (4:1)	85850-92-6	0	0.02	0
313	3-[(6-Ethyl-5-phenyl-11,12-dihydrodibenzo [*a* ,*e*]cycloocten-2-yl)oxy]-1,2-propanediol	85850-91-5	0	0.1	0
314	Eugenol	97-53-0	1	0	2
315	Fenvalerate	51630-58-1	1	0	2
316	2-(2-Fluorophenyl)-3-phenyl-6-hydroxyindene		0	49	0
317	2-Fluoroestratrien-17β-ol	101772-22-9	0	2	0
318	4-Fluoroestratrien-17β-ol	96607-54-4	0	8	0
319	Fisetin	528-48-3	1	0.005	2
320	Flavanone	17002-31-2	0	0	0
321	Flavone	525-82-6	1	0	0

322	Fluoranthene	206-44-0	1	0	0
323	Fluorene	86-73-7	1	0	0
324	Fluorotamoxifen	73617-96-6	1	0	0
325	Folic acid	59-30-3	1	0	2
326	Formononetin	485-72-3	1	0.0007	0
327	Furfural	98-01-1	1	0	1
328	Genistein	446-72-0	1	1.7	1
329	Genistin	529-59-9	1	0	1
330	Glyceollin	66241-09-6	0	0.2	2
331	Glycitein	40957-83-3	1	0.02	1
332	Glycitin	40246-10-4	0	0.0008	2
333	Heptachlor	76-44-8	1	0	0
334	2,2',3,3',4',5,5'-Heptachloro-4- biphenylol	158076-64-3	0	0.05	0
335	2,2',3,3',4,5,6-Heptachlorobiphenyl	68194-16-1	1	0	0
336	2,2',3,3',4',5,6-Heptachlorobiphenyl	52663-70-4	1	0	0
337	2,2',3,3',5,5',6-Heptachlorobiphenyl	52663-64-6	1	0	0
338	2,2',3',4,4',5,5'-Heptachloro-3- biphenylol	158076-69-8	0	0.06	0
339	2,2',3,4,4',5',6-Heptachlorobiphenyl	52663-69-1	1	0	0
340	2,2',3,4,4',6,6'-Heptachlorobiphenyl	74472-48-3	1	0.02	0
341	2,2',3,4',5,5',6-Heptachloro-4- biphenylol	158076-68-7	0	0.1	0
342	2,2',3,4',5,5',6-Heptachlorobiphenyl	52663-68-0	1	0	0
343	2,2',3,4',5,6,6'-Heptachlorobiphenyl	74487-85-7	1	0.02	0
344	2,3,3',4,4',5,6-Heptachlorobiphenyl	41411-64-7	1	0	0
345	2,3,3',4',5,5',6-Heptachlorobiphenyl	69782-91-8	1	0	0
346	4-(Heptyloxy)phenol	13037-86-0	1	0.001	2
347	Heptanal	111-71-7	1	0	1
348	Heptyl 4-paraben	1085-12-7	1	0.008	0
349	Hesperetin	520-33-2	1	0	2
350	Hexachlorobenzene	118-74-1	1	0	2
351	2,2',3,3',4,4'-Hexachlorobiphenyl	38380-07-3	1	0	1
352	2,2',3,4,4',5'-Hexachlorobiphenyl	35065-28-2	1	0	0
353	2,2',3,4,5,6'-Hexachlorobiphenyl	68194-15-0	1	0	0
354	2,2',3,4',5',6-Hexachlorobiphenyl	38380-04-0	1	0	0
355	2,2',3,5,5',6-Hexachlorobiphenyl	52663-63-5	1	0	0
356	2,2',4,4',5,5'-Hexachlorobiphenyl	35065-27-1	1	0	0
357	2,2',4,4',6,6'-Hexachlorobiphenyl	33979-03-2	1	0.1	0
358	2,3,3',4,4',6-Hexachlorobiphenyl	74472-42-7	1	0	0
359	2,3',4,4',5',6-Hexachlorobiphenyl	59291-65-5	1	0	0
360	3,3',4,4',5,5'-Hexachlorobiphenyl	32774-16-6	1	0	1
361	2,2',3,3',4',5-Hexachloro-4-biphenylol	158076-62-1	0	0.04	0
362	2,2',3,4',5,5'-Hexachloro-4-biphenylol	145413-90-7	0	0.02	0
363	2',3,3',4',5,5'-Hexachloro-4-biphenylol	158076-63-2	0	3.2	0
364	*n* -Hexanol	111-27-3	1	0	2
365	Hexestrol	84-16-2	1	215	0
366	*DL* -Hexestrol	5776-72-7	1	3.6	0
367	Hexestrol monomethyl ether	13026-26-1	1	9.4	1
368	3-Hydroxybenzo[*b*]naphtho[2,1- d]thiophene		0	1.8	2
369	2-Hydroxybenzo[*c*]phenanthrene	22717-94-8	1	2.2	2
370	3-Hydroxybenzo[*b*]phenanthro[2,3- d]thiophene		0	2.4	0
371	4-Hydroxychalcone	20426-12-4	1	0.003	2
372	4'-Hydroxychalcone	2657-25-2	1	0.004	0
373	4'-Hydroxychalcone (*cis*- and *trans*-)	38239-52-0	1	0.004	0
374	2-Hydroxychrysene	65945-06-4	1	5.8	0
375	4'-Hydroxy-2,3-diphenylindenone-1		0	0.5	0
376	6'-Hydroxy-2,3-diphenylindenone-1		0	59	0
377	11α-Hydroxyestradiol	1464-61-5	0	3.1	0
378	11β-Hydroxyestradiol	5444-22-4	1	7.2	0
379	14β-Hydroxyestradiol	60183-66-6	0	10	0
380	2-Hydroxyestradiol	362-05-0	1	7	1
381	4-Hydroxyestradiol	5976-61-4	1	13	0
382	2-Hydroxyestratrien-17β-ol	2259-89-4	1	18	0
383	4-Hydroxyestratrien-17β-ol	17592-89-1	0	0.8	1
384	3-Hydroxyestra-1,3,5(10)-trien-16-one	3601-97-6	1	0.5	1

385	2-Hydroxyestrone	362-06-1	1	2	0
386	2-Hydroxyethyl methacrylate	868-77-9	1	0	1
387	3'-Hydroxyflavanone	92496-65-6	0	0.002	0
388	4'-Hydroxyflavanone	135413-27-3	0	0.002	0
389	6-Hydroxyflavanone	4250-77-5	0	0.0009	0
390	7-Hydroxyflavanone	6515-36-2	0	0.0002	1
391	6-Hydroxyflavone	6665-83-4	1	0.0004	0
392	7-Hydroxyflavone	6665-86-7	1	0	0
393	Hydroxyflutamide	52806-53-8	1	0.0002	2
394	2-Hydroxy-4-methoxybenzophenone	131-57-7	1	0	0
395	6-Hydroxy-2'-methoxyflavone	61546-59-6	0	0	1
396	2-Hydroxy-5-methylchrysene		0	20	0
397	8-Hydroxy-5-methylchrysene		0	3.1	0
398	16β-Hydroxy-16-methyl-17β-estradiol 3-methyl ether	3434-79-5	1	0.03	1
399	4-Hydroxytamoxifen	68047-06-3	1	152	0
400	6-Hydroxytetralin	1125-78-6	1	0	1
401	ICI 164384	98007-99-9	1	69	1
402	ICI 182780	129453-61-8	1	251	2
403	Imiprothrin	72963-72-5	1	0	2
404	Indanestrol	71855-45-3	1	2.3	2
405	Indanyldiethylstilbestrol		0	80	0
407	(*R*)-Indenestrol A	115217-03-3	1	13	2
409	(*S*)-Indenestrol A	115217-04-4	1	285	0
411	(*R*)-Indenestrol B		0	100	0
413	(*E*)-17α-Iodovinylestradiol	82123-96-4	1	12	1
414	(*S*)-Indenestrol B	115217-06-6	1	143	0
415	(*Z*)-17α-Iodovinylestradiol	177159-09-0	0	57	0
416	11-Keto-9β-estradiol		0	0	1
417	16-Ketoestradiol	566-75-6	1	5.4	2
418	16α-Iodoestradiol	71765-94-1	1	100	0
419	6-Ketoestradiol	571-92-6	1	20	1
420	9α-Methyl-14-dehydroestrone 3- methyl ether		0	0	0
421	9α-Methylestradiol-17β	66463-44-3	0	35	0
422	9α-Methylestradiol-17β 3-methyl ether	51242-32-1	0	0	1
423	9α-Methylestrone	71563-77-4	0	5	0
424	9α-Methylestrone 3-methyl ether	31266-41-8	0	0	0
425	Indeno[1,2,3-*cd*]pyrene	193-39-5	1	20	0
426	Indole[3,2-*b*]carbazole		0	0.0008	1
427	Ipriflavone	35212-22-7	1	0	2
428	Isoeugenol	97-54-1	1	0	1
429	Kaempferol	520-18-3	1	0.06	0
430	Kepone	143-50-0	1	0.06	2
431	Lindane	58-89-9	1	0.0002	1
432	Melatonin	73-31-4	1	0	1
433	MER-25	67-98-1	1	0.001	2
434	Mestilbol	18839-90-2	0	20	0
435	Mestranol	72-33-3	1	3.5	0
436	Methoxychlor olefin	2132-70-9	1	0	2
437	Methoxytamoxifen		0	0.7	0
438	Methyl methacrylate	80-62-6	1	0	1
439	Methyl paraben	99-76-3	1	0.0004	2
440	Methyltamoxifen	73617-95-5	0	0.3	0
441	*o,p'* -Methoxychlor	30667-99-3	1	0.01	2
442	*p,p'* - Methoxychlor	72-43-5	1	0.002	0
443	9α-Methyl-14-dehydroestrone	88598-67-8	0	6	0
444	9α-Methyl-14-dehydroestradiol-17β 3- methyl ether	88598-64-5	0	0.1	2
445	9α-Methyl-14-dehydroestradiol-17β	88598-63-4	0	41	0
446	7α-Methylestrone 3-methyl ether	10449-00-0	1	0	1
447	7α–Methylestrone	10448-96-1	1	68	0
448	7α-Methylestradiol-17β 3-methyl ether	15506-01-1	0	5.3	0
449	7α-Methylestradiol-17β	10448-97-2	0	104	1
450	7α-Methyl-14-dehydroestrone 3-methyl ether	35644-57-6	0	0	1
451	7α-Methyl-14-dehydroestrone	88958-66-7	0	52	0

452	7α-Methyl-14-dehydroestradiol-17β 3- methyl ether	35644-59-8	0	3.1	0
453	7α-Methyl-14-dehydroestradiol-17β	88598-62-3	0	74	0
454	4,4'-Methylenedianiline	101-77-9	1	0	1
455	4,4'-Methylenebis(N,N-dimethylaniline)	101-61-1	1	0	1
456	3-Methoxyestriol	1474-53-9	1	0.02	0
457	3-Methoxy-10-methyl-11- phenyldibenzo[*b,f]* thiepin (16β)	85807-06-1	0	0.005	1
458	2-(2-Methylphenyl)-3-phenyl-6- hydroxyindene		0	100	1
459	1-Methyl-6-hydroxy-2,3- diphenylindene		0	12	1
461	11β-Methylestrone 3-methyl ether	13667-04-4	0	0	2
462	11β-Methylestrone	13667-06-6	0	47	0
463	11β-Methylestradiol-17β 3-methyl ether	18046-75-8	0	5.1	1
464	11β-Methylestradiol-17β	23637-93-6	1	112	0
465	11β-Methyl-14-dehydroestrone 3-methyl ether	88598-69-0	0	0	0
466	11β-Methyl-14-dehydroestradiol-17β 3-methyl ether	88598-65-6	0	1.2	1
467	(*Z*)-11β-Methoxy-17α- iodovinylestradiol	177159-11-4	0	36	0
468	(*E*)-11β-Methoxy-17α- iodovinylestradiol	90857-55-9	1	14	2
469	Metolachlor	51218-45-2	1	0	1
470	Mirex	2385-85-5	1	0	0
471	Mono-*m* -acetoxy-1,1,2-triphenylbut-1-ene	82333-69-5	0	1	0
472	Monohydroxymethoxychlor	28463-03-8	0	0.1	0
473	Monohydroxymethoxychlor olefin	75938-34-0	1	0.2	0
474	Monohydroxytamoxifen	68392-35-8	1	52	0
475	Morin	480-16-0	1	0.0005	0
476	Moxestrol	34816-55-2	1	20	0
477	Myricetin	529-44-2	1	0.002	0
478	Nafoxidine	1845-11-0	1	17	2
479	Naringenin	480-41-1	1	0.004	0
480	Naringin	10236-47-2	1	0	1
481	Nerolidol	7212-44-4	1	0	1
482	2-Nitroestratriene-3,17β-diol	6298-51-7	1	1	1
483	4-Nitroestratriene-3,17β-diol	6936-94-3	1	13	0
484	2-Nitroestratrien-3-ol-17-one	5976-73-8	1	0.1	0
485	4-Nitroestratrien-3-ol-17-one	5976-74-9	1	6	2
486	Nitromifene	10448-84-7	1	0.1	2
487	*cis* -Nonachlor	5103-73-1	1	0	0
488	*trans* -Nonachlor	39765-80-5	1	0	0
489	Nonylbenzene	1081-77-2	1	0	1
490	*p* -Nonylphenol	104-40-5	1	0.1	0
491	*n* -Nonylphenol	25154-52-3	1	0.01	0
493	Nordihydroguariaretic acid	500-38-9	1	0.03	0
494	Norethindrone	68-22-4	1	0.07	2
495	Norethynodrel	68-23-5	1	0.4	0
496	19-Nortestosterone	434-22-0	1	0.06	1
497	2,2',3,3',4,4',5,5'-Octachlorobiphenyl	35694-08-7	1	0	0
498	1,8-Octanediol	629-41-4	1	0	0
499	4-*n* -Octylphenol	1806-26-4	1	0.3	0
500	4-*tert* -Octylphenol	140-66-9	1	0.2	2
501	2,2',3,3',6-Pentachlorobiphenyl	52663-60-2	1	0	0
502	2,2',3',4',5'-Pentachloro-4-biphenylol	150304-12-4	0	0.06	1
503	2,2',3,4,5'-Pentachlorobiphenyl	38380-02-8	1	0	0
504	2,2',3',4',6'-Pentachloro-4-biphenylol	150304-10-2	0	0.2	0
505	2,2',3,4',6-Pentachlorobiphenyl	68194-05-8	1	0	0
506	2,2',3',5',6'-Pentachloro-4-biphenylol	150304-11-3	0	0.08	0
507	2,2',3,5',6-Pentachlorobiphenyl	38379-99-6	1	0	0
508	2,2',4,4',5-Pentachlorobiphenyl	38380-01-7	1	0	0
509	2,2',4,5,5'-Pentachlorobiphenyl	37680-73-2	1	0	0
510	2,2',4',6,6'-Pentachloro-4-biphenylol		0	21	0
511	2,2',4,6,6'-Pentachlorobiphenyl	56558-16-8	1	0.5	0
512	2',3,3',4,4'-Pentachloro-2-biphenylol	150975-80-7	0	0.004	1
513	2',3,3',4',5'-Pentachloro-4-biphenylol	149589-55-9	0	0.09	0
514	2,3,3',4',5-Pentachloro-4-biphenylol	152969-11-4	0	0.3	0
515	2',3,3',4',5-Pentachloro-4-biphenylol	192190-09-3	0	0.07	0
516	2',3,3',4',6'-Pentachloro-4-biphenylol	192190-10-6	0	0.06	0

517	2',3,3',5',6'-Pentachloro-4-biphenylol	189578-02-7	0	0.05	0
518	2,3,3',5,6-Pentachlorobiphenyl	74472-36-9	1	0	0
519	2,3',4,4',5-Pentachloro-3-biphenylol	170946-11-9	0	1	0
520	2',3',4,4',5-Pentachloro-3-biphenylol	150975-81-8	0	2	0
521	2,3,4,4',6-Pentachlorobiphenyl	74472-38-1	1	0	0
522	2',3,4',5,5'-Pentachloro-4-biphenylol	149589-56-0	0	0.8	0
523	3,3',4,4',5-Pentachlorobiphenyl	57465-28-8	1	0	0
524	3,3',4',5,5'-Pentachloro-4-biphenylol	130689-92-8	1	0.02	0
525	4-Phenethylphenol	6335-83-7	1	0.002	0
526	Pentolame	150748-24-6	1	0.04	2
527	Permethrin	52645-53-1	1	0	0
528	Phenanthrene	85-01-8	1	0	0
529	Phenol, 4,4'-[1,2-bis(methylene)-1,2- ethanediyl]bis-	107144-81-0	0	20	1
530	Phenol, 4-[3-(2dimethylamino)ethoxy]-11- ethyldibenzo[*b,f*]thioctin-12-yl)	85850-81-3	0	51	1
531	Phenol, 4-[7-(2-dimethylamino)ethoxy]-11- ethyldibenzo[*b,f*]thiepin-10-yl]-	85850-74-4	0	63	2
532	Phenol, 4-[2-(2 dimethylamino)-ethoxy]-6-ethyl-11,12-dihydro-dibenzo[*a,e*]cycloocten-5-yl]-	85850-75-5	0	50	0
533	Phenol, 3-[2-dimethylaminoethoxy]-10-ethyl- 4-hydroxyphenyl dibenzo-[*b,f*]oxepin	85850-80-2	0	6.1	0
534	Phenol, 4-[1-[4-[2- (dimethylamino)ethoxy]phenyl]-2- phenyl-1-butenyl]-3-methyl-, (*E*)-	96474-35-0	1	100	2
535	Phenol, 4-(1, 2-diphenyl-1-butenyl)-	69967-79-9	0	15	0
536	Phenol, 4-(1*Z*)-1,2-diphenyl-1-butenyl)-	69967-80-2	0	1	0
537	Phenol, 4-[2-Nitro-2-phenyl-1-[4-[2-(1pyrrolidinyl)ethoxy]phenyl]ethenyl]phenyl, (*E*)-	107144-84-3	0	100	0
538	Phenol, 4,4'-(2-phenyl-1-butenylidene)bis-	91221-46-4	1	100	2
539	Phenolphthalein	77-09-8	1	0.01	1
540	Phenolphthalin	81-90-3	1	0.0002	2
541	Phenol Red	143-74-8	1	0.001	1
542	*d* -Phenothrin	26002-80-2	1	0	2
543	2-Phenyl-3-(2-fluoro-4- hydroxyphenyl)-6-hydroxyindene		0	9.6	2
544	2-Phenyl-3-(2-fluorophenyl)-6- hydroxyindene		0	12	0
545	3-Phenyl-4'-hydroxy-2-phenylindene		0	0.2	0
546	3-Phenyl-6-hydroxy-2-phenylindene		0	8.9	0
547	2-Phenyl-3-(2-methylphenyl)-6- hydroxyindene		0	11	0
548	2-Phenyl-3-(4-methylphenyl)-6- hydroxyindene		0	1.7	0
549	2-Phenylphenol	90-43-7	1	0	2
550	3-Phenylphenol	580-51-8	1	0.0004	2
551	4-Phenylphenol	92-69-3	1	0.0005	0
552	Phloretin	60-82-2	1	0.1	2
553	Prallethrin	23031-36-9	1	0	0
554	Progesterone	57-83-0	1	0.00005	1
555	Prolame	99876-41-2	1	0.1	2
556	Promegesterone	34184-77-5	1	0.07	2
557	Prometon	1610-18-0	1	0	2
558	Propazine	139-40-2	1	0	1
559	Propyl paraben	94-13-3	1	0.0006	0
560	Propylpyrazoletriol		0	49	2
561	Prunetin	552-59-0	1	0.002	0
562	Pseudodiethylstilbestrol	39011-86-4	0	118	1
563	Pyrene	129-00-0	1	0	0
564	Pyrrolidine, 1-[2-[4-[1-(4-methoxyphenyl)-2-nitro-2- phenylethenyl]phenoxy]ethyl]-, (*E*)	77413-87-7	0	11	0
565	Quercetin	117-39-5	1	0.002	0
566	7-Quinolinol, 1-ethyl-1,2-dihydro-3-(4 hydroxyphenyl)-4-methyl-	107144-83-2	0	9	2
567	6-Quinolinol, 1-ethyl-1,2-dihydro-3-(4 hydroxyphenyl)-4-methyl-	107144-82-1	0	33	0
568	Raloxifene	84449-90-1	1	60	2
570	Resveratrol	501-36-0	1	300	2
571	Rutin	153-18-4	1	0.00008	1
572	Simazine	122-34-9	1	0	0
573	β-Sitosterol	83-46-5	1	0	0
574	4,4'-Stilbenediol	659-22-3	1	0.3	1
575	4-Stilbenol	3839-46-1	1	0	0
576	Suberic acid	505-48-6	1	0	2
577	Tamoxifen	10540-29-1	1	4.3	0
579	Taxifolin	480-18-2	1	0	1
580	Testosterone	58-22-0	1	0.007	0
581	2,2',3,3'-Tetrachlorobiphenyl	3844-93-8	0	0	0

582	2,2',3,4-Tetrachlorobiphenyl	52663-59-9	1	0	0
583	2,2',3,6-Tetrachlorobiphenyl	41464-47-5	1	0	0
584	2,2',4,4'-Tetrachlorobiphenyl	2437-79-8	1	0	0
585	2,2',4,5'-Tetrachlorobiphenyl	41464-40-8	1	0	0
586	2,2',4,6'-Tetrachlorobiphenyl	68194-04-7	1	0	0
587	2,2',5,5'-Tetrachlorobiphenyl	35693-99-3	1	0	0
588	2,2',6,6'-Tetrachlorobiphenyl	15968-05-5	1	0	1
589	2,3,3',5'-Tetrachlorobiphenyl	41464-49-7	1	0	1
590	2,3,4,4'-Tetrachlorobiphenyl	33025-41-1	1	0	0
591	2,3',4,5'-Tetrachlorobiphenyl	73575-52-7	1	0	0
592	2,3',4',5-Tetrachlorobiphenyl	32598-11-1	1	0	0
593	2,4,4',5-Tetrachlorobiphenyl	32690-93-0	1	0	0
594	3,3',4,4'-Tetrachlorobiphenyl	32598-13-3	1	0	0
595	3,3',4,5-Tetrachlorobiphenyl	70362-49-1	1	0	0
596	2',3',5',6'-Tetrachloro-4,4'-biphenyldiol	100702-98-5	0	0.02	0
597	3,3',5,5'-Tetrachloro-4,4'-biphenyldiol	13049-13-3	1	0.2	0
598	2,2',4',6'-Tetrachloro-4-biphenylol	150304-08-8	1	0.1	0
599	2,2',6,6'-Tetrachloro-4-biphenylol	219952-18-8	0	0.3	0
600	2',3',4',5'-Tetrachloro-3-biphenylol	67651-37-0	0	0.1	1
601	2',3',4',5'-Tetrachloro-4-biphenylol	67651-34-7	1	2.4	0
602	2',3,4',6'-Tetrachloro-4-biphenylol	189578-00-5	0	0.06	1
603	2,3,7,8-Tetrachlorodibenzo-p-dioxin	1746-01-6	1	0	1
604	Tetrahydrochrysene	104460-72-2	0	3	1
607	2,2',4,4'-Tetrahydroxybenzil	5394-98-9	1	0.2	1
608	Tetramethylhexestrol	74385-27-6	1	1.5	0
609	Thalidomide	50-35-1	1	0	2
610	Toremifene citrate	89778-27-8	1	1.4	0
611	Tosyl nonylphenol (mixed branched isomers)		0	0.006	2
612	Toxaphene	8001-35-2	1	0.0001	1
613	Triaryl-pyrazole		0	60	2
614	2,2',5-Trichlorobiphenyl	37680-65-2	1	0	0
615	2,4,6-Trichlorobiphenyl	35693-92-6	1	0	0
616	2',4',6'-Trichloro-4-biphenylol	14962-28-8	1	2.9	0
617	3,3',4'-Trichloro-4-biphenylol	124882-64-0	0	0.3	1
618	3,4',5-Trichloro-4-biphenylol	4400-06-0	1	0.1	0
619	2,4,5-Trichlorophenoxyacetic acid	93-76-5	1	0	0
620	Triethylamine, 2-[*p* -[6-methoxy-2-phenyl-3-inden-3-yl)phenoxy] hydrochloride	64-96-0	1	0.0006	1
621	Triethylene glycol dimethacrylate	109-16-0	1	0	0
622	4,2',4'-Trihydroxychalcone	961-29-5	1	0.05	1
623	3,6,4',-Trihydroxyflavone	253195-19-6	0	0.5	0
624	6,7,4'-Trihydroxyisoflavone	17817-31-1	0	0	0
625	7,3',4'-Trihydroxyisoflavone	485-63-2	0	0.005	0
626	1,1,2-Triphenylbut-1-ene	63019-13-6	1	0.01	1
627	Triphenylethylene	58-72-0	1	0.002	2
628	Triphenyl phosphate	115-86-6	1	0	2
629	Tris(4-chlorophenyl)methane	27575-78-6	1	0.1	2
630	Tris(4-chlorophenyl)methanol	30100-80-8	0	0.1	1
631	Vanillin	121-33-5	1	0	2
632	Vinclozolin	50471-44-8	1	0.0002	1
633	α–Zearalanol	26538-44-3	1	30	0
634	β–Zearalanol	42422-68-4	1	8.3	0
635	Zearalanone	5975-78-0	1	2.1	1
636	α-Zearalenol	36455-72-8	1	48	2
637	Zearalenone	17924-92-4	1	17	1
638	β–Zearalenol	71030-11-0	1	8.1	0

[a] the prefix “tox-” is added to the name in the SD file.
[b] Chemical Abstracts Service registry numbers.
[c] “1” indicates that the structure was taken from the ChemIDplus or NCI-3D databases using the given CAS registry numbers. The databases are available at http://sis.nlm.nih.gov/Chem/ChemMain.html.
[d] relative binding affinity values. See below.
[e] training set, test set and unused compounds are indicated as 1, 2, and 0, respectively.

Notes on the ER-tox set

A compilation of binding affinities for 638 substances has been prepared by the National Toxicology Program at the National Institute of Environmental Health Sciences, with the purpose of evaluating the performance of various *in vitro* ER binding assays (the document "Current Status of Test Methods for Detecting Endocrine Disruptors: In Vitro Estrogen Receptor Binding Assays" is available at http://iccvam.niehs.nih.gov/methods/endodocs/final/erbndbrd/erbndall.pdf). It was found that there is reasonable correspondence between binding affinities determined by the various assays. While it is probably not wise to use this mixed-source data for conventional QSAR, it is ideal for classification analysis. From this collection, we selected 616 substances that were non-redundant and consisted of a single chemical compound (i.e. some agrochemicals and PCBs are mixtures of compounds). There are two subtypes of estrogen receptors, ER-α and ER-β. Because of the higher concentration of ER-α in breast and uterine tissues, most assays are considered to give affinities for ER-α. The data have been reported using the relative binding affinity (RBA) scale. RBA values are calculated as the ratio of IC_{50} values of 17β-estradiol and the test substance, multiplied by 100. On this scale, estradiol has an RBA value of 100. Some compounds have been tested by multiple assays; an average RBA value was calculated, excluding assays that use ER-β or ER from the lizard, chicken and rainbow trout. The latter sources were excluded because they are reported for few substances, making the evaluation of their compatibility with other assays more difficult. Values determined with ER from intact MCF-7 cells were only used for 16 compounds not tested with other methods (9 of which have no measurable binding affinity).

The data in table 4.1 are organised in the order given in Appendix C of the NTP document (Information sorted by substance name). The MDL SD file also includes the chemical class given in Appendix C and the sources of ER used in the assays. The abbreviations for ER sources are indicated below:

RUC, rat uterine cytosol; MUC, mouse uterine cytosol; RBC, rabbit uterine cytosol; MCF-7 cytosol, cytosol from human adenocarcinoma MCF-7 cells; MCF-7 cells, intact MCF-7 cells; hERa, semi-purified human ERα protein; hERa-FP, semi-purified human ERα as measured by FP; GST-hERadef; glutathione-S-transferase fusion proteins consisting of the def domains of the human ERα receptor; GST-mERαdef, glutathione-S-transferase fusion proteins consisting of the def domains of the mouse ERα receptor.

Table 5.1.1. Structures (families) of ER ligands from 2 literature compilations (ER-lit set)

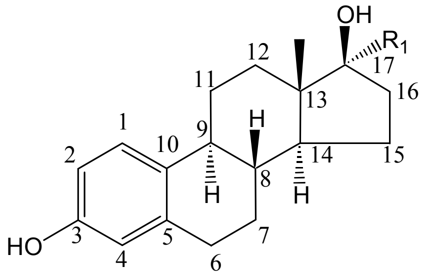
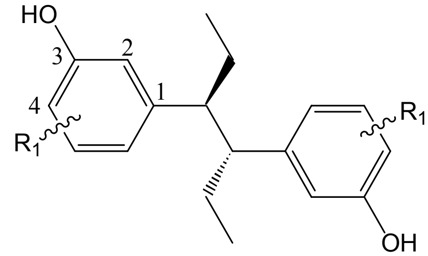
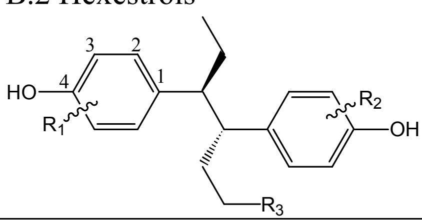
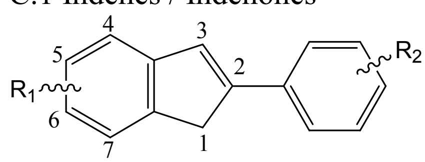
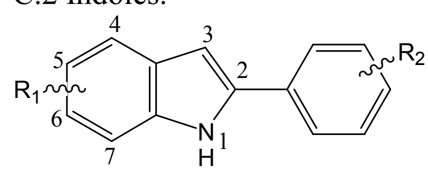
A.1 Estradiols 	A.2 Misc. steroids (see below)	B.1 Metahexestrols 
B.2 Hexestrols 	C.1 Indenes / Indenones 	C.2 Indoles. 

Table 5.1.2. Miscellaneous steroids

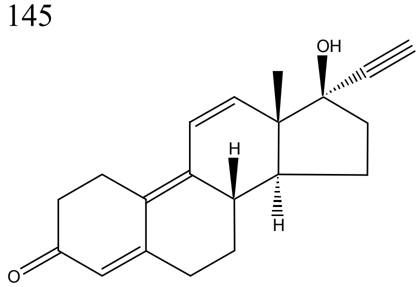
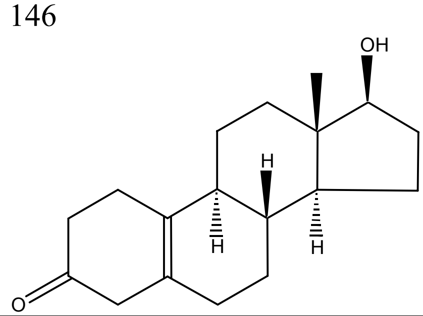
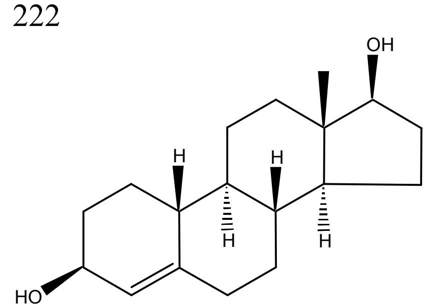
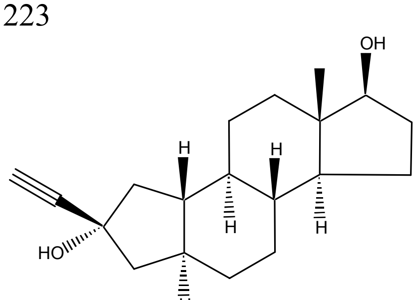
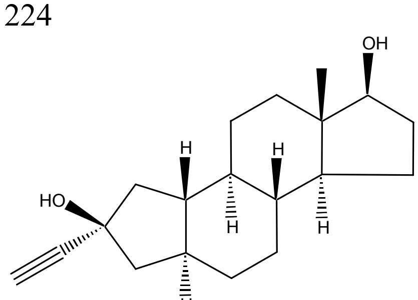
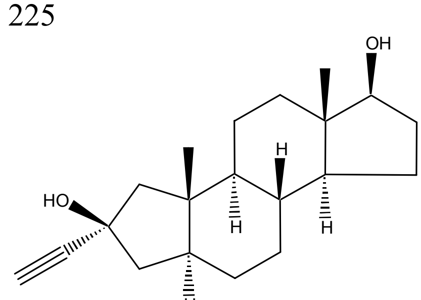
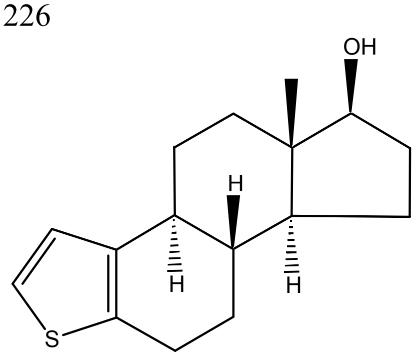
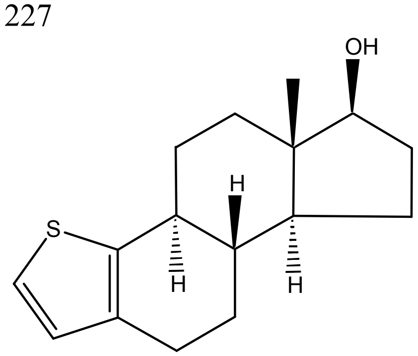
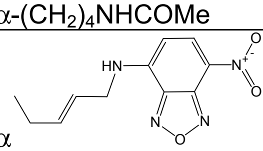
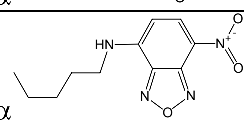
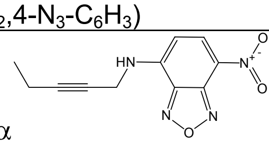
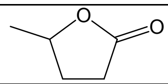
145 	146 	222 
223 	224 	225 
226 	227 	

Table 5.2. ER ligands from 2 literature compilations (ER–lit set)

Name[a]	R1	R2[b]	R3	Family	RBA[c]	set[d]
1	-H	16-α-Br		A.1	105	0
2	-H			A.1	100	0
3	-H	16-α-Cl		A.1	100	0
4	-H	16-α-CH_2Br		A.1	93	0
5	-H	16-α-F		A.1	81	0
6	-H	16-α-I		A.1	79	0
7	-H	16-α-CH_2I		A.1	76	0
8	-H	16-α-CH_2N_3		A.1	65	1
9	-H	16-α-CN		A.1	63	2
10	-H	16-α-CH_2Cl		A.1	55	0
11	-H	16-α-$CH_2C{\equiv}CH$		A.1	44	1
12	-H	16-α-$CH_2CH{=}CH_2$		A.1	38	0
13	-H	16-α-OH		A.1	20	0
14	-H	16-α-$CH_2CH(Me)F$		A.1	7.1	1
15	-H	16-α-$CH_2CH{=}CHCH_2OPh$		A.1	7.1	2
16	-H	16-α-n-Pr		A.1	5	0
17	-H	16-α-CH_2Ph		A.1	5	2
18	-H	16-α-$CH_2C{\equiv}CCH_2NH$-(2-NO_2,4-N_3-C_6H_3)		A.1	3.8	1
19	-H	16-α-CH_2OH		A.1	2.4	1
20	-H	16-α-$CH_2CH_2CH_2CN$		A.1	0.89	2
21	-H	16-α-$(CH_2)_4NHCOMe$		A.1	0.79	1
22	-H	 16-α		A.1	0.3	0
23	-H	 16-α		A.1	0.2	2
24	-H	16-α-$CH_2CH{=}CHCH_2NH$-(2-NO_2,4-N_3-C_6H_3)		A.1	0.2	2
25	-H	 16-α		A.1	0.1	0
26	-$C{\equiv}CH$			A.1	107	0
27	-$C{\equiv}CH$	11-β-nPr		A.1	95	0
28	-$C{\equiv}CH$	11-β-$CH_2CH{=}CH_2$		A.1	93	0
29	-$C{\equiv}CH$	11-β-$CHMe_2$		A.1	83	0
30	-$C{\equiv}CH$	11-β-CH_2Ph		A.1	78	0
31	-$C{\equiv}CH$	11-β-(2-thienyl)		A.1	76	2
32	-$C{\equiv}CH$	11-β-C_6H_4-4-OMe		A.1	62	1
33	-$C{\equiv}CH$	11-β-C_6H_4-2-OMe		A.1	15	0
34	-Me			A.1	65	1
35	-$CH_2C{\equiv}CH$			A.1	49	1
36	-$C{\equiv}CMe$			A.1	40	0
37	-$C{\equiv}CI$			A.1	31	2
38	-$CH_2C{\equiv}CI$			A.1	28	1
39	-$CH_2CH{=}CH_2$			A.1	18	1
40	-Ph			A.1	21	2

41				A.1	8.3	2
42	-2-oxirane			A.1	7.9	1
43	-C≡CC≡CPh			A.1	7.1	0
44	-C≡CCH$_2$OH			A.1	6.5	1
45	-C≡CPh			A.1	5.8	1
46	-n-Pr			A.1	4.9	0
47	-n-Bu			A.1	4.7	1
48	-C≡C(CH$_2$)$_4$I			A.1	4.7	1
49	-CH$_2$Ph			A.1	4.3	1
50	-C≡CCH$_2$O-4-tetrahydropyran			A.1	4	1
51	-C≡C-cyclohexyl			A.1	3.2	0
52	-CH$_2$-2-ozirane			A.1	2.1	2
53	-CH$_2$CH$_2$CHOH			A.1	1.4	0
54	-C≡CC≡CC$_5$H$_{11}$			A.1	2.5	0
55	-CH$_2$C≡CC$_5$H$_{11}$			A.1	1.1	2
56	-CH$_2$C≡CC≡CC$_4$H$_9$			A.1	1.1	0
57	-C≡CC$_6$H$_{13}$			A.1	0.89	0
58	-CH$_2$CH=CHC$_5$H$_{11}$			A.1	0.5	1
59	-CH$_2$C≡CC≡CPh			A.1	0.4	1
60	-CH$_2$OC$_6$H$_{13}$			A.1	0.2	1
61	-CH$_2$CH$_2$CO-4-tetrahydropyran			A.1	0.11	2
62	-CH=CHI(Z)			A.1	127	0
63	-CH=CHBr(Z)			A.1	195	0
64	-CH=CHCl(Z)			A.1	126	0
65	-CH=CHSPh(Z)			A.1	102	1
66	-CH=CHCl(E)			A.1	102	0
67	-CH=CHBr(E)			A.1	78	0
68	-CH=CHI(E)			A.1	59	0
69	-CH=CHSPh(E)			A.1	25	0
70	-H	11-β-Et		A.1	180	1
71	-CH=CHMe(Z)			A.1	100	1
72	-C≡CH	11-β-Et		A.1	87	1
73	-Me	11-β-Et,16-α-OH		A.1	85	0
74	-Me	11-β-Et		A.1	83	0
75	-C≡CH	11-β-Et,16-α-OH		A.1	79	0
76	-C≡CMe	11-β-Et		A.1	66	0
77	-H	11-β-Et,16-α-OH		A.1	28	0
78	-CH=CHMe(Z)	11-β-OMe		A.1	26	1
79	-C≡CH	16-α-OH		A.1	20	1
80	-C≡CMe	11-β-Et,16-α-OH		A.1	18	1
81	-C≡CMe	11-β-OMe		A.1	18	2
82	-Ph	11-β-OMe		A.1	15	0
83	-Me	16-α-OH		A.1	14	1
84	-C≡CH	11-β-OMe		A.1	14	0
85	-Ph	16-α-OH		A.1	7.9	1
86	-H	11-β-OMe		A.1	8.3	0
87	-Me	11-β-OMe		A.1	6	1
88	-C≡CMe	16-α-OH		A.1	5	0
89	-Ph	11-β-OMe,16-α-OH		A.1	5	2
90	-Me	11-β-OMe,16-α-OH		A.1	4	0
91	-H	11-β-OMe,16-α-OH		A.1	1	0
92	-C≡CH	11-β-OMe,16-α-OH		A.1	1	0
93	-C≡CMe	11-β-OMe,16-α-OH		A.1	0.79	0

94	-H	7-α–$(CH_2)_{11}OH$		A.1	2	1
95	-H	7-α–$(CH_2)_{11}F$		A.1	1.5	1
96	-H	7-α–$(CH_2)_{11}Cl$		A.1	0.69	0
97	-H	7-α–$(CH_2)_{11}Br$		A.1	0.2	0
98	-H	7-α–$(CH_2)_{11}I$		A.1	0.1	0
99	-H	7-α–$(CH_2)_{11}OPh$		A.1	0.1	1
100	-C≡CH	4-F		A.1	120	0
101	-$CH=CH_2$	4-F		A.1	74	0
102	-C≡CH	7-α–Me		A.1	72	0
103	-CH=CHI(Z)	4-F		A.1	68	0
104	-C≡CH	2-F		A.1	65	0
105	-H	11-β-Me		A.1	65	0
106	-C≡CH	4-F,7-α–Me		A.1	63	2
107	-CH=CHI(E)	4-F		A.1	56	0
108	-CH=CHI(Z)	2-F		A.1	54	0
109	-C≡CCl			A.1	54	0
110	-CH=C(I)Cl (E)			A.1	49	0
111	-CH=CHI(Z)	7-α–Me		A.1	45	0
112	-CH=CHI (E)	7-α–Me		A.1	44	0
113	-CH=CHI (E)	11-β-Et		A.1	43	1
114	-CH=CHI (Z)	4-F,11-β-OMe		A.1	38	0
115	-C≡CCl	7-α–Me		A.1	38	0
116	-CH=CHI (Z)	4-F,7-α–Me		A.1	37	1
117	-CH=C(I)Cl (E)	7-α–Me		A.1	37	2
118	-CH=CHI (E)	4-F,7-α–Me		A.1	36	0
119	-CH=C(I)Cl (E)	11-β-OMe		A.1	35	0
120	-C≡CH	2-F,7-α–Me		A.1	33	0
121	-C≡CCl	11-β-OEt		A.1	33	1
122	-CH=CHI (Z)	11-β-OMe		A.1	32	0
123	-CH=C(I)Cl (E)	11-β-OEt		A.1	32	0
124	-C≡CCl	11-β-OMe		A.1	30	0
125	-CH=CHI (Z)	2-F,7-α–Me		A.1	29	0
126	-CH=CHI (Z)	2-F,11-β-OMe		A.1	28	0
127	-CH=CHI (E)	11-β-OMe		A.1	28	0
128	-CH=CHI (E)	4-F,11-β-OMe		A.1	28	0
129	-C≡CH	11-β-OEt		A.1	16	0
130	-$CH=CH_2$	2-F		A.1	18	2
131	-CH=CHI (E)	11-β-OEt		A.1	17	0
132	-C≡CH	4-F,11-β-OMe		A.1	17	1
133	-C≡CH	2-F,11-β-OMe		A.1	16	0
134	-C≡CH	11-β-OMe		A.1	15	0
135	-CH=CHI (E)	2-F,11-β-OMe		A.1	13	0
136	-CH=CHI (E)	2-F,7-α–Me		A.1	12	0
137	-H	11-β-OEt		A.1	10	2
138	-CH=CHI (E)	2-F		A.1	6.2	0
139	-H	4-Br		A.1	5	0
140	-H	2-Br		A.1	1.2	0
141	-H	2-I		A.1	0.4	0
142	-H	4-I		A.1	0.1	0
143	-H	3-F		A.1	0.43	0
144	-H	3-OMe		A.1	1.3	0
147	-H	17-β-CH_2OH		A.1	62	0
148	-H	17-β-NH_2		A.1	82	1
149	-H	17-β-NH_3^+		A.1	0.5	0

150	-H	17-β-NHOH		A.1	13	1
151	-H	17-β-NHOMe		A.1	1.7	1
152	-H	17-β-NHCHO		A.1	0	1
153	R_2	17=NNH_2		A.1	81	1
154	R_2	17=NOH		A.1	13	1
155	R_2	17=NOMe		A.1	2.2	2
156	R_2	17=$N^+(Me)O^-$		A.1	0	1
157	R_2	17=NNH_3^+		A.1	0	0
158	R_2	17=C=CH_2		A.1	0	1
159	R_2	17=CH_2		A.1	63	0
160	-H	17-β-C≡CH		A.1	0	2
161	-H	17-β-Et		A.1	6.3	0
162	-H	17-β-H		A.1	30	0
163	-H	3-H		A.1	4.9	0
164	-H	17-β-OMe		A.1	11	1
165	-H	17-β-OC(O)Me		A.1	12	0
166	-OH	17-β-H		A.1	27	0
167	-H	16-α-OH,17-β-H		A.1	54	0
168	R_2	17=O		A.1	29	0
170	-H	16-β-OH		A.1	42	0
171	R_2	16-α-OH,17=O		A.1	7	0
172	-H	15-α-OH,16-α-OH		A.1	0.5	0
173	-C≡CCH_2CH_2OH			A.1	3.9	0
174	-CH=CHPh (Z)			A.1	48	2
175	-CH=CHPh (E)			A.1	15	0
176	-C≡CCH_2OH			A.1	3.6	0
177	-C≡CCH_2Cl			A.1	20	1
178	-CH=CH_2			A.1	63	0
179	-CO_2Me			A.1	22	2
180	-COMe			A.1	36	1
181	-C≡C-t-Bu			A.1	1.5	0
182	-CH=CH-t-Bu (Z)			A.1	3.2	1
183	-C≡CCH_2F			A.1	84	0
184	-Et			A.1	19	0
185	-CH_2I			A.1	9	2
186	-CH_2Br			A.1	38	0
187	-CH_2CN			A.1	98	2
188	-CH_2N_3			A.1	93	2
189	-CH_2OH			A.1	38	1
190	-CH_2OMe			A.1	31	1
191	-CH_2SCN			A.1	0	2
192	-H	1-Me		A.1	15	0
193	-H	1-CH_2CH_2-11		A.1	22	0
194	-H	2-Me		A.1	41	0
195	-H	2-F		A.1	76	0
196	-H	2-Cl		A.1	10	0
197	-H	2-Br		A.1	9.3	0
198	-H	2-I		A.1	0	0
199	-H	2-N_3		A.1	2.8	2
200	-H	2-NO_2		A.1	0.52	0
201	-H	2-NH_2		A.1	10	0
202	-H	2-OEt		A.1	0	0
203	-H	2-NHEt		A.1	0.35	2
204	-H	2-SEt		A.1	0.29	1
205	-H	2-CH=CHMe (E)		A.1	0	2
206	-H	4-Me		A.1	10	0

207	-H	4-F		A.1	155	0
208	-H	4-Cl		A.1	75	0
209	-H	4-Br		A.1	21	0
210	-H	4-I		A.1	0	0
211	-H	4-N_3		A.1	1	0
212	-H	4-NO_2		A.1	9.6	0
213	-H	4-NH_2		A.1	8.2	0
214	-H	1-OH		A.1	19	0
215	-H	2-OH		A.1	23	0
216	-H	4-OH		A.1	42	0
217	-H	1-OH,3-H		A.1	0.2	0
218	-H	2-OH,3-H		A.1	18	0
219	-H	3-H,4-OH		A.1	0.7	0
220	-H	2-OMe		A.1	0.08	0
221	-H	4-OMe		A.1	1.3	1
228	-H	11-β-Me		A.1	95	0
229	-H	11-β-CH_2OMe		A.1	37	1
230	-H	11-β-OH		A.1	4.7	0
231	-H	11-β-CH_2CH_2CN		A.1	18	1
232	-H	11-β-$CH{=}CH_2$		A.1	121	0
233	-H	11-β-$(CH_2)_3CN$		A.1	31	0
234	-H	11-β-C≡CMe		A.1	25	0
235	-H	11-β-C≡CH		A.1	51	0
236	-H	11-β-CH=CHI (Z)		A.1	63	2
237	-H	11-β-CH=CHI (E)		A.1	54	0
238	-H	11-β-CH_2CH_2F		A.1	77	0
239	-H	11-β-$(CH_2)_3F$		A.1	57	1
240	-H	11-β-$(CH_2)_4F$		A.1	112	0
241	-H	11-β-CH_2CH_2OH		A.1	13	0
242	-H	11-β-CH_2Cl		A.1	110	0
243	-H	16-β-I		A.1	57	0
244	-H	16-β-F		A.1	38	0
245	-H	7-α–Me		A.1	103	0
246	-H	7-α–OH		A.1	0.8	0
247	-H	7-α–$(CH_2)_3CH{=}CH_2$		A.1	132	1
248	-H	7-α–$(CH_2)_5OH$		A.1	36	0
249	-H	7-α–$(CH_2)_5OS(Me)O_2$		A.1	74	1
250	-H	7-α–$(CH_2)_5F$		A.1	63	0
251	-H	6=O		A.1	21	0
252	-H	6-β-OH		A.1	0.78	0
253	-H	6-β-NH_2		A.1	0.02	2
254	-H	6-β-Me		A.1	3.6	0
255	-H	6-α-Me		A.1	31	0
256	-H	18-β-Et		A.1	59	1
257	-H	18-β-n-Pr		A.1	40	0
258	-H	18-β-n-Bu		A.1	13	2
259	-H	15-α-Me		A.1	26	1
260	-H	15-β-Me		A.1	29	0
261	-H	9-α-Me		A.1	37	0
262	-C≡CH	12-β-Me		A.1	111	1
263	-H	14-α-CH_2-15α		A.1	100	0
264	-H	14-α-O-15α		A.1	41	0
265	-H	14-β-CH_2-15β		A.1	95	0
266	-H	14-β-O-15β		A.1	5	0
145				A.2	0	1

146				A.2	0	0
222				A.2	5	2
223				A.2	0.1	2
224				A.2	12	0
225				A.2	6	2
226				A.2	2.4	2
227				A.2	0.8	0
267				B.1	10	0
268	6-Cl			B.1	4.2	1
269	6-F			B.1	6.9	0
270	4-F			B.1	1.1	1
271	4-Me			B.1	0.72	0
272	4-NH_2			B.1	0.52	1
273	4-Cl			B.1	0.07	0
274	4-Et			B.1	0.04	1
275	4-Br			B.1	0.03	0
276	4-CH_2OMe			B.1	0.03	1
277	4-CH_2OH			B.1	0.03	0
278	5-Cl			B.1	0.03	0
279	5-OH			B.1	0.01	0
280	4-NO_2			B.1	0.02	2
281	2-OH	2-OH	-H	B.2	32	1
282			-H	B.2	165	0
283	3-OH	3-OH	-H	B.2	20	0
284	3-F	3-F	-H	B.2	16	0
285	2-Me	2-Me	-H	B.2	8.5	0
286	2-F	2-F	-H	B.2	6.5	0
287	3-Me	3-Me	-H	B.2	8.1	0
288	3-NH_2	3-NH_2	-H	B.2	5.6	0
289	2-Et	2-Et	-H	B.2	3.7	0
290	2-Cl	2-Cl	-H	B.2	2.2	0
291	2-Br	2-Br	-H	B.2	1.8	0
292	3-Cl	3-Cl	-H	B.2	1.6	0
293	3-Et	3-Et	-H	B.2	1.2	0
294	3-CH_2OH	3-CH_2OH	-H	B.2	0.71	0
295	3-CH_2OMe	3-CH_2OMe	-H	B.2	0.55	0
296	3-Br	3-Br	-H	B.2	0.25	0
297	3-CH_2OMe	3-CH_2OMe	-H	B.2	0.18	2
298	3-I	3-I	-H	B.2	0.17	0
299	3-NO_2	3-NO_2	-H	B.2	0.14	0
300	3-CH_2NMe_2	3-CH_2NMe_2	-H	B.2	0.04	2
302		2-F,4-OH	-H	B.2	204	0
303		4-OH	-F	B.2	129	0
304		4-OH	-Cl	B.2	112	0
305		4-NH_2	-H	B.2	95	1
306		4-OH	-Br	B.2	71	0
307	3-F	2-F,4-OH	-H	B.2	65	2
308		4-OH	-I	B.2	60	0
309		4-CH_2OH	-H	B.2	40	0
310		4-OCH_2CH_2OH	-H	B.2	21	0
311		4-OMe	-H	B.2	19	0
312		2-Br,4-OH	-H	B.2	16	0
313		2-I,4-OH	-H	B.2	14	0
314		4-NO_2	-H	B.2	11	1
315		4-CH_2Br	-H	B.2	9.3	1
316		4-CH_2I	-H	B.2	7.9	0
317		4-$OCH_2CH(OH)CH_2OH$	-H	B.2	6.9	2

318		4-N_3	-H	B.2	5.8	2
319		4-$OCH_2CH=CH_2$	-H	B.2	5.4	1
320		4-CN	-H	B.2	5.2	1
321		4-CH_2OEt	-H	B.2	4.9	1
322		4-$OCH_2CH(OH)Me$	-H	B.2	4.3	0
323		4-COMe	-H	B.2	4.1	2
324		4-$OCH_2CH(OH)CH_2Cl$	-H	B.2	4	1
325	3-I	2-I,4-OH	-H	B.2	3.2	0
326	3-Br	2-Br,4-OH	-H	B.2	2.6	0
327		4-SO_2N_3	-H	B.2	1.8	1
328		4-CO_2H	-H	B.2	1.8	0
329		4-$COCH_2Br$	-H	B.2	1.1	1
330		4-$OCH_2CH_2OCH_2CH_2Cl$	-H	B.2	0.89	1
331		4-$COCH_2Cl$	-H	B.2	0.44	0
332		4-SO_2F	-H	B.2	0.33	1
333	6-OH,3-C_6H_4-4-OH	2-CF_3		C.1	38	1
334	6-OH,3-C_6H_4-4-OH	2-Me		C.1	30	0
335	6-OH,3-Ph	4-OH		C.1	28	0
336	6-OH,3-C_6H_4-4-OH	4-OH		C.1	15	0
337	6-OH,1-Me,3-Ph			C.1	12	0
338	6-OH,3-Ph			C.1	8.9	0
339	6-OH,3-C_6H_4-4-OH			C.1	4.7	0
340	6-OH,3-C_6H_4-4-OH	4-CN		C.1	1	1
341	6-OH,3-C_6H_4-4-OH	4-Br		C.1	0.76	1
342	6-OH,3-C_6H_4-4-OH	4-NO_2		C.1	0.69	2
343	6-OH,3-Ph	4-NO_2		C.1	0.46	0
344	6-OH,3-C_6H_4-4-OH	3-NO_2		C.1	0.6	0
345	6-OH,1-Me,3-Et	4-OH		C.1	81	0
346	6-OH,1-Et,3-Et	4-OH		C.1	78	0
347	6-OH,1=O,3-Ph			C.1	78	0
348	6-OH,1-Me,3-Ph			C.1	12	0
349	6-OH,3-Et	4-OH		C.1	16	0
350	1-Et,3-Et	4-OH		C.1	9.3	0
351	6-OH,3-Ph			C.1	8.9	0
352	1=O,3-Et	4-OH		C.1	4.6	0
353	3-Et	4-OH		C.1	2.3	0
354	6-OH,1-Et,3-Et			C.1	2.2	0
355	6-OH,1=O,3-Et			C.1	1.2	0
356	6-OH,3-Et			C.1	0.58	0
357	1=O,3-Ph	4-OH		C.1	0.45	0
358	3-Ph	4-OH		C.1	0.36	0
359	3-C_6H_4-4-OH			C.1	0.02	0
360	1=O,3-Ph			C.1	0.01	0
361	6-OH,1-Et,3-Me	4-OH		C.2	33	0
362	5-OH,1-Et,3-Et	4-OH		C.2	23	1
363	6-OH,1-Et,3-Et	4-OH		C.2	21	0
364	6-OH,1-n-Pr,3-Et	4-OH		C.2	19	0
365	5-OH,1-n-Pr	4-OH		C.2	18	0
366	6-OH,1-Et	4-OH		C.2	16	0
367	5-OH,1-n-Pr,3-Me	4-OH		C.2	16	0
368	6-OH,1-n-Pr,3-Me	4-OH		C.2	13	1
369	6-OH,1-i-Pr,3-Me	4-OH		C.2	13	0
370	6-OH,1-Me,3-Et	4-OH		C.2	5.9	0
371	5-OH,1-Et,3-Me	4-OH		C.2	9.5	0
372	6-OH,1-n-Pr	4-OH		C.2	8.5	0
373	5-OH,1-n-Pr,3-Me	3-OH		C.2	7.4	0
374	6-OH,1-Me,3-Me	4-OH		C.2	10	2

375	5-OH,1-Et	4-OH		C.2	5.8	1
376	5-OH,1-Me,3-Me	4-OH		C.2	4.6	0
377	5-OH,1-n-Bu,3-Me	4-OH		C.2	4.6	0
378	6-OH,1-n-Bu	4-OH		C.2	4.3	2
379	6-OH,1-Me	4-OH		C.2	3.8	1
380	5-OH,1-i-Pr,3-Me	4-OH		C.2	3.5	0
381	6-OH,1-n-Pr,3-Me	3-OH		C.2	3.5	0
382	6-OH,1-Et,3-Me	3-OH		C.2	3	1
383	5-OH,1-C_5H_{11},3-Me	4-OH		C.2	2.3	1
384	5-OH,1-Et,3-Me	3-OH		C.2	2.2	0
385	5-OH,1-n-Pr,3-n-Pr	4-OH		C.2	1.7	0
386	6-OH,1-Et	3-OH		C.2	1.7	0
387	5-OH,1-Et	3-OH		C.2	1.7	2
388	5-OH,1-Me,3-Me	3-OH		C.2	0.6	0
389	6-OH,1-Me,3-Me	3-OH		C.2	0.55	0
390	6-OH,3-Et	4-OH		C.2	0.13	1
391	6-OH,3-Me	4-OH		C.2	0.06	0
392	5-OH,3-Me	4-OH		C.2	0.06	1
393	7-OH,1-Et,3-Me	4-OH		C.2	0.02	1
394	6-OH	4-OH		C.2	0.01	2
395	5-OH	4-OH		C.2	0.01	0

[a] the prefix “lit-” is added to the name in the SD file.

[b] compounds in family A.1 always have an –OH group at ring position 3, and a β-OH group at ring position 17, except where indicated otherwise. Compounds in family B.2 always have an –OH group at position 4 on the second phenyl ring, except where indicated otherwise. α-groups lie below the plane, β-groups lie above the plane in table 5.1.1.

[c] RBA values were determined between 0°C and 4°C; most used ER from the uterine cytosol of rats, mice, lambs, cows and rabbits. Where multiple RBA values were available, the value given is the average. The ER-lit set is probably suitable for conventional QSAR, as the assays are compatible and many values have been reported by the group of J. Katzenellenbogen. Their compatibility is supported by the analysis at the NTP, described with the ER-tox set. The SD file contains the individual values for each compounds, and the source of the ER (abbreviations used for the source of ER: RUC, rat uterine cytosol; MUC, mouse uterine cytosol; RBC, rabbit uterine cytosol; CUC, calf uterine cytosol; LUC, lamb uterine cytosol; MCF-7, cytosol from human adenocarcinoma MCF-7 cells; MMEC, mouse mammary epithelial cytosol).

[d] training set, test set and unused compounds are indicated as 1, 2, and 0, respectively. Note that there are duplicates between the ER-tox and ER-lit sets. The duplication is removed by the coverage-design algorithm.

ER-lit references

All compounds except 143-266 are from reference 1; compounds 143-266 are from reference 2.

(1) Gao, H.; Katzenellenbogen, J. A.; Garg, R.; Hansch, C. Comparative QSAR analysis of estrogen receptor ligands. *Chem. Rev.* **1999**, *99*, 723-744.

(2) Anstead, G. M.; Carlson, K. E.; Katzenellenbogen, J. A. The estradiol pharmacophore: Ligand structure-estrogen receptor binding affinity relationships and a model for the receptor binding site. *Steroids* **1997**, *62*, 268-303.

Table 6.1. Definition of MAO training and test sets

training set[a]

93, 94, 194, 195, 196, 245, 260, 264, 265, 273, 274, 275, 326, 345, 522, 614, 616, 617, 636, 727, 807, 898, 904, 955, 956, 1012, 1127, 1140, 1224, 1273, 1279, 1293, 1329, 1348, 1469, 1553, 1557, 1649, 1713, 1725, 1741, 1752, 1868, 1936, 1945, 1951, 2071, 2185, 2297, 2311, 2359, 2363, 2527, 2545, 2549, 2649, 2657, 2715, 2717, 2735, 2786, 2805, 2835, 2883, 2884, 2885, 2886, 2887, 2978, 3079, 3109, 3112, 3144, 3145, 3155, 3170, 3174, 3189, 3199, 3282, 3377, 3405, 3417, 3454, 3455, 3457, 3486, 3491, 3492, 3494, 3497, 3500, 3501, 3529, 3540, 3541, 3603, 3674, 3696, 3697, 3731, 3734, 3737, 3763, 3772, 3778, 3784, 3831, 3842, 3858, 3865, 3888, 3902, 3925, 3926, 3928, 3930, 3933, 3934, 3935, 3966, 3977, 3979, 3981, 3984, 4020, 4021, 4030, 4032, 4041, 4046, 4054, 4060, 4067, 4120, 4120, 4154, 4165, 4204, 4223, 4234, 4240, 4250, 4263, 4364, 4466, 4610, 4664, 4667, 4675, 4676, 4677, 4678, 4680, 4684, 4686, 4687, 4691, 4694, 4695, 4696, 4698, 4700, 4702, 4704, 4706, 4711, 4840, 4848, 4849, 4862, 4883, 4938, 4950, 4951, 4963, 5034, 5059, 5116, 5130, 5135, 5154, 5175, 5176, 5177, 5187, 5195, 5215, 5216, 5217, 5373, 5417, 5426, 5461, 5508, 5575, 5586, 5620, 5753, 5797, 5815, 5816, 5826, 5859, 5868, 5880, 5925, 5931, 5994, 6056, 6076, 6089, 6090, 6097, 6138, 6159, 6217, 6282, 6326, 6366, 6386, 6386, 6391, 6409, 6422, 6454, 6481, 6485, 6497, 6534, 6535, 6626, 6628, 6631, 6651, 6660, 6664, 6769, 6808, 6900, 7012, 7013, 7046, 7083, 7085, 7086, 7088, 7089, 7119, 7132, 7133, 7166, 7171, 7173, 7223, 7241, 7281, 7311, 7320, 7369, 7375, 7376, 7385, 7427, 7460, 7566, 7680, 7729, 7761, 7791, 7862, 8101, 8222, 8224, 8281, 8310, 8378, 8459, 8485, 8490, 8491, 8497, 8515, 8597, 8620, 8645, 8667, 8704, 8706, 8752, 8758, 8809, 8987, 9091, 9116, 9138, 9143, 9145, 9153, 9213, 9221, 9258, 9311, 9320, 9366, 9459, 9554, 9564, 9576, 9629, 9665, 9668, 9720, 9722, 9748, 9750, 9756, 9781, 9794, 9795, 9837, 9838, 9881, 9884, 9936, 9946, 9996, 10043, 10057, 10058, 10065, 10068, 10093, 10100, 10102, 10136, 10141, 10248, 10309, 10405, 10460, 10541, 10555, 10556, 10566, 10605, 10655, 10657, 10676, 10703, 10707, 10712, 10715, 10722, 10726, 10744, 10754, 10755, 10757, 10758, 10764, 10768, 10769, 10772, 10815, 10816, 10849, 10855, 10859, 10876, 10883, 10906, 10915, 10924, 10972, 11012, 11052, 11085, 11097, 11104, 11114, 11227, 11269, 11381, 11397, 11423, 11439, 11511, 11542, 11588, 11589, 11593, 11716, 11717, 11719, 11720, 11721, 11786, 11789, 11825, 11845, 11878, 11887, 12069, 12102, 12103, 12106, 12151, 12222, 12363, 12429, 12508, 12562, 12605, 12606, 12608, 12652, 12657, 12671, 12672, 12697, 12718, 12726, 12779, 12891, 12894, 12919, 12956, 13002, 13020, 13064, 13094, 13176, 13179, 13186, 13189, 13196, 13197, 13198, 13201, 13202, 13205, 13206, 13214, 13221, 13222, 13234, 13340, 13341, 13361, 13371, 13427, 13486, 13613, 13616, 13623, 13663, 13668, 13672, 13699, 13704, 13706, 13710, 13737, 13742, 13786, 13792, 13794, 13810, 13858, 13931, 13932, 13943, 13944, 13950, 13954, 14003, 14027, 14029, 14030, 14031, 14113, 14117, 14222, 14246, 14255, 14270, 14329, 14339, 14357, 14449, 14450, 14456, 14525, 14540, 14544, 14545, 14640, 14652, 14697, 14938, 14959, 14969, 15028, 15048, 15067, 15106, 15112, 15129, 15214, 15441, 15485, 15551, 15556, 15659, 15663, 15741, 15772, 15776, 15802, 15804, 15825, 15832, 15874, 15883, 15884, 15970, 16084, 16110, 16112, 16198, 16391, 16410, 16412, 16432, 16502, 16602, 16603, 16604, 16607, 16612, 16616, 16622, 16623, 16670, 16690, 16713, 16786, 16860, 16865, 16868, 16908, 16924, 16971, 16981, 16997, 17001, 17043, 17109, 17224, 17260, 17261, 17271, 17339, 17365, 17382, 17391, 17392, 17394, 17399, 17400, 17405, 17406, 17414, 17415, 17446, 17498, 17534, 17535, 17545, 17550, 17551, 17564, 17580, 17604, 17627, 17628, 17660, 17665, 17676, 17678, 17679, 17680, 17710, 17715, 17718, 17862, 17869, 17875, 17954, 17981, 18106, 18107, 18121, 18260, 18262, 18264, 18274, 18285, 18286, 18294, 18295, 18298, 18310, 18391, 18472, 18498, 18566, 18567, 18580, 18601, 18611, 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test set[a]

89, 92, 191, 224, 225, 334, 357, 447, 470, 615, 635, 808, 841, 883, 886, 899, 954, 1020, 1039, 1044, 1082, 1105, 1190, 1294, 1327, 1551, 1559, 1703, 1726, 1746, 1782, 1934, 2118, 2195, 2236, 2349, 2419, 2478, 2487, 2539, 2617, 2678, 2701, 2712, 2882, 2899, 2903, 3247, 3256, 3322, 3343, 3413, 3451, 3493, 3499, 3539, 3653, 3662, 3708, 3726, 3780, 3795, 3812, 3873, 3914, 3929, 3938, 3962, 3982, 3983, 4016, 4026, 4036, 4100, 4169, 4219, 4235, 4579, 4615, 4629, 4679, 4688, 4690, 4692, 4693, 4697, 4709, 4717, 4719, 4720, 4777, 4860, 4872, 4909, 4923, 5035, 5085, 5104, 5174, 5179, 5198, 5260, 5312, 5361, 5425, 5471, 5478, 5538, 5569, 5604, 5688, 5689, 5705, 5724, 5795, 5809, 5837, 5850, 5870, 5871, 5872, 5954, 6035, 6049, 6070, 6189, 6207, 6216, 6239, 6276, 6357, 6382, 6435, 6449, 6452, 6453, 6537, 6617, 6743, 6756, 6952, 6993, 7023, 7036, 7092, 7142, 7158, 7165, 7170, 7179, 7266, 7312, 7373, 7395, 7421, 7423, 7461, 7538, 7647, 7657, 7679, 7727, 7737, 7789, 7807, 7859, 7860, 7929, 8064, 8127, 8214, 8259, 8260, 8357, 8445, 8514, 8552, 8567, 8649, 8672, 8679, 8701, 8713, 8768, 8769, 8846, 8864, 8865, 8874, 8997, 8998, 9016, 9022, 9039, 9059, 9090, 9095, 9114, 9137, 9144, 9197, 9211, 9291, 9340, 9368, 9403, 9442, 9587, 9663, 9699, 9746, 9853, 9878, 10027, 10028, 10056, 10073, 10105, 10109, 10123, 10134, 10137, 10145, 10151, 10152, 10163, 10186, 10205, 10219, 10297, 10298, 10368, 10370, 10474, 10480, 10481, 10488, 10489, 10560, 10611, 10613, 10624, 10654, 10672, 10688, 10723, 10749, 10751, 10756, 10838, 10845, 10846, 10847, 10877, 10985, 11015, 11025, 11044, 11050, 11051, 11201, 11212, 11284, 11396, 11466, 11470, 11477, 11497, 11631, 11791, 11817, 11818, 11973, 11997, 12003, 12027, 12100, 12108, 12110, 12153, 12216, 12272, 12371, 12410, 12428, 12653, 12676, 12694, 12719, 12747, 12794, 12852, 12902, 12960, 13010, 13066, 13140, 13141, 13203, 13204, 13213, 13216, 13263, 13354, 13412, 13469, 13551, 13597, 13601, 13622, 13649, 13651, 13667, 13670, 13718, 13720, 13736, 13813, 13829, 13833, 13864, 13882, 13906, 13909, 13911, 13913, 13969, 13974, 13979, 13981, 14084, 14104, 14191, 14219, 14261, 14294, 14318, 14336, 14447, 14448, 14500, 14559, 14824, 14859, 14907, 14960, 15030, 15043, 15108, 15113, 15114, 15220, 15238, 15269, 15415, 15430, 15439, 15523, 15553, 15651, 15683, 15809, 15810, 15971, 16106, 16218, 16294, 16431, 16433, 16469, 16546, 16547, 16549, 16569, 16598, 16632, 16722, 16861, 16910, 16967, 16992, 16994, 17000, 17116, 17119, 17145, 17258, 17259, 17262, 17333, 17381, 17395, 17398, 17408, 17423, 17521, 17555, 17556, 17581, 17582, 17641, 17657, 17674, 17677, 17687, 17692, 17709, 17712, 17767, 17787, 17793, 17844, 17845, 17855, 17968, 18045, 18118, 18139, 18205, 18267, 18403, 18495, 18561, 18581, 18599, 18610, 18612, 18615, 18618, 18620, 18627, 18628, 18629, 18630, 18633, 18634, 18638, 18660, 18663, 18725, 18728, 18729, 18736, 18745, 18749, 18785, 18802, 18816, 18857, 18874, 18878, 18891, 18898, 18915, 18927, 18942, 18966, 18978, 18988, 19011, 19024, 19025, 19050, 19053, 19099, 19112, 19130, 19133, 19135, 19136, 19141, 19148, 19154, 19220, 19312, 19498, 19500, 19512, 19533, 19606, 19657, 19672, 19756, 19757, 19877, 19884, 19896, 19928, 19944, 19957, 19985, 19991, 20016, 20046, 20059, 20068, 20098, 20100, 20105, 20190, 20191, 20201, 20203, 20204, 20253, 20291, 20297, 20298, 20302, 20307, 20317, 20322, 20324, 20352, 20360, 20362, 20388, 20422, 20446, 20456, 20540, 20548, 20560, 20567, 20568, 20580, 20619, 20620, 20637, 20652, 20654, 20669, 20671, 20673, 20690, 20693, 20708, 20830, 20943, 20945, 21011, 21039, 21251, 21275, 21288, 21360, 21454, 21457, 21598, 21736

[a] The compound names are those used in the RP demo distributed with Cerius2 (Cerius2-Resources/COMBICHEM/demos/rp/mao.tbl). While we used only the pre-computed descriptors and ISIS fingerprints in mao.tbl, the structures are available from Y.C. Martin at Abbott Laboratories.

B. Optimal-complexity classification models

For SIMCA, the “modelling power” (M) and “discriminating power” (D) output from Sybyl is indicated below each descriptor.

For RP, the notation *x*: *y* (*z*) indicates that terminal node *x* with activity designation *y* contains *z* compounds.

For SFGA, the terms *a* * <desc – *b*> are splines, described in the article.

COX-2-SIMCA

	S_sCH3	S_ssCH2	S_aaCH	S_dssC	S_aasC	S_ssssC	S_aaN	S_aasN
M	0.729	0.820	0.823	0.789	0.869	0.880	0.447	0.466
D	1.639	5.873	1.717	4.642	2.012	1.676	2.359	2.818

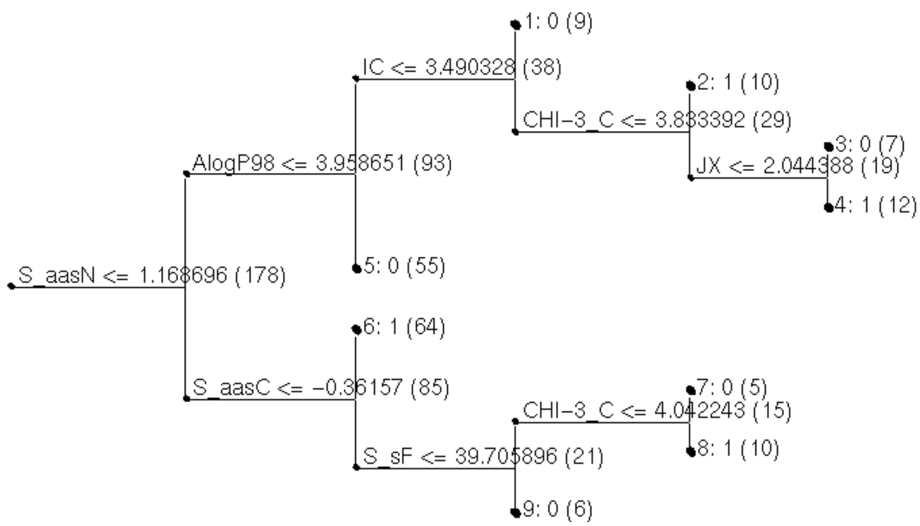
	S_dO	S_ssO	S_ddssS	S_sF	S_sCl	IC	SIC	IAC-Total
M	0.783	0.443	0.879	0.913	0.146	0.871	0.819	0.940
D	5.299	1.456	9.728	1.563	1.179	1.967	2.023	2.158

	MW	Rotlbonds	Hbond_acceptor	Hbond_donor	AlogP98	MolRef	JX
M	0.936	0.749	0.899	0.719	0.803	0.966	0.790
D	1.486	1.732	1.547	1.820	3.870	1.195	2.955

	Kappa-1-AM	Kappa-2-AM	Kappa-3-AM	CHI-1	CHI-2	CHI-3_P	CHI-3_C
M	0.978	0.941	0.843	0.947	0.960	0.897	0.938
D	2.224	2.282	1.342	1.613	1.551	1.366	1.640

	CHI-V-0	CHI-V-1	CHI-V-2	CHI-V-3_P	CHI-V-3_C
M	0.982	0.956	0.956	0.901	0.859
D	1.310	2.223	3.447	2.623	2.381

COX-2-RP



COX-2-SFGA

0.604221 + 0.169408 * <5 - Rotlbonds> - 0.079704 * <S_aaN - 3.73777> - 24.8116 * <S_ddssS + 3.69987> + 0.400231 * <0.969153 - S_aasN> - 0.171235 * <5.2043 - AlogP98> + 24.6254 * <S_ddssS + 3.6905>

BZR -SIMCA

	S_sCH3	S_ssCH2	S_aaCH	S_sssCH	S_dssC	S_aasC	S_aaaC	S_ssNH
M	0.284	0.502	0.751	0.163	0.603	0.793	0.885	0.645
D	1.390	1.593	2.744	1.104	1.615	2.642	2.966	2.403

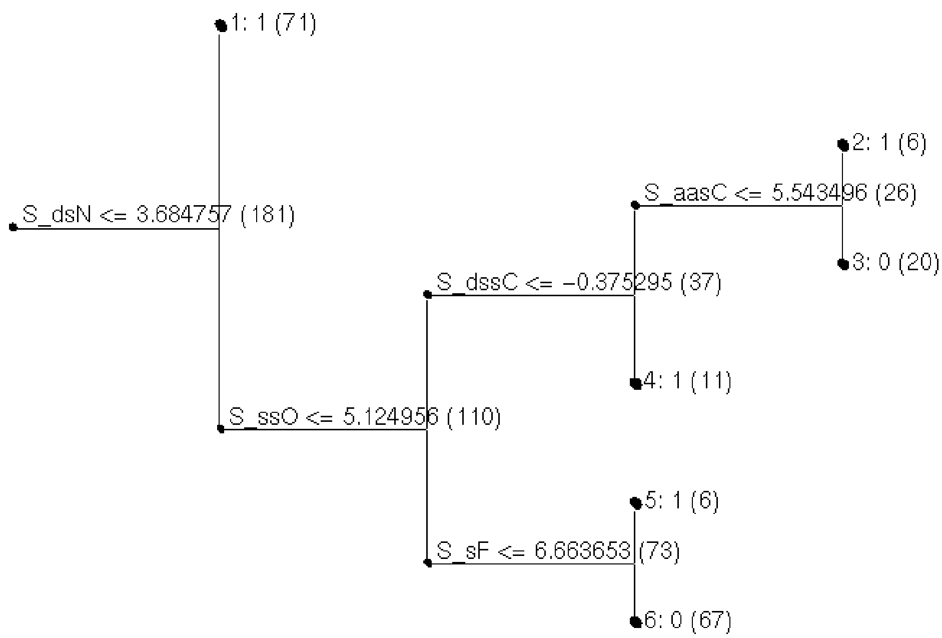
	S_aaNH	S_dsN	S_aaN	S_sssN	S_aasN	S_sOH	S_dO	S_ssO
M	0.833	0.711	0.722	0.542	0.433	0.246	0.595	0.496
D	2.406	2.480	2.010	1.670	1.399	1.336	1.413	1.315

	S_sF	S_sCl	IC	SIC	MW	Rotlbonds	Hbond_acceptor
M	0.751	0.576	0.815	0.753	0.961	0.818	0.811
D	3.705	1.747	2.739	1.641	2.487	2.425	2.051

	Hbond_donor	AlogP98	MolRef	SC-3_C	CHI-3_P	CHI-3_C	CHI-V-2
M	0.575	0.729	0.946	0.953	0.897	0.881	0.961
D	2.037	1.696	2.609	3.165	2.082	2.224	2.783

	CHI-V-3_P	CHI-V-3_C	Kappa-2	Kappa-3	JX	Wiener
M	0.887	0.715	0.946	0.892	0.811	0.938
D	1.872	1.372	2.249	1.875	1.462	1.519

BZR-RP



BZR-SFGA

0.557937 + 0.042256 * <21.6702 - S_aaCH> + 4.63969 * <4.58747 - S_dsN> - 0.08908 * <4.37304 - S_aaN> - 4.80036 * <4.5095 - S_dsN> + 1.58626 * <S_aaN - 9.72321> - 0.305735 * <3.287 - AlogP98>

DHFR-SIMCA

	S_sCH3	S_ssCH2	S_aaCH	S_dssC	S_aasC	S_aaaC	S_sNH2	S_ssNH
M	0.909	0.859	0.964	0.869	0.890	0.827	0.892	0.887
D	1.628	1.742	1.777	1.622	1.444	1.431	2.288	1.417

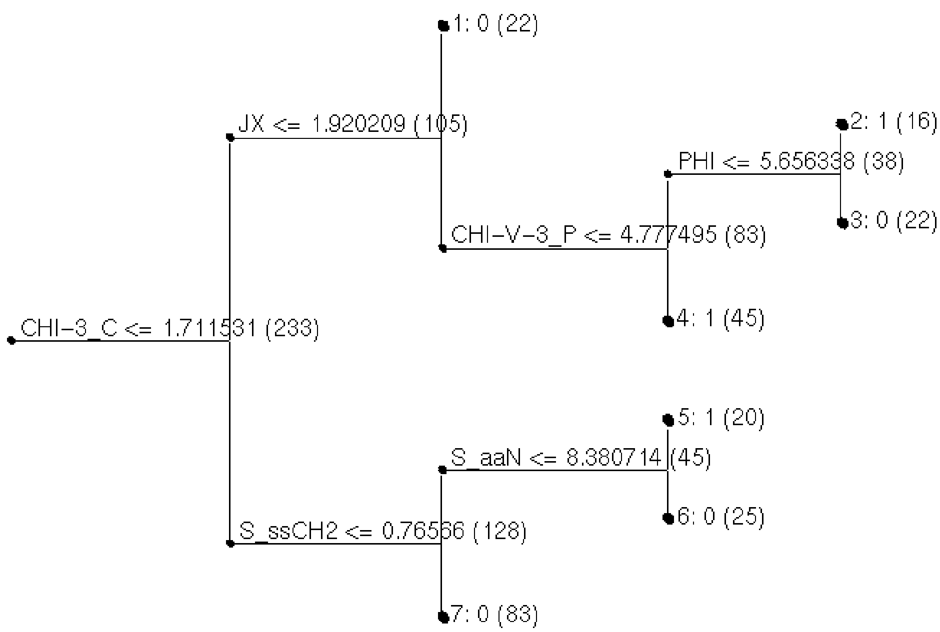
	S_aaN	S_sssN	S_dO	S_ssO	S_sCl	IC	SIC	IAC-Total
M	0.836	0.858	0.831	0.886	0.951	0.964	0.955	0.991
D	1.474	2.187	1.253	1.753	1.445	1.587	1.905	1.760

	MW	Rotlbonds	Hbond_acceptor	Hbond_donor	AlogP98	JX	PHI
M	0.985	0.941	0.950	0.962	0.847	0.867	0.970
D	1.327	1.630	1.749	1.672	2.065	1.437	2.065

	SC-3_C	CHI-1	CHI-2	CHI-3_P	CHI-3_C	CHI-V-1	CHI-V-2	CHI-V-3_P
M	0.962	0.990	0.987	0.963	0.955	0.980	0.978	0.947
D	1.519	2.392	1.970	1.951	1.646	1.783	2.449	2.504

	CHI-V-3_C	Wiener
M	0.915	0.960
D	1.935	1.611

DHFR-RP



DHFR-SFGA

1.67863 - 0.053301 * <12.2737 - S_aaN> - 0.151926 * <Hbond_acceptor - 3> + 0.126429 * <Rotlbonds - 4> - 0.346226 * <CHI-V-3_P - 4.1683> - 0.204482 * <2.7169 - AlogP98> - 0.512405 * <3.07988 - CHI-3_C>

ER -SIMCA

	S_sCH3	S_ssCH2	S_dsCH	S_aaCH	S_sssCH	S_dssC	S_aasC	S_ssssC
M	0.452	0.700	0.241	0.618	0.773	0.461	0.776	0.323
D	1.711	1.144	1.384	0.931	3.414	1.483	1.167	1.348

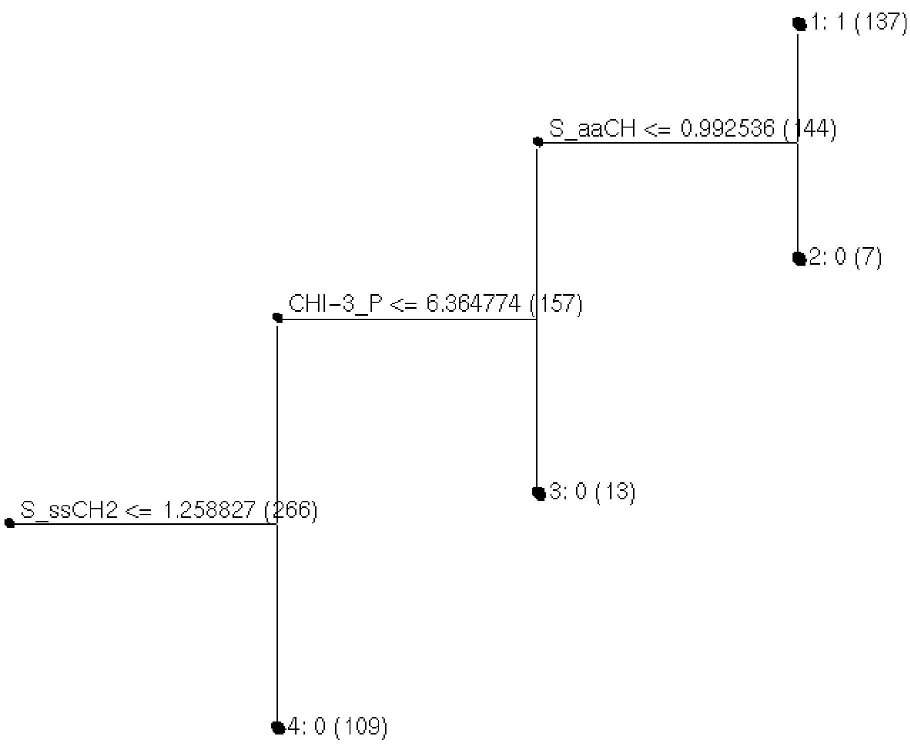
	S_sOH	S_dO	S_ssO	S_sCl	IC	SIC	IAC-Total
M	0.833	0.843	0.492	0.597	0.818	0.746	0.965
D	1.375	1.782	1.568	2.936	2.601	2.716	2.108

	E-DIST-mag	MW	Rotlbonds	Hbond_acceptor	Hbond_donor	AlogP	MolRef
M	0.935	0.947	0.878	0.836	0.789	0.599	0.967
D	2.455	1.962	1.296	1.886	1.198	1.468	1.237

	SC-3_C	CHI-2	CHI-3_P	CHI-3_C	CHI-V-1	CHI-V-2	CHI-V-3_P	CHI-V-3_C
M	0.960	0.968	0.949	0.885	0.975	0.951	0.941	0.767
D	1.725	1.331	1.799	1.292	1.238	1.412	1.178	1.522

	Kappa-1	Kappa-2	Kappa-3	PHI	JX	Wiener
M	0.980	0.982	0.905	0.907	0.749	0.902
D	1.667	2.824	2.211	1.270	1.127	2.845

ER -RP



ER-SFGA

0.68804 + 0.051625 * <"AlogP" - 1.76> - 0.208596 * <2.09214 - "S_ssCH2"> - 0.215077 * <2 - "Hbond_donor">

MAO-SIMCA

	S_sCH3	S_ssCH2	S_aaCH	S_sssCH	S_tsC	S_dssC	S_aasC	S_ssssC
M	0.789	0.619	0.517	0.433	0.243	0.326	0.595	0.145
D	4.520	1.264	1.449	1.654	1.315	1.304	1.227	0.943

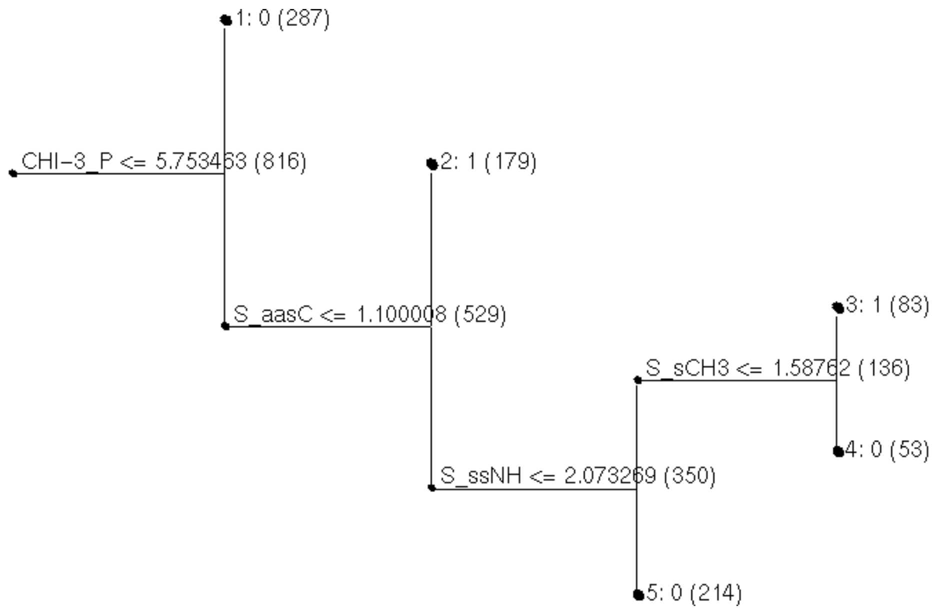
	S_sNH2	S_ssNH	S_aaN	S_sssN	S_sOH	S_dO	S_ssO	S_sCl
M	0.300	0.176	0.050	0.289	0.281	0.381	0.241	0.132
D	1.115	1.179	1.099	0.958	1.181	0.958	1.124	0.849

	IC	CIC	SIC	MW	Rotlbonds	Hbond_acceptor	Hbond_donor
M	0.562	0.678	0.537	0.769	0.797	0.536	0.604
D	0.926	0.922	0.886	0.961	1.516	1.189	1.144

	AlogP	JX	Kappa-1-AM	Kappa-2-AM	Kappa-3-AM	SC-3_C	CHI-1
M	0.653	0.691	0.961	0.868	0.823	0.879	0.959
D	1.271	2.434	1.551	1.235	2.150	1.286	1.757

	CHI-2	CHI-3_P	CHI-3_C	CHI-V-1	CHI-V-2	CHI-V-3_P	CHI-V-3_C	Wiener
M	0.958	0.946	0.846	0.959	0.913	0.838	0.773	0.835
D	1.156	2.561	2.328	1.577	1.832	1.201	3.129	1.414

MAO-RP



MAO-SFGA

1.18857 + 0.108836 * <Hbond_acceptor - 7> - 0.12213 * <1.58625 - S_sssN> - 0.056132 * <8.08219 - S_ssNH> - 0.078639 * <CHI-1 - 6.3637> - 0.047778 * <6.70926 - S_aaCH> + 0.464864 * <2.29802 - CHI-V-1>